

1. Inventory of Server Hosting NADCM

Steps: -

1. Record the number of servers hosting NADCM.
2. Verify & record FRU data of servers hosting NADCM by logging in BMC WEBGUI.

System info → FRU information

3. Verify & record hardware inventory of servers hosting NADCM by login to BMC WEBUI.

System info → System inventory

4. Verify & record FRU information, system and hardware inventory of servers hosting in NADCM by login in NADCM GUI.

Result: -

1. FRU data of servers hosting NADCM is recorded and valid.

The screenshot displays the BMC WebGUI interface with three panels: Chassis Information, Board Information, and Product Information. Each panel contains a table of system details.

Chassis Information Area Format Version 1	
Chassis Type	Rack Mount Chassis
Chassis Part Number	155BN000173
Chassis Serial Number	QTFCT98280020
Chassis Extra	

Board Information Area Format Version 1	
Language	English
Manufacture Date Time	Thu Jul 12 20:49:00 2018
Board Manufacturer	Nokia Solutions and Networks
Board Product Name	AR-D52B1-D/AF0315.01
Board Serial Number	8T982300119
Board Part Number	31558MB00G0
FRU File ID	V0.29
Board Extra	00

Product Information Area Format Version 1	
Language	English
Product Manufacturer	Nokia Solutions and Networks
Product Name	AR-S5001A-2A/AF1279.01
Product Part Number	155BN000173
Product Version	N/A
Product Serial Number	QTFCT98280020
Asset Tag	Reserved
FRU File ID	-
Product Extra	

The bottom of the image shows a Windows taskbar with a search bar, task icons, and system tray information including the date and time (9:41 AM 7/21/2026).



2. Sign in using Web GUI

Steps: -

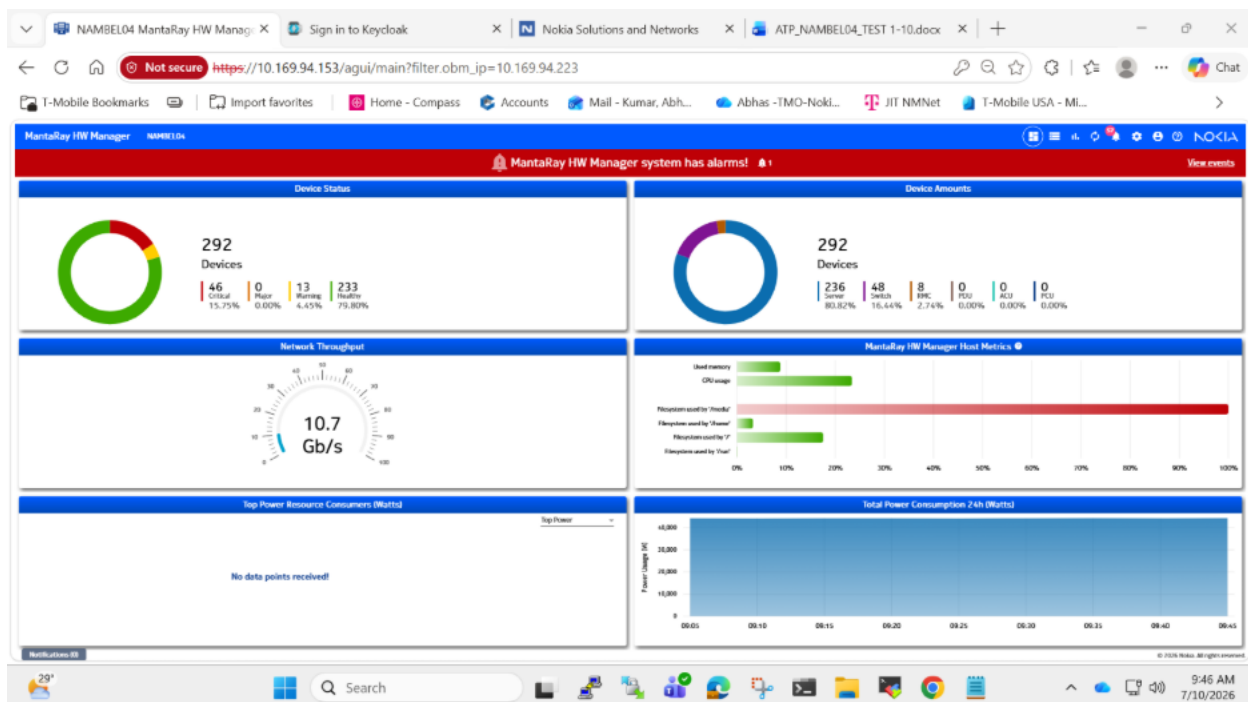
1. Enter NADCM address to the internet browser address bar
2. Example:

https://<NADCM_IP>/

3. Enter valid username and password to Nokia Airframe Data Center Manager login page.

(Note: The first administrator username and password are set during NADCM installation.)

Result: - Successfully logged in NADCM

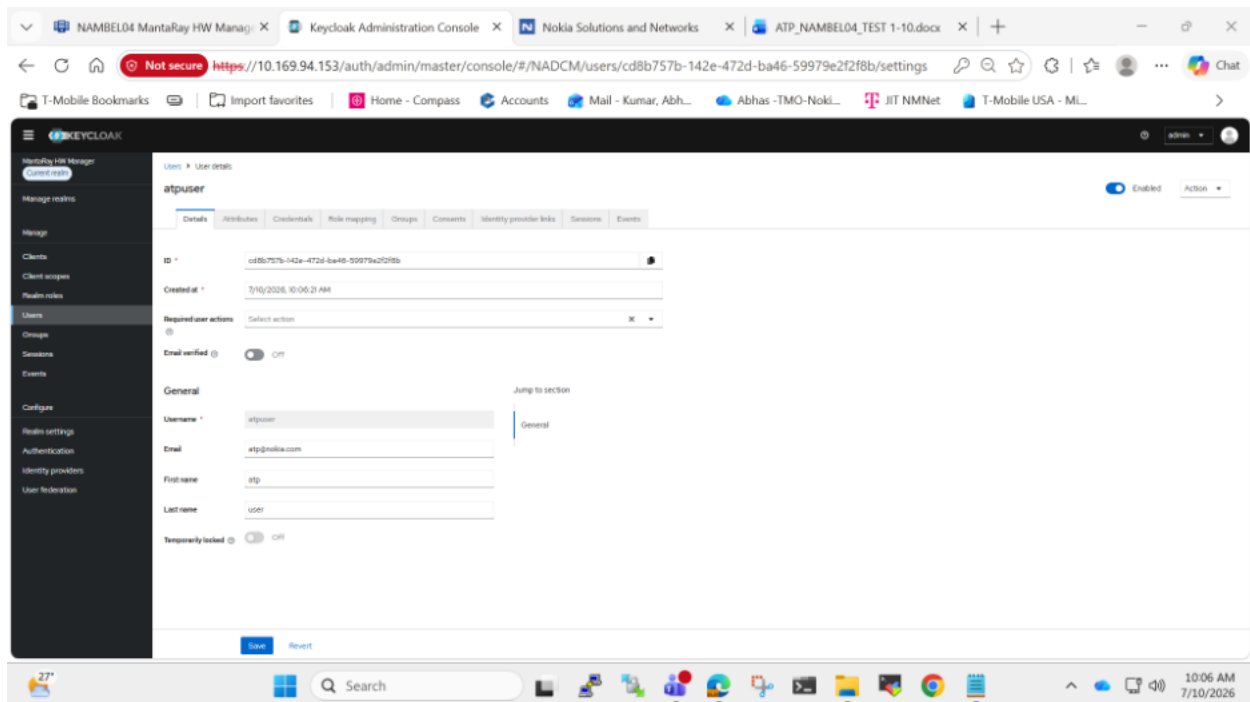


3. Adding User into Keycloak

Steps: -

1. Enter Keycloak address to the internet browser
e.g., [http:// <NADCM_IP>/auth/admin](http://<NADCM_IP>/auth/admin)
2. Login into Keycloak using username and password given in the NADCM installation phase
3. Select Add **user** under **Users** tab and enter data for new user and **save**.

Result: - Successfully created atpuser with mail id of nadcmuser@nokia.com



NAMBEL04 MantaRay HW Manag... x Keycloak Administration Console x Nokia Solutions and Networks x ATP_NAMBEL04_TEST 1-10.docx x

Not secure https://10.169.94.153/agui/main

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MantaRay HW Manager NAMBEL04

MantaRay HW Manager system has alarms! View events

Device Status

293 Devices

45 Critical 15.36% 0 Near 0.00% 13 Warning 4.44% 235 Healthy 80.20%

Device Amounts

293 Devices

237 Server 80.89% 48 Switch 16.38% 8 BMC 2.73% 0 FQD 0.00% 0 FCU 0.00%

Network Throughput

9.6 Gb/s

MantaRay HW Manager Host Metrics

Top Power Resource Consumers (Watt/s)

Total Power Consumption 24h (Watt/s)

User Management

ATP User

Roles Administrator

HW Roles

Change user password

Log out

Close

29° Search 10:13 AM 7/10/2026

1.Add license to MantaRay HW Manager

Steps: -

1. Login to NADCM GUI
2. Go to Administration → Licenses
3. Click on “Add Licenses” and select the XML file and upload it.

Result: -

License has been successfully imported to Mantaray HW Manager.

The screenshot displays the MantaRay HW Manager web interface. The browser address bar shows the URL: <https://10.169.94.153/agui/admin/licenses?filter.dynamicstate.alarmlighest=critical>. The interface has a blue header bar with the title "MantaRay HW Manager" and a sidebar on the left containing various navigation options like Files, Roles, Backup, Activity Logs, SNMP, Hardware Scan, etc. The main content area is divided into two sections: "Features" and "Licenses".

Features Section: This section displays a table with 3 features. The table has columns for Feature code, Description, Capacity, Used, Free, and Usage %.

Feature code	Description	Capacity	Used	Free	Usage %
0000018244	Integration-Express-NE Capacity	0	0	0	Not %
0000018242	Integration-Express-App SW	0	0	0	Not %
0000018172	Base Capacity	700	291	409	40 %

Licenses Section: This section displays a table with 2 licenses. The table has columns for License name, Serial number, Start time, End time, Capacity, and Actions.

License name	Serial number	Start time	End time	Capacity	Actions
MantaRay HW self-renew HW and AND LK	23871719137E7	a year ago	Expires in 4 years	0	
HW Management LK	7001024155113	a year ago	Expires in 4 years	700	

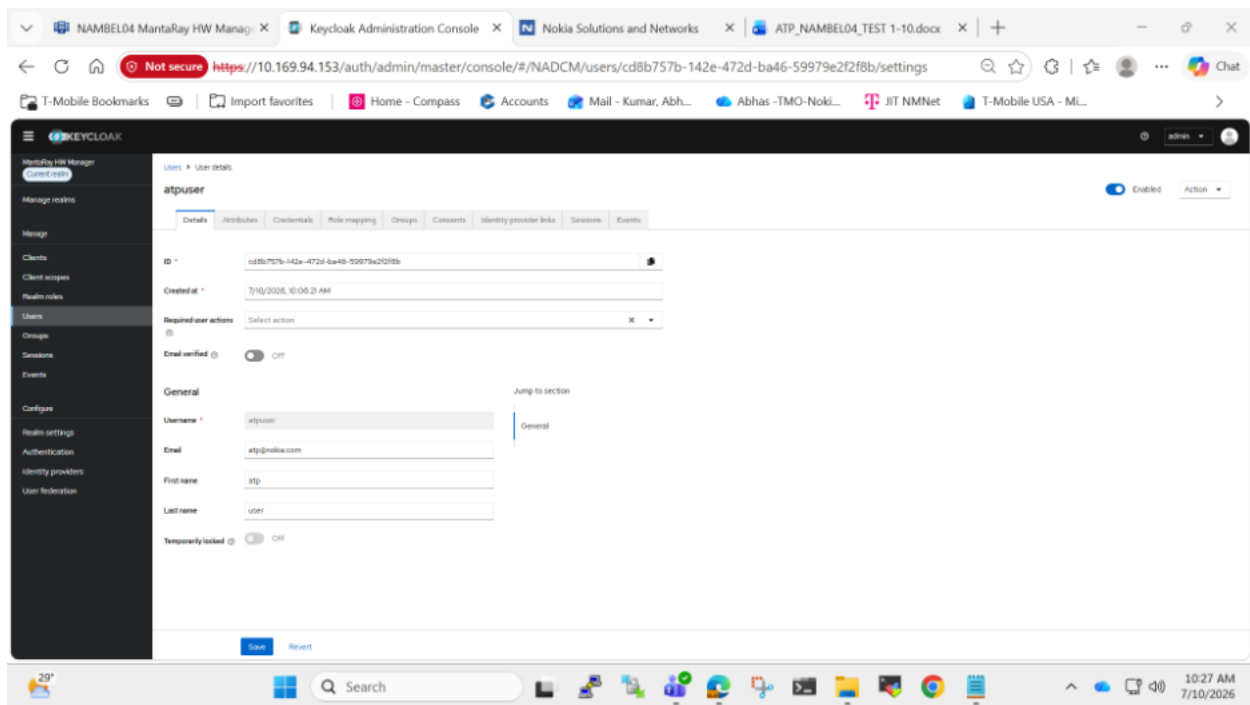
The bottom of the screenshot shows a Windows taskbar with the date and time as 3:19 PM on 7/10/2026.

5. Provision user account - admin access rights

Steps: -

1. Navigate to NADCM Keycloak GUI. <http:// <NADCM_IP>/auth/admin>
2. Click on **User** then **Add new user** option. Enter the required fields and press Submit.
3. Set password for created user & Select Temporary to OFF and click on **Credentials** → **enter the password**.
4. Click on **Role Mapping** Select proper role for the user e.g., **administrator** or auditor and press Save.
5. Click on **Groups** and grant admin rights.
6. Login NADCM using the new username/password.

Result: - Successfully created one user 'atpuser' and given admin rights access.



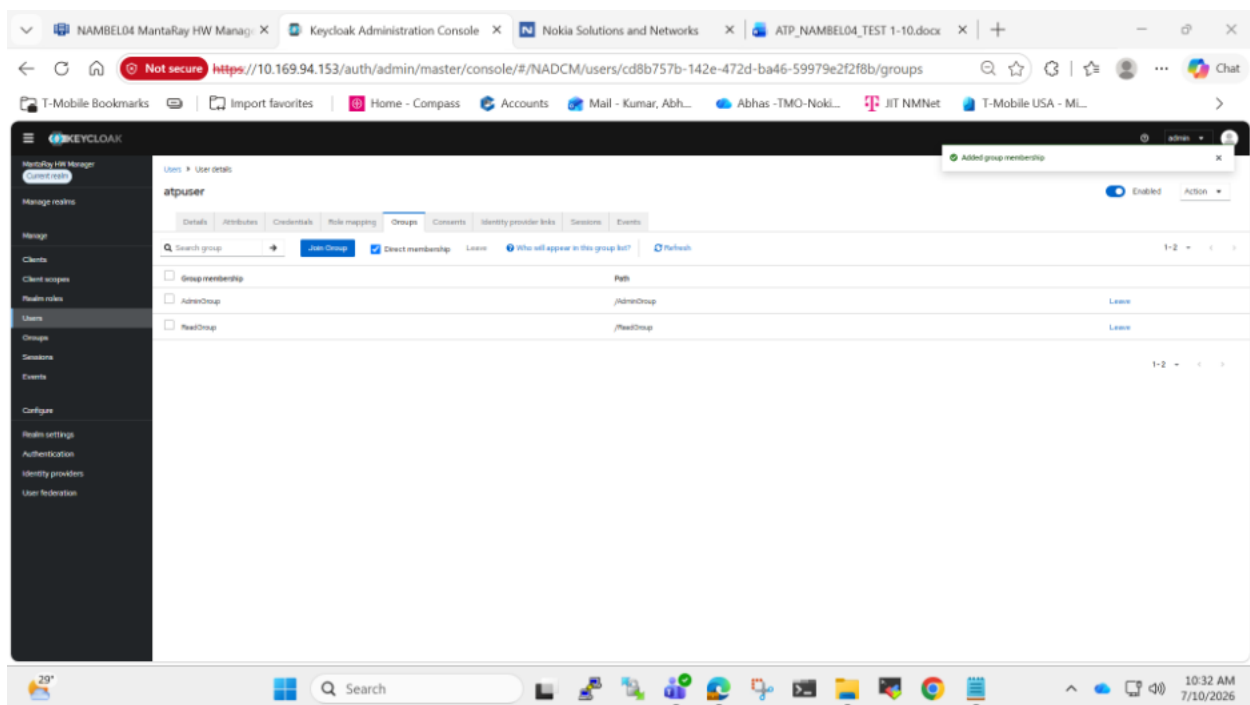
The screenshot shows the Keycloak Administration Console interface. The browser address bar displays the URL: <https://10.169.94.153/auth/admin/master/console/#/NADCM/users/cd8b757b-142e-472d-ba46-59979e2f2f8b/groups>. The interface is for the 'atpuser' realm, and the 'Groups' tab is selected. A notification banner at the top right states 'Added group membership'. The left sidebar contains navigation links: Management, Clients, Client scopes, Realm roles, Users, Groups, Sessions, Events, Configure, Realm settings, Authentication, Identity providers, and User federation. The main content area shows a table of groups with columns for search, group membership, path, and actions. The table has two rows: 'Group membership' with path '/' and 'AdminGroup' with path '/adminGroup'. The 'Group membership' row has a checkbox for 'Group membership' and a 'Leave' button. The 'AdminGroup' row has a checkbox for 'AdminGroup' and a 'Leave' button. The bottom of the screen shows the Windows taskbar with the search bar and various application icons.

6. Create and test read-only user

Steps: -

1. Navigate to NADCM Keycloak GUI. <http:// <NADCM_IP>/auth/admin>
2. Click on **User** then **Add new user** option. Enter the required fields and press Submit.
3. Set password for created user & Select Temporary to OFF and click on Credentials → **Enter the password.**
4. Click on **Role Mapping** Select proper role for the user e.g., **read-only** or whatever permission is required as per the request and press Save.
4. Click on **Groups** and select read only option.

Result: - Successfully created one user 'atpuser' and given read-only access.

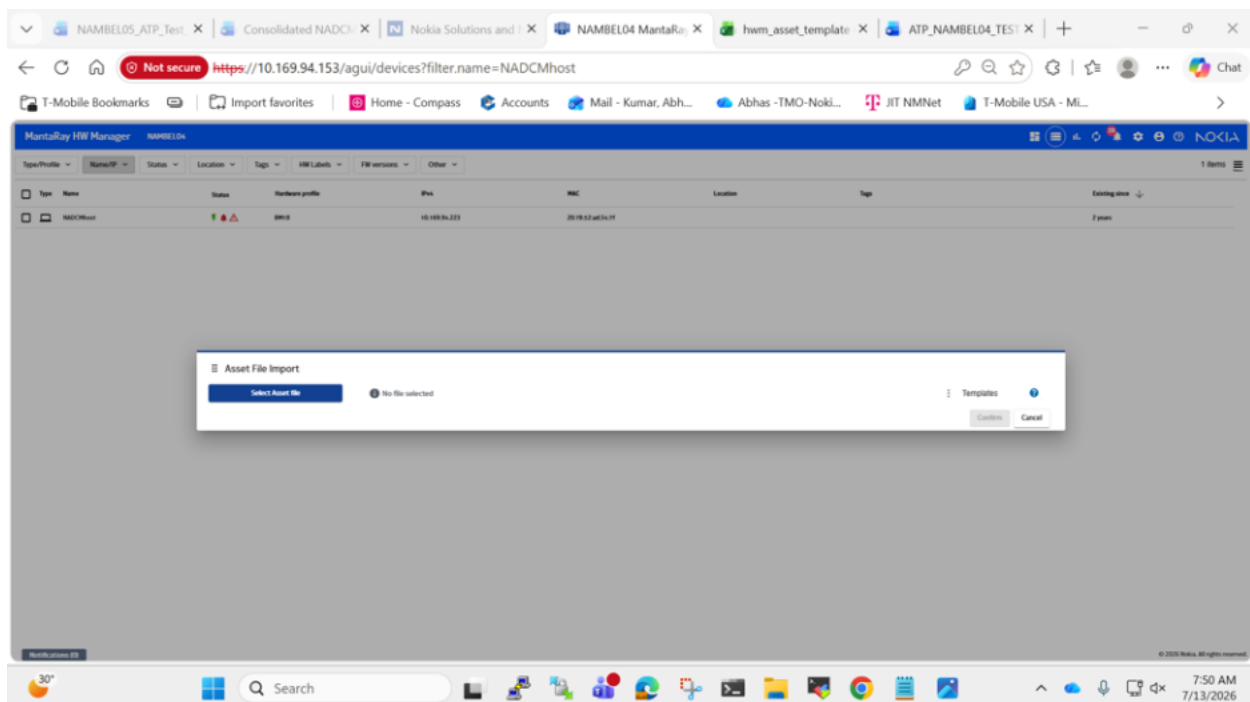


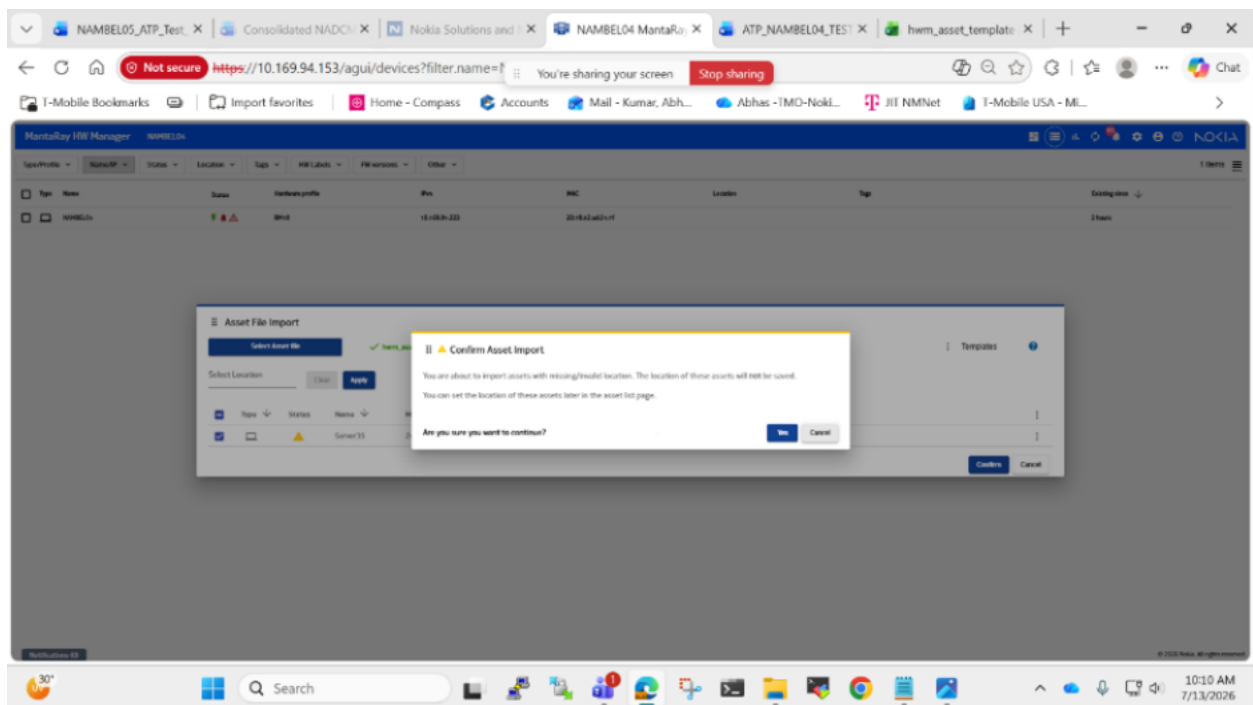
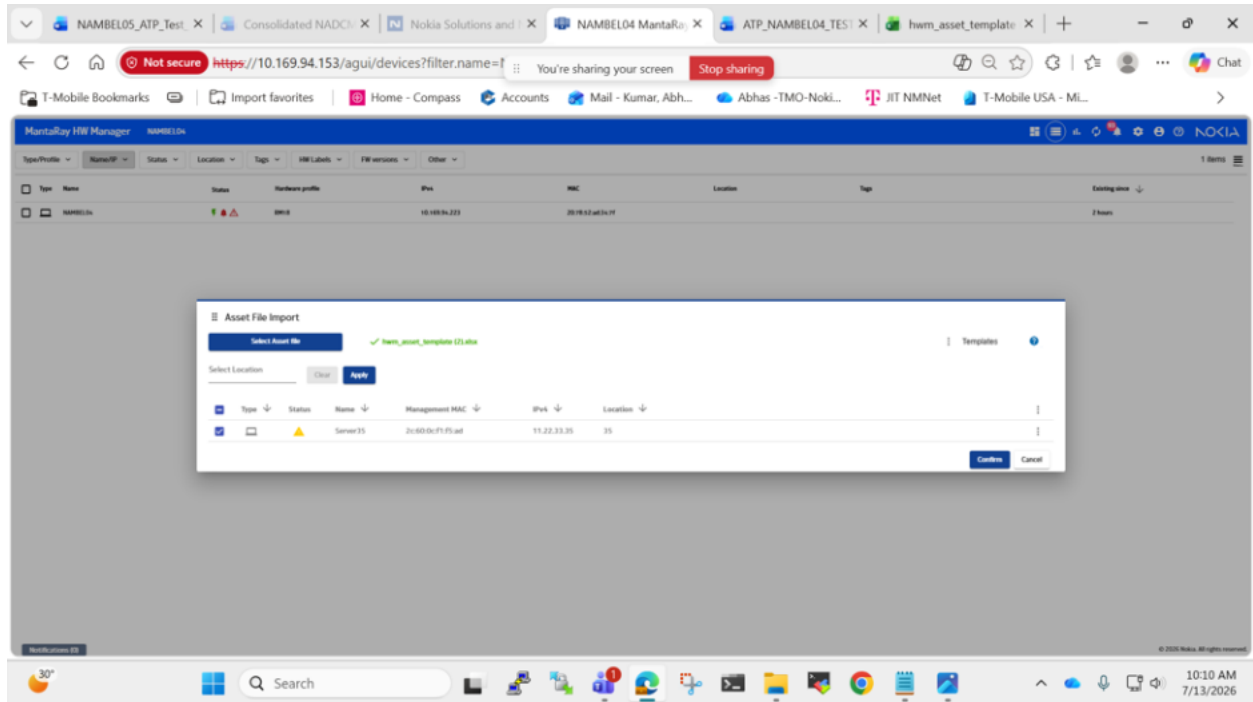
7. Adding new hardware using rack asset file

Steps: -

1. Login to NADCM and Click on Devices
2. Menu → Asset Import → Asset file Import
3. Download the Template and update the asset details
4. Upload the Asset file

Result: - New hardware has been added to the NADCM successfully using the rack asset file.





NAMBEL05_ATP_Test x Consolidated NADCh x Nokia Solutions and x NAMBEL04 MantaRay x ATP_NAMBEL04_TES hwm_asset_template x

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MantaRay IWM Manager NAMBEL04

Type Profile Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware profile	IPs	MAC	Location	Tags	Deleting status
Device	NAMBEL04	OK	HW01	10.169.94.153	20:84:ad:00:00:00			2 items

Asset File Import

Select Asset File hwm_asset_template (2.0.0.0) Templates

Select Location Close Apply

Type	Status	Name	Management MAC	IPs	Location
Device	OK	Server35	20:84:ad:00:00:00	10.169.94.153	35

Continue Cancel

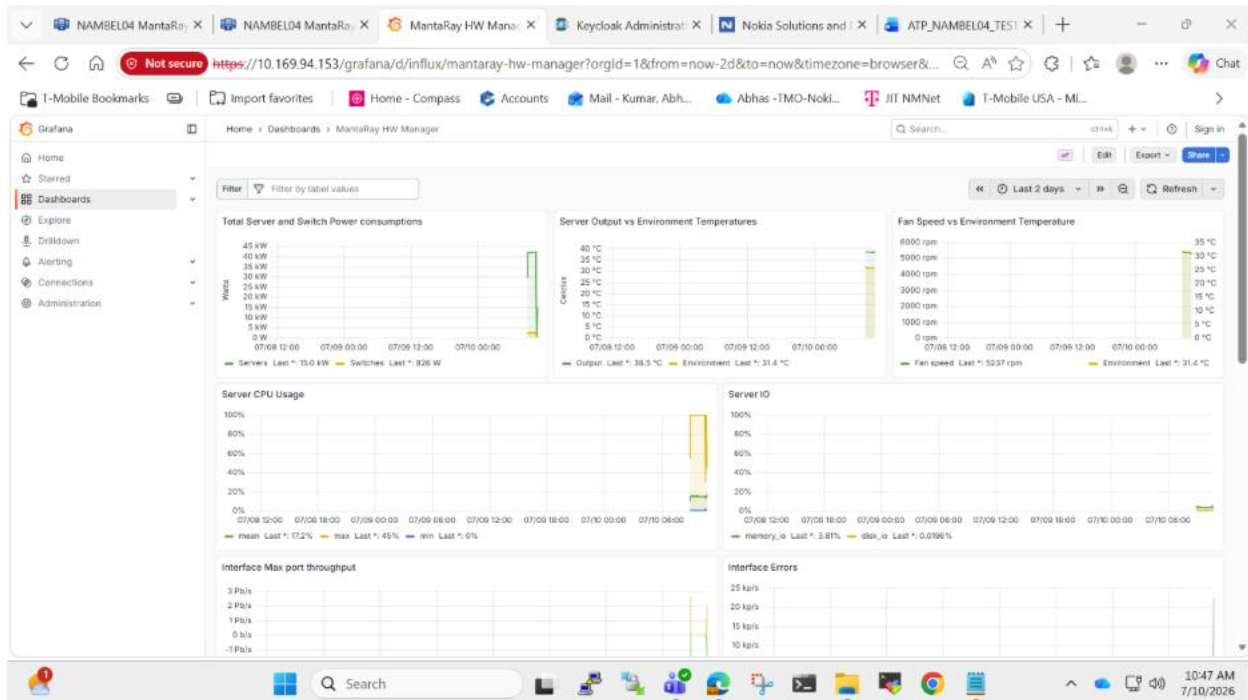
30° Search 10:11 AM 7/13/2026

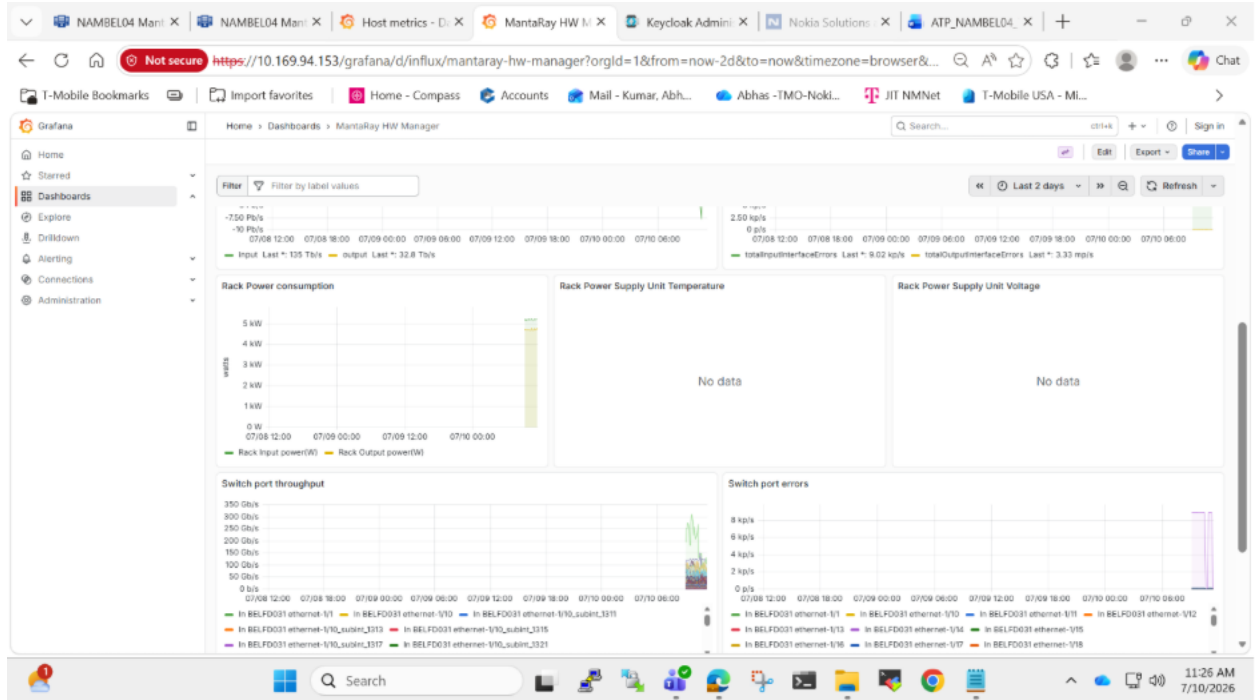
8. Check FAN Cooling, sensor readings, power usage

Steps: -

1. Go to NADCM GUI → Grafana
2. Different dashboard panels are visible and show data from e.g., server power consumption, fan speed etc.

Result: - Below screenshot showing fan cooling sensor readings and power usage



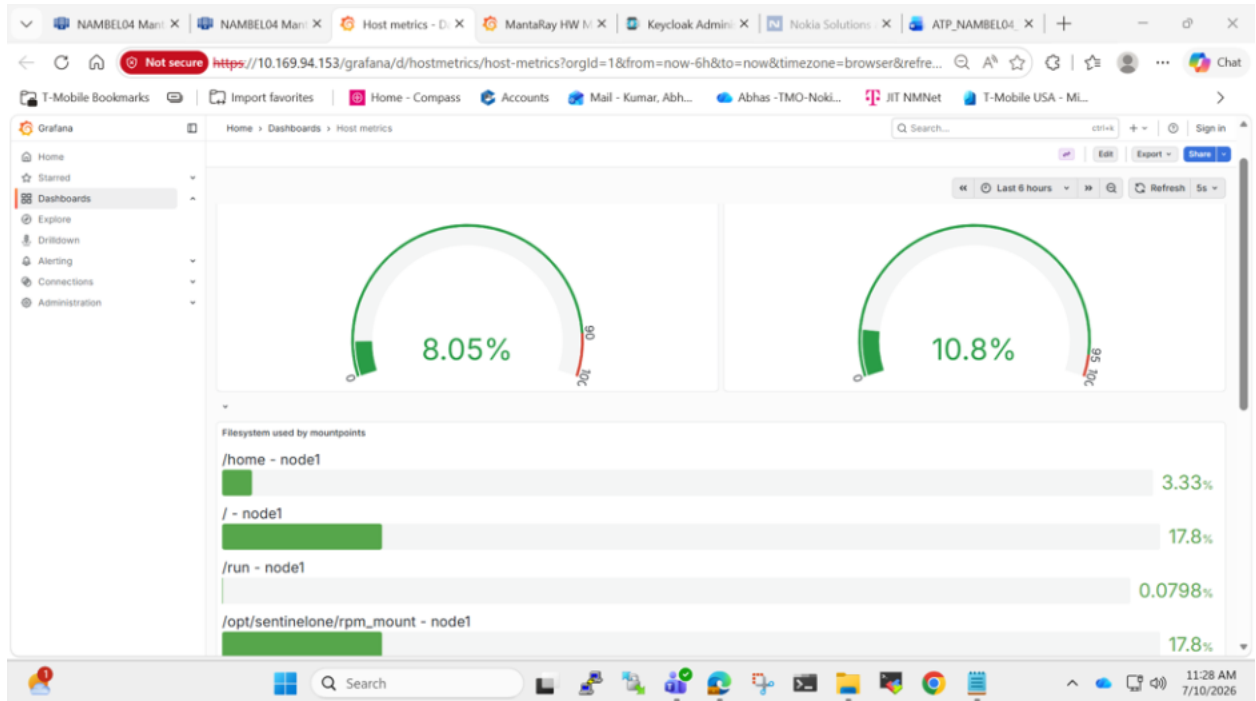


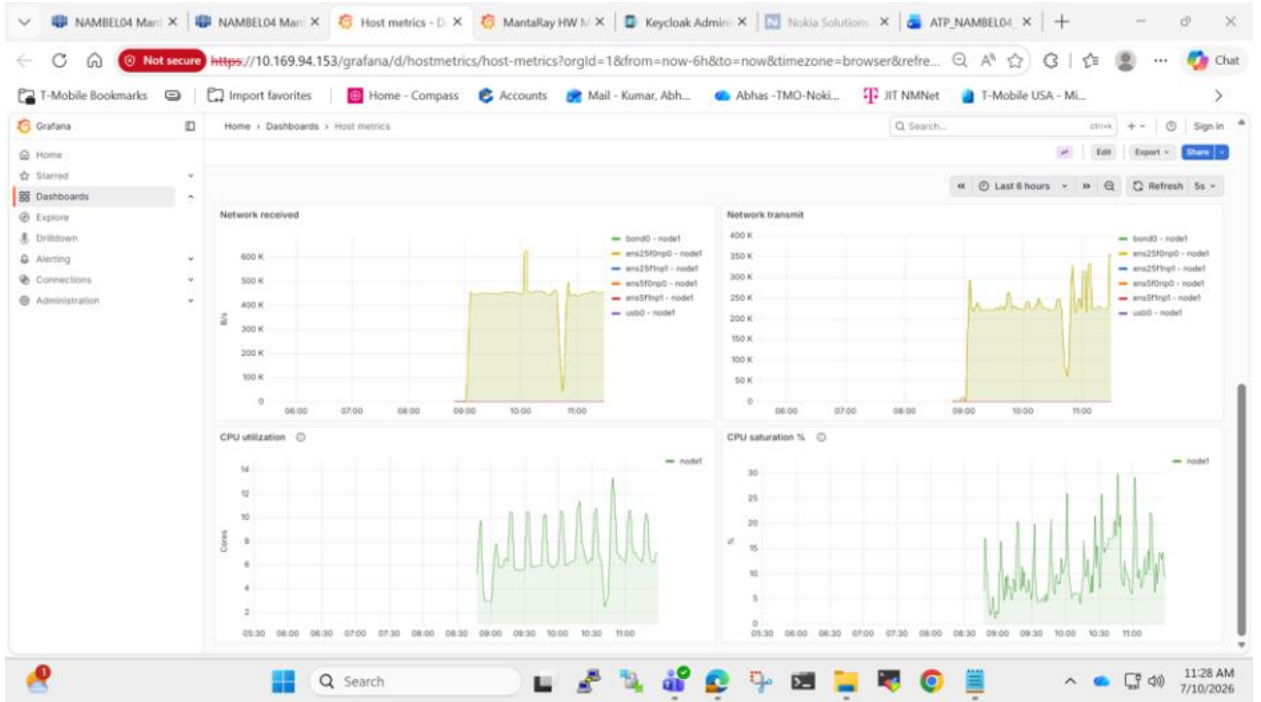
9. Check CPU, Memory, and I/O

Steps: -

1. Go to NADCM GUI → Grafana.
2. We can see the details as below

Result: -





10. Check event logs

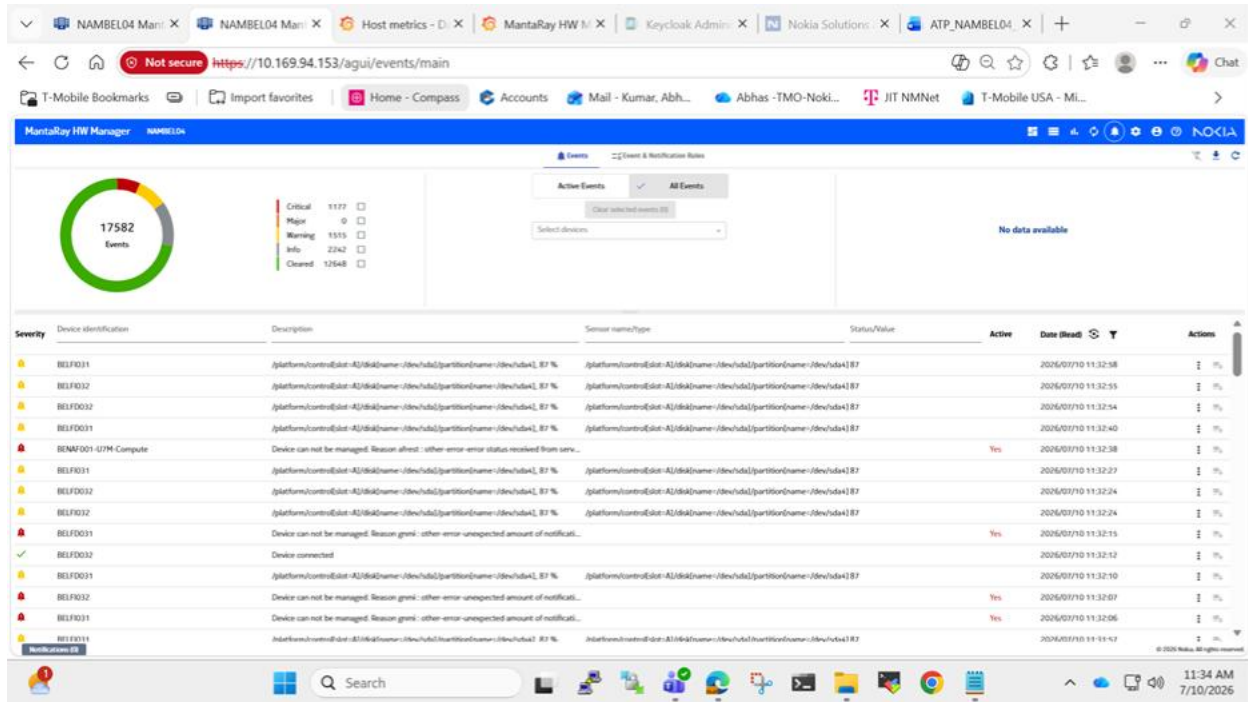
Steps: -

1. Login NADCM GUI.
2. Go to **Events** → **Active Events**. Active events are flagged as active according to defined rules.

The screenshot shows the MantaRay HW Manager NADCM GUI. The top navigation bar includes 'Events' and 'Event & Notification Rules'. The main content area displays a summary of 775 events, categorized by severity: Critical (751), Major (0), Warning (24), Info (0), and Cleared (0). Below the summary is a table of active events.

Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Read)	Actions
Critical	BELFI031	Device can not be managed. Reason gnm1 : other-error-unexpected ...			2026/07/10 11:30:16	
Critical	BELFD031	Device can not be managed. Reason gnm1 : other-error-unexpected ...			2026/07/10 11:30:09	
Critical	BENAF002-U32R-RMC02	Device can not be managed, snmpv3 used			2026/07/10 10:47:48	
Warning	NADCMhost	Login failed from IP:10.147.179.2	Login Info		2026/07/10 10:47:00	
Critical	BENEF001-U14L-Comp	Device can not be managed. Reason redfish : other-error-GetRoot er...			2026/07/10 10:44:19	
Critical	OPARF001_Master-1-U25	Device can not be managed. Reason ipmi : cred-error-Unauthorized			2026/07/10 10:44:16	
Critical	SIARF004-U14	Device can not be managed. Reason ipmi : cred-error-Unauthorized			2026/07/10 10:44:15	
Critical	BENAF001-U7R-Compute	Device can not be managed. Reason redfish : other-error-GetRoot er...			2026/07/10 10:44:10	
Critical	BENEF001-U14R-Un_cloud	Device can not be managed. Reason redfish : cred-error-Unauthorized			2026/07/10 10:43:46	

3. Go to Events->All Events. All Events show the history of all events.



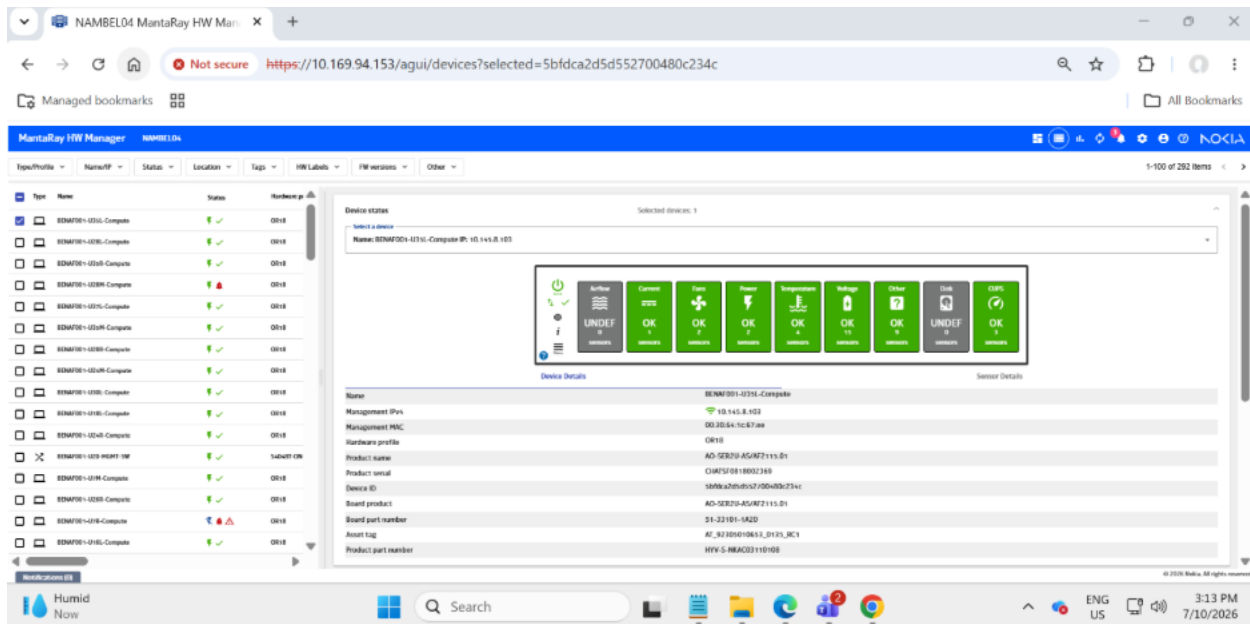
4. If no events are visible a new threshold rule can be created* for the system to raise an alarm event. Go to Event Rules. (**Events -> Event Rules**)
5. New Threshold Rule can be created using the Add rule button.

11. Check BIOS settings

Steps: -

1. In NADCM GUI go to **Devices**.
2. Select **the device** using checkbox.
 - a. Select Set BIOS parameters from **Menu** → **Set BIOS parameters**.
3. Available BIOS parameters and their current values are visible in Set BIOS Parameters panel.

Result: - Below screenshot showing BIOS settings of a server.

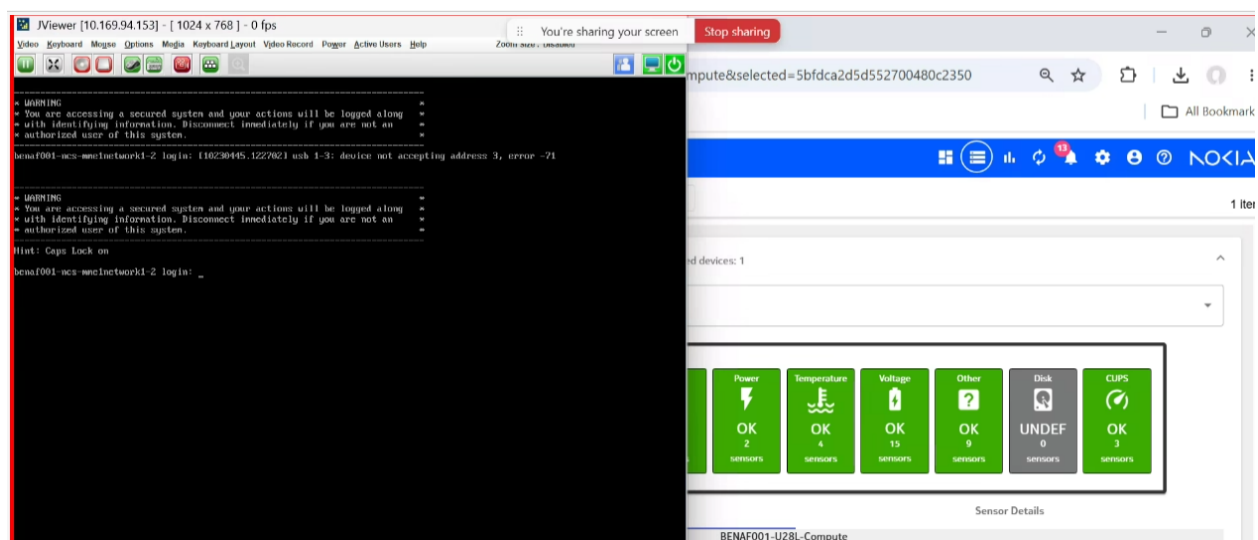
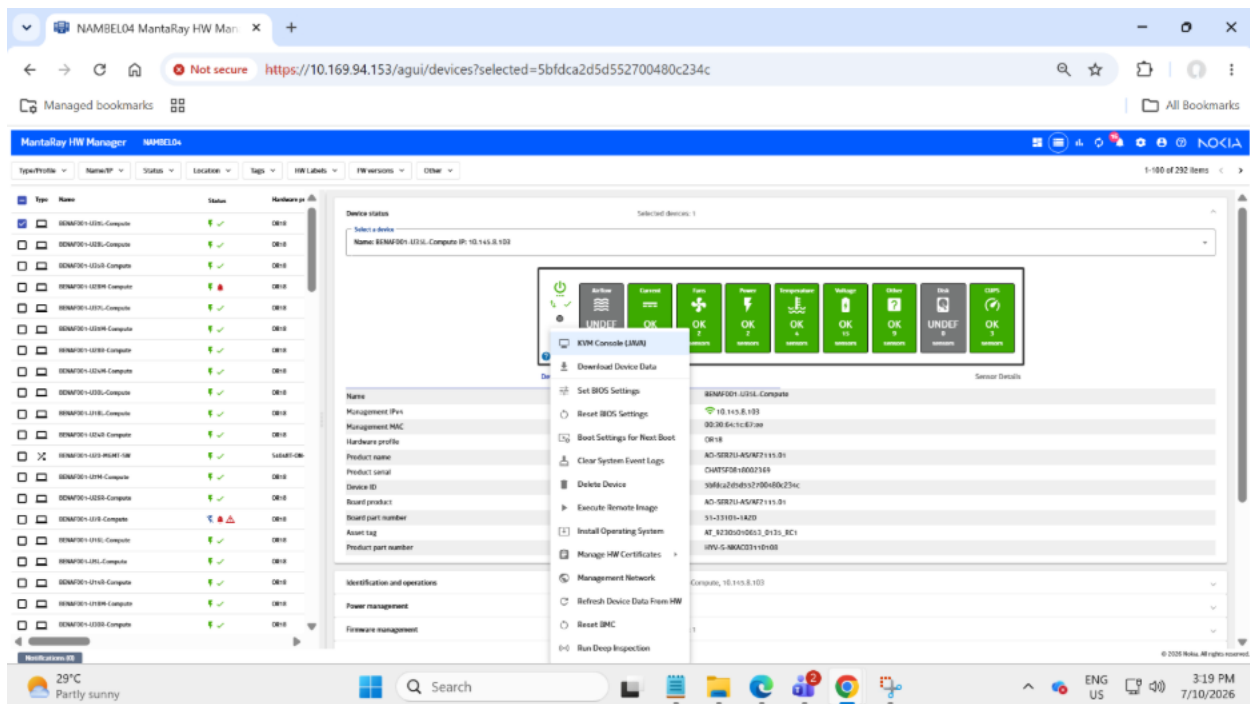


12. Check Remote Console for servers

Steps: -

1. Go to **Devices** → Select the server
2. At right side **Menu** → Select **KVM (Kernel based Virtual Machine) Console (JAVA)**
3. Console connection is set up when java application is executed.

Result: - KVM Console opened.

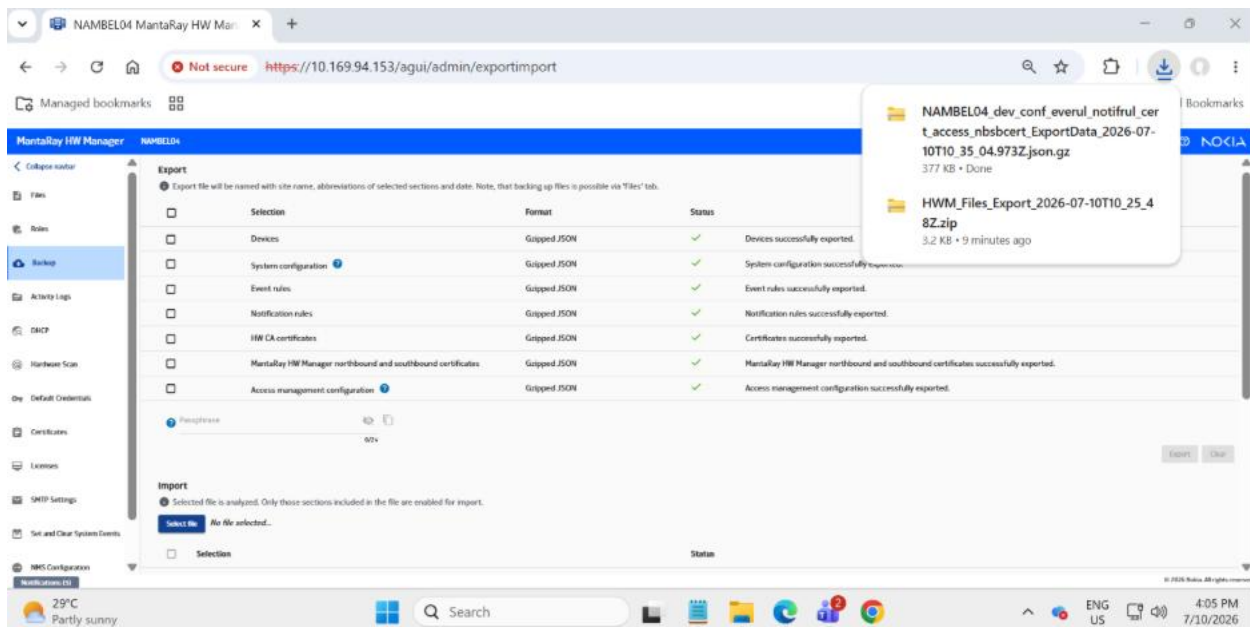


13. Backup and Restore datacentre inventory

Steps: -

1. Select the Administration → Backup → In Export, select “Devices” and enter passphrase.
2. Select the export and hit Yes.
3. Delete the hardware inventory completely.
4. Import hardware inventory back from saved JSON file.
5. Choose file saved from step 2 and click on save.

Result: - Below screenshot shows the backup of hardware inventory and once deletion of hardware we can restore it from backup Successfully.



← Collapse sidebar

Files

Rules

Backup

Activity Logs

DMCP

Hardware Scan

Default Credentials

Certificates

Licenses

SMTP Settings

Set and Clear System Events

NMS Configuration

Hardware Allocations

Managing HW Manager northbound and southbound certificates

Imported JSON

Access management configuration

Imported JSON

Importance

92x

Export

Clear

Import

Clear

Imported

Clear

Import

Selected file is analyzed. Only those sections included in the file are enabled for import.

Select file

No file selected...

Selection

Devices

System configuration

Event rules

Notification rules

HW CA certificates

Managing HW Manager northbound and southbound certificates

Access management configuration

Set default user password?

92x

User federation password?

92x

Import

Clear

Notifications: 0

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14. Power off/Power on OR18 servers using NADCM

Steps: -

1. Login NADCM and Go to Devices
2. Select OR18 server
3. Click on Power button to Power off
4. Check the power status and Power on again

Result: -

Power off/on has been completed successfully.

The screenshot displays the MantaRay HW Manager web interface. The browser address bar shows a URL: https://10.169.94.153/agui/devices?filter.inventory_node_type=server&filter.name=OPNAF004_Compute-U18R&selected=6.... The interface has a blue header with the 'MantaRay HW Manager' logo and a navigation bar with tabs for 'Type/Profile', 'Name/IP', 'Status', 'Location', 'Tags', 'HW Labels', 'FW versions', and 'Other'. A sidebar on the left lists device types, with 'OPNAF004_Compute-U18R' selected. The main panel shows the 'Device status' for the selected device, 'OPNAF004_Compute-U18R IP: 10.145.217.90'. Below this, a row of status indicators is displayed: 'Power' (ON), 'Current' (OK), 'Fan' (OK), 'Power' (OK), 'Temperature' (OK), 'Voltage' (OK), 'Clock' (OK), and 'CUPS' (OK). The 'Device Details' section on the right lists various attributes: Name (OPNAF004_Compute-U18R), Management IPv4 (10.145.217.90), Management MAC (00:30:64:3d:a3:34), Hardware profile (OR18), Product name (AO-SER2U-AS/A/2115.01), Product serial (CH75N1022014793), Device ID (64217264c811fb82c74498d2), Board product (AO-SER2U-AS/A/2115.01), and Reset count number (1/1-11101-1A20). The bottom of the screen shows a Windows taskbar with the search bar and various application icons.

NAMBEL04 MantaRay HW Manager x ATP_NAMBEL04.docx x +

Not secure https://10.169.94.153/aguui/devices?filter.invento... You're sharing your screen Stop sharing 4_Compute-U18R&selected=6...

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004_Compute-U18R		OR18

Identification and operations OPNAF004_Compute-U18R, 10.145.217.90

Power management ON

Device (Server)	Power	State	Connectivity	Details
10.145.217.90 OPNAF004_Compute-U18R		ON	ok	ok

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Credential management Selected devices: 1

Location management TMO Overland Park / OPNAF004 / 6600 / 6624 / 18 / Right

Interface management Interface count: 1

Tag management Selected devices: 1

Notifications DB

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6:40 AM 7/21/2026

NAMBEL04 MantaRay HW Manager x ATP_NAMBEL04.docx x +

Not secure https://10.169.94.153/aguui/devices?filter.invento... You're sharing your screen Stop sharing 4_Compute-U18R&selected=6...

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004_Compute-U18R		OR18

Identification and operations OPNAF004_Compute-U18R, 10.145.217.90

Power management ON

Device (Server)	Power	State	Connectivity	Details
10.145.217.90 OPNAF004_Compute-U18R		OFF "GRACEFUL SHUTDOWN"	ok	ok

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Credential management Selected devices: 1

Location management TMO Overland Park / OPNAF004 / 6600 / 6624 / 18 / Right

Interface management Interface count: 1

Tag management Selected devices: 1

Notifications DB

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6:40 AM 7/21/2026

ATP_NAMBEL04.docx x NAMBEL04 MantaRay HW Manager x +

Not secure https://10.169.94.153/aguui/devices?filter.name=OPNAF004_Compute-U18R&selected=64217266c611fb82c74498d2 Chat

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas - TMO-Noki... JIT NMNet T-Mobile USA - ME... NITAC RAN OSS ...

MantaRay HW Manager NAMBEL04

Type/Profile v Name/TP v Status v Location v Tags v HW Labels v FW versions v Other v 1 Items

Type	Name	Status	Hardware
☒	OPNAF004_Compute-U18R		OR18

Asset tag: AT67649274993_0307

Product part number: HYV-5-NKAC03300151

Identification and operations: OPNAF004_Compute-U18R, 10.145.217.90

Power management: ON

Device (Server)	Power	State	Connectivity	Details
10.145.217.90 OPNAF004_Compute-U18R		ON	ok	

Save Clear

Firmware management: Selected devices: 1

Bank selection: Selected devices: 1

Credential management: Selected devices: 1

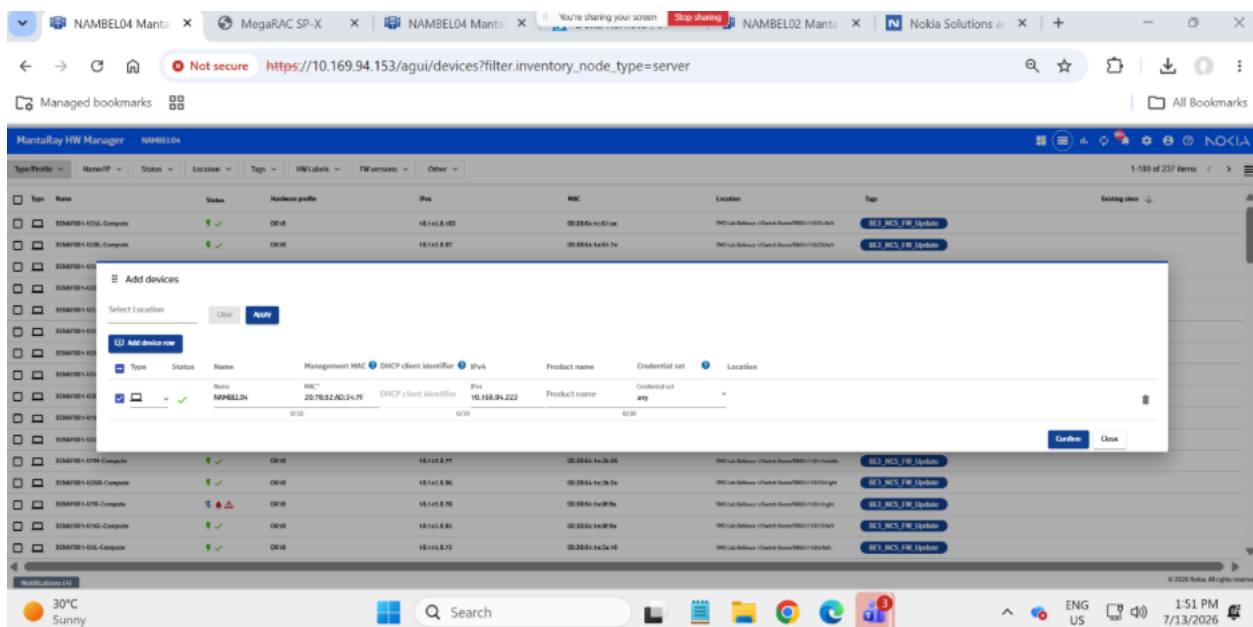
Location management: TMO Overland Park / OPNAF004 / 6600 / 6624 / 18 / Right

Interface management: Interface count: 1

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Notifications 0

You're sharing your screen Stop sharing Search 6:50 AM 7/21/2026



16. Check hardware connectivity

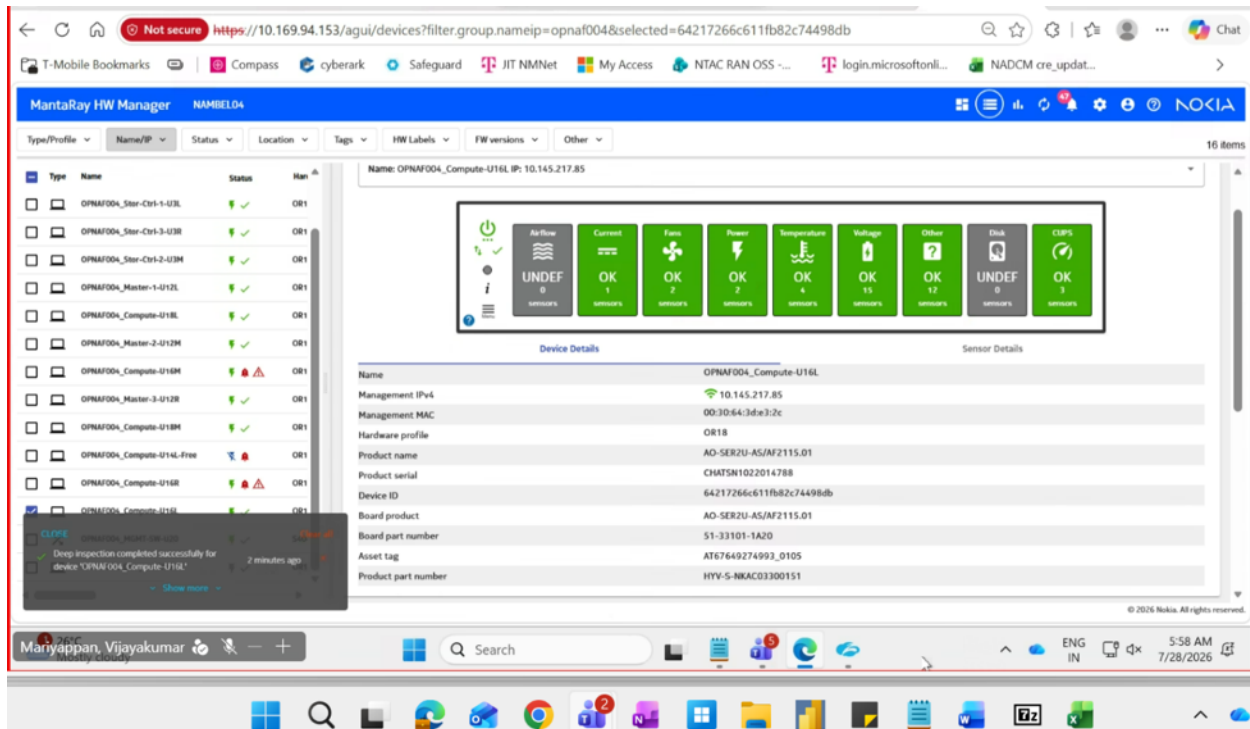
Steps: -

1. Go to Devices.
2. Check device status from Status column.
3. (Possible hardware states are NoConnection, PowerON and PowerOFF.)

Result: Below screenshot shows the hardware connectivity status Successfully.

The screenshot displays the MantaRay HW Manager web interface. The browser address bar shows the URL <https://10.169.94.153/agui/devices>. The interface includes a navigation bar with tabs for 'Type/Profile', 'Name/IP', 'Status', 'Location', 'Tags', 'HW Labels', 'FW versions', and 'Other'. Below the navigation bar, a table lists various devices. Each row contains a checkbox, a device name, a status indicator (green checkmark for 'Success', red X for 'Error'), a hardware profile, an IP address, a MAC address, a location, and a 'Type' column with a 'Link' button. The status column shows that most devices are in a 'Success' state, while one device, 'B3M4F101-1038-MG45-DW', is marked as 'Error'.

Type	Name	Status	Hardware profile	IPs	MAC	Location	Type
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.103	00:30:84:14:57:07	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.97	00:30:84:14:51:79	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.105	00:30:84:14:57:09	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.101	00:30:84:14:50:76	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.106	00:30:84:14:54:42	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.104	00:30:84:14:51:77	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.99	00:30:84:14:56:44	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.92	00:30:84:14:50:64	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.100	00:30:84:14:51:64	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.88	00:30:84:14:50:45	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.83	00:30:84:14:52:06	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☒	B3M4F101-1030-MG45-DW	Error	Su-001-001-001-001	10.145.8.85	00:30:84:14:52:06	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.77	00:30:84:14:50:06	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.96	00:30:84:14:50:64	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.101	00:30:84:14:50:76	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link
☐	B3M4F101-1030-Complete	Success	CR-10	10.145.8.81	00:30:84:14:50:45	1902 Lab-Belmonte-VL-Campus-Rover/190211/190211/01	Link



17. Deep Inspection for Servers

Steps: -

1. From Devices view select the device and then right-side Menu, select Run Deep
2. Inspection Check the status in Activity Logs Page in Administration- Activity Logs.

Result: Deep inspection has been completed for server as below.

Home | Magi | X | NADCM cre... | T-Mo_Engine... | My Files | You're sharing your screen | Stop sharing | NAMBEL04 | NAMBEL04 | + | - |

Not secure https://10.169.94.153/agui/admin/dhcpd

T-Mobile Bookmarks | Compass | cyberark | Safeguard | JIT NMNet | My Access | NTAC RAN OSS... | login.microsoft... | NADCM cre.updat...

MantaRay HW Manager NAMBEL04

Files Roles Backup Activity Logs DHCP Hardware Scan Default Credentials Certificates

Notifications 0/8

Modify IPv4 Subnet 10.145.217.64

Subnet parameters

Subnet ☒ ENABLED Instant hardware scan ☒ ENABLED

Options

☐ Deny unknown clients ☐ Deny unknown host ports

Subnet *

Subnet* 10.145.217.64 Network* 255.255.255.192

IP address pools (Hint: Allowed IP range is 10.145.217.64 - 10.145.217.127)

First address 10.145.217.66 Last address 10.145.217.100

DHCP options

Gateway IP TFTP server

Next Save Cancel

26°C Mostly cloudy

5:38 AM 7/28/2026

Home | Magi | X | NADCM cre... | T-Mo_Engine... | My Files | You're sharing your screen | Stop sharing | NAMBEL04 | NAMBEL04 | + | - |

Not secure https://10.169.94.153/agui/devices?filter.group.nameip=opnaf004&selected=64217266c611fb82c74498db

T-Mobile Bookmarks | Compass | cyberark | Safeguard | JIT NMNet | My Access | NTAC RAN OSS... | login.microsoft... | NADCM cre.updat...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other

16 items

Type Name Status Location

OPNAF004_Master-1-U12L OR1

OPNAF004_Compute-U18L OR1

OPNAF004_Master-2-U12M

OPNAF004_Compute-U16M

OPNAF004_Master-3-U12R

OPNAF004_Compute-U18M

OPNAF004_Compute-U14L-free OR1

OPNAF004_Compute-U16R OR1

OPNAF004_Compute-U16L OR1

OPNAF004_MGMT-SW-U20 S40

OPNAF004_Compute-U18R OR1

Device status

Select a device

Name: OPNAF004_Compute-U16L IP: 10.145.217.85

Run Deep Inspection

Note that deep inspection will reboot the node

Yes Cancel

CLIPS OK 3 sensors

Device Details

Name OPNAF004_Compute-U16L

Management IP4 10.145.217.85

Management MAC 00:30:64:3d:e3:2c

Hardware profile OR18

Sensor Details

AO-SF8211-AS/AF2115.01

26°C Mostly cloudy

5:48 AM 7/28/2026

Not securehttps://10.169.94.153/agui/devices?filter.group.name=ip=opnaf004&selected=64217266c611fb82c74498db

T-Mobile BookmarksCompasscyberarkSafeguardJIT NMNetMy AccessNTAC RAN OSS...login.microsoftonli...NADCM cre_updat...

MantaRay HW ManagerNAME104

Type/ProfileNameStatusLocationTagsHW LabelsFW versionsOther16 items

Type	Name	Status	Man
☐	OPNAF004_Ster-Ctrl-1-U3L		OR1
☐	OPNAF004_Ster-Ctrl-3-U3R		OR1
☐	OPNAF004_Ster-Ctrl-2-U3M		OR1
☐	OPNAF004_Master-1-U12L		OR1
☐	OPNAF004_Compute-U18L		OR1
☐	OPNAF004_Master-2-U12M		OR1
☐	OPNAF004_Compute-U16M		OR1
☐	OPNAF004_Master-3-U12R		OR1
☐	OPNAF004_Compute-U18M		OR1
☐	OPNAF004_Compute-U14L-Free		OR1
☐	OPNAF004_Compute-U16R		OR1
☒	OPNAF004_Compute-U16L		OR1

Close

Deep inspection completed successfully for device "OPNAF004_Compute-U16L" 2 minutes ago

Show more

Name: OPNAF004_Compute-U16L IP: 10.145.217.85

Airflow

UNDEF

0

sensors

Current

OK

1

sensors

Fans

OK

2

sensors

Power

OK

2

sensors

Temperature

OK

4

sensors

Voltage

OK

15

sensors

Other

OK

12

sensors

Disk

UNDEF

0

sensors

CLIPS

OK

3

sensors

Device Details

Sensor Details

Name	OPNAF004_Compute-U16L
Management IPv4	10.145.217.85
Management MAC	00:30:64:3de3:2c
Hardware profile	OR18
Product name	AO-SER2U-AS/AF2115.01
Product serial	CHATSNI022014788
Device ID	64217266c611fb82c74498db
Board product	AO-SER2U-AS/AF2115.01
Board part number	51-33101-1A20
Asset tag	AT67649274993_0105
Product part number	HYV-5-NKAC03300151

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Mariyappan, Vijayakumar

26°C

Search

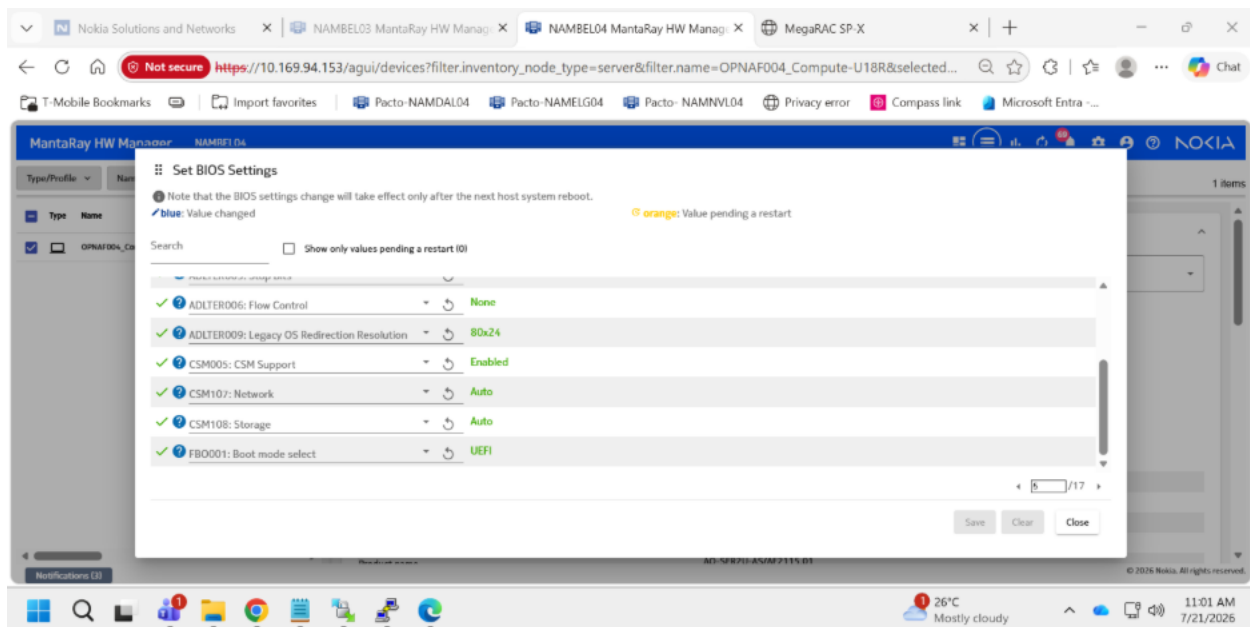
ENG IN 5:58 AM 7/28/2026

18. Set BIOS Parameters using NADCM

Steps: -

1. In NADCM GUI go to Devices → Select Server
2. Click on Menu at right side and Select Set BIOS Settings
3. Change and save the required BIOS setting value.

Result: - BIOS parameter settings has been changed.



Nokia Solutions and Networks | NAMBEL03 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | MegaRAC SP-X

Not secure https://10.169.94.153/agui/devices?filter.inventory_node_type=server&filter.name=OPNAF004_Compute-U18R&selected...

T-Mobile Bookmarks | Import favorites | Pacto-NAMDAL04 | Pacto-NAMELG04 | Pacto-NAMNVL04 | Privacy error | Compass link | Microsoft Entra ...

MantaRay HW Manager

Set BIOS Settings

Note that the BIOS settings change will take effect only after the next host system reboot.

blue: Value changed | orange: Value pending a restart

Search ☐ Show only values pending a restart (0)

ADLTER006: Flow Control	None
ADLTER009: Legacy OS Redirection Resolution	80x24
CSM005: CSM Support	Enabled
CSM107: Network	Auto
CSM108: Storage	Auto
FR0001: Boot mode select	LEGACY

5 / 17

Save Clear Close

26°C Mostly cloudy 11:01 AM 7/21/2026

Nokia Solutions and Networks | NAMBEL03 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | MegaRAC SP-X

Not secure https://10.169.94.153/agui/devices?filter.inventory_node_type=server&filter.name=OPNAF004_Compute-U18R&selected...

T-Mobile Bookmarks | Import favorites | Pacto-NAMDAL04 | Pacto-NAMELG04 | Pacto-NAMNVL04 | Privacy error | Compass link | Microsoft Entra ...

MantaRay HW Manager

Set BIOS Settings

Note that the BIOS settings change will take effect only after the next host system reboot.

blue: Value changed | orange: Value pending a restart

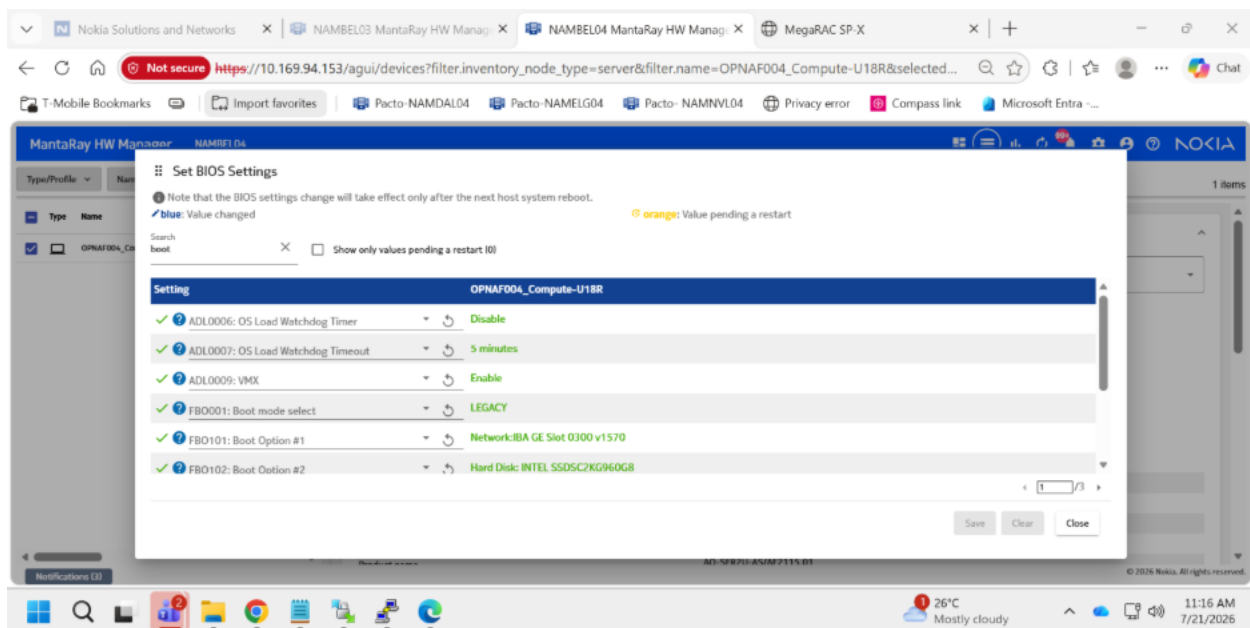
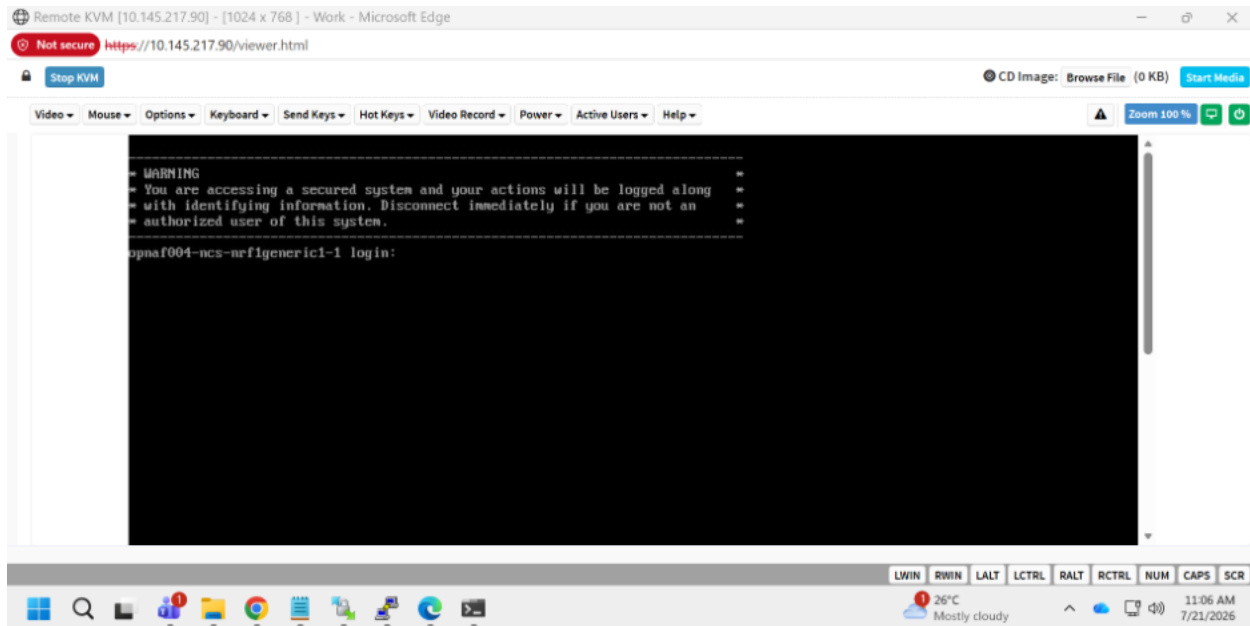
Search ☐ Show only values pending a restart (1)

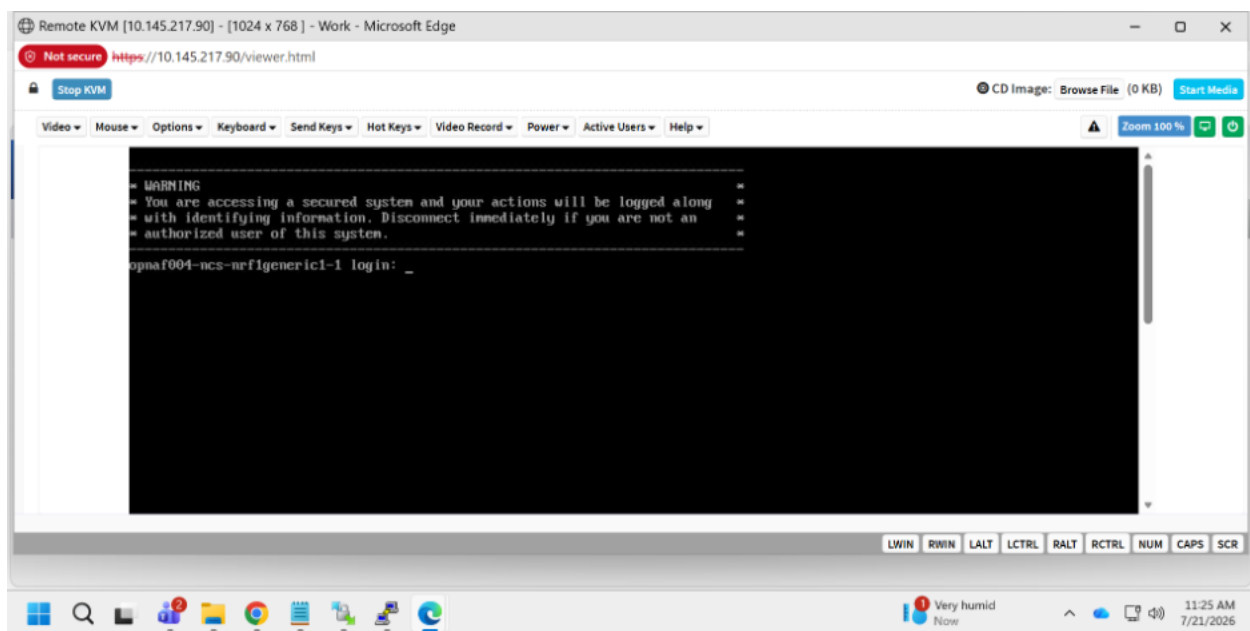
ADLTER006: Flow Control	None
ADLTER009: Legacy OS Redirection Resolution	80x24
CSM005: CSM Support	Enabled
CSM107: Network	Auto
CSM108: Storage	Auto
FR0001: Boot mode select	LEGACY

5 / 17

Save Clear Close

26°C Mostly cloudy 11:02 AM 7/21/2026



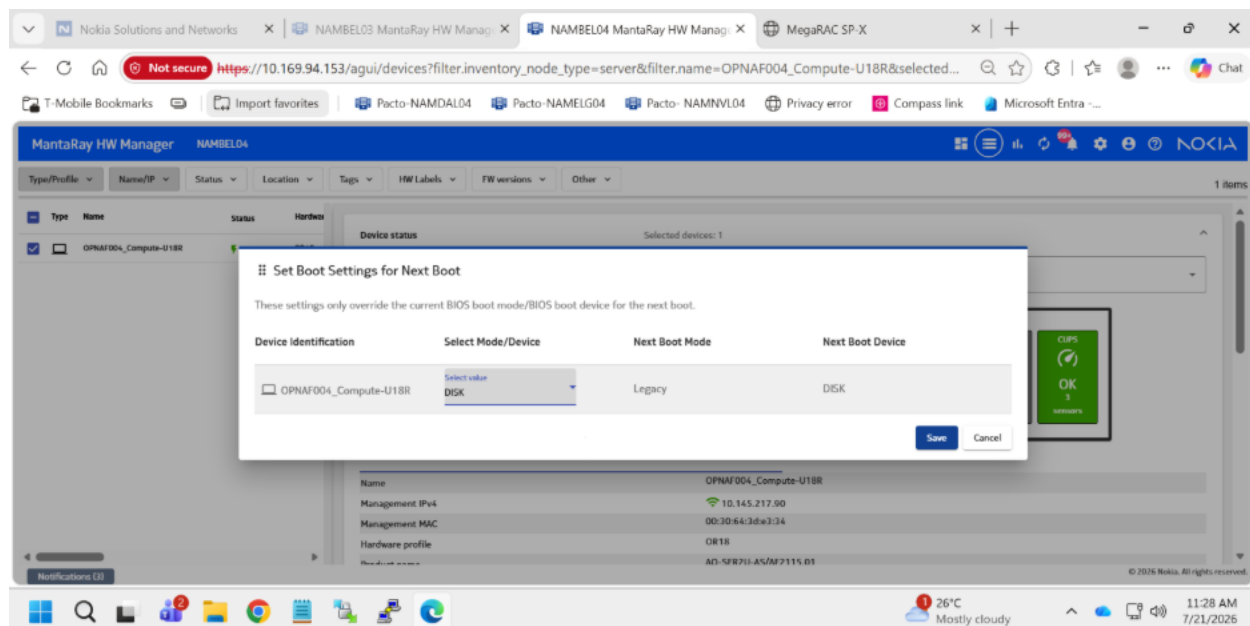


19. Set boot from parameter using NADCM

Steps: -

1. In NADCM GUI go to Devices → Select Server
2. Click on Menu at right side and Select **Set boot Settings** for next boot.
3. Change and save the required boot mode/device value.

Result: - Boot parameter changes has been completed as per the above steps.



20. Exporting and importing MantaRay HW Manager files

Steps: -

1. Select Administration → Files.
2. Select any file and click on Export button/Select Backup files and download the file.
3. For Import, select Add file/in Backup files-Select Import files and add the required file.

Result: -

File export and import have been completed successfully.

Export:

The screenshot shows the MantaRay HW Manager web interface. The browser address bar indicates the URL is <https://10.169.94.153/agui/admin/files>. The interface has a blue header with the title 'MantaRay HW Manager' and a navigation menu on the left. The main content area displays a table of files. The file 'AD-530480N-A-NO-84854-A-102_C_017.conf.tpl' is highlighted. A Windows taskbar is visible at the bottom, showing the date and time as 3:56 PM on 7/10/2026.

Type	File name	Hardware profiles	Release IDs	File size	Upload date	Actions
switch_conf_templates	AD-291000N-X-NO-84854-A-102_C_017SP1.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-X-NO-84854-A-102_C_017SP1.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-84854-A-102_C_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-84854-A-102_C_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-83953-A-102_H_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-83953-A-102_H_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-812512-A-102_S_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-291000N-A-NO-812512-A-102_S_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-54048T-X-NO-84854-A-102_S_017SP1.conf.tpl			4.52 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-54048T-X-NO-84854-A-102_H_017SP1.conf.tpl			7.74 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-54048T-X-NO-84854-A-102_C_017SP1.conf.tpl			9 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-530480N-A-NO-84854-A-102_C_017.conf.tpl			6.87 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-530480N-A-NO-84854-A-102_C_017.conf.tpl			6.88 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-530480N-A-NO-83953-A-102_H_017.conf.tpl			7.15 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]
switch_conf_templates	AD-530480N-A-NO-812512-A-102_S_017.conf.tpl			6.13 KB	Jul 10, 2026	[Edit] [Delete] [Download] [Share]

Web browser window showing the MantaRay HW Manager interface. The address bar displays <https://10.169.94.153/agi/admin/activity>. The interface includes a sidebar with navigation options: Files, Roles, Backup, Activity Logs, DHCP, Hardware Scan, Default Credentials, Certificates, Licenses, SMTP Settings, Set and Clear System Events, and NMS Configuration. The main content area shows a table of activity logs with columns: Result, Action, Details, Device, User, and Time. The table lists various actions such as EXPORT, DEVICE REMOVE FROM INVENTORY, DEVICE ADD TO INVENTORY, CHANGE DEVICE USERS, and OPEN KVM CONSOLE (UVM). The bottom status bar shows the temperature as 29°C, weather as Partly sunny, and the time as 3:57 PM on 7/10/2026.

Import:

Web browser window showing the MantaRay HW Manager interface. The address bar displays <https://10.169.94.153/agi/admin/files>. The interface includes a sidebar with navigation options: Files, Roles, Backup, Activity Logs, DHCP, Hardware Scan, Default Credentials, Certificates, Licenses, SMTP Settings, Set and Clear System Events, and NMS Configuration. The main content area shows a table of files with columns: File name, File size, Upload date, and Actions. A modal dialog titled "Confirm Import of Files" is displayed, providing instructions on the import process and a "Select import file" button. The bottom status bar shows the temperature as 29°C, weather as Partly sunny, and the time as 3:59 PM on 7/10/2026.

NAMBELO4 MantaRay HW Man x +

Not secure https://10.169.94.153/agu/admin/files

Managed bookmarks All Bookmarks

MantaRay HW Manager NAMBELO4

Colapsar toolbar + Add file Delete files Backup files 15 files

Files

Type	File name	Hardware profile	Release ID	File size	Upload date	Actions
switch_conf_templates	AD-291000N-X_AO-8ACX420-B_U02_X_0175P1.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-X_AO-8ACX420-B_U02_X_0175P1.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-84854-A_U02_C_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-84854-A_U02_C_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-83953-A_U02_H_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-83953-A_U02_H_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-812512-A_U02_S_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-291000N-A_AO-812512-A_U02_S_017.conf.tpl			2.09 KB	Jul 10, 2026	
switch_conf_templates	AD-54648T-X_AO-8ACX420-B_U02_S_0175P1.conf.tpl			4.32 KB	Jul 10, 2026	
switch_conf_templates	AD-54648T-X_AO-8ACX420-B_U02_S_0175P1.conf.tpl			7.24 KB	Jul 10, 2026	
switch_conf_templates	AD-530480N-A_AO-84854-A_U02_C_0175P1.conf.tpl			9 KB	Jul 10, 2026	
switch_conf_templates	AD-530480N-A_AO-84854-A_U02_C_017.conf.tpl			6.87 KB	Jul 10, 2026	
switch_conf_templates	AD-530480N-A_AO-84854-A_U02_C_017.conf.tpl			6.88 KB	Jul 10, 2026	
switch_conf_templates	AD-530480N-A_AO-83953-A_U02_H_017.conf.tpl			7.15 KB	Jul 10, 2026	
switch_conf_templates	AD-530480N-A_AO-812512-A_U02_S_017.conf.tpl			6.13 KB	Jul 10, 2026	

Import (Batched): 1 files imported already, 0 files imported. 0 files removed. 0 files imported.

29°C Partly sunny

Search

ENG US 4:00 PM 7/10/2026

NAMBELO4 MantaRay HW Man x +

Not secure https://10.169.94.153/agu/admin/activity

Managed bookmarks All Bookmarks

MantaRay HW Manager NAMBELO4

Colapsar toolbar Result Action Device 8

Activity Logs

Result	Action	Details	Device	User	Time
✓	IMPORT	Targets: files. Format: import.zip	NADCHSystem	automatic@nokia.com	2 minutes ago
✓	EXPORT	Targets: files. Format: export.zip	NADCHSystem	automatic@nokia.com	6 minutes ago
✓	DEVICE REMOVE FROM INVENTORY	device removed (id: 6a5d6a3146d742e7b67946)	NAMBELO4	automatic@nokia.com	10 minutes ago
✓	DEVICE ADD TO INVENTORY	device added to inventory (MAC: aa5b0cc11.22.33)	NAMBELO4	automatic@nokia.com	19 minutes ago
✓	CHANGE DEVICE USERS	Add Management device user: "testuser"	BENAF001-US3LL-Compute	automatic@nokia.com	24 minutes ago
✓	EXPORT	Targets: inventory, eventrules, systemconfig, notificationrules, certificates, localizations, events. Format: export.json	NADCHSystem	automatic@nokia.com	36 minutes ago
✗	OPEN KVM CONSOLE (JRU)	Failed to open console: no working SSH credentials found. (Step: 1/1)	10.145.8.235	automatic@nokia.com	40 minutes ago
✗	OPEN KVM CONSOLE (JRU)	Failed to open console: no working SSH credentials found. (Step: 1/1)	10.145.8.235	automatic@nokia.com	41 minutes ago
✗	OPEN KVM CONSOLE (JRU)	Failed to open console: no working SSH credentials found. (Step: 1/1)	10.145.8.106	automatic@nokia.com	41 minutes ago
✗	OPEN KVM CONSOLE (JRU)	Failed to open console: no working SSH credentials found. (Step: 1/1)	10.145.8.103	automatic@nokia.com	42 minutes ago
✓	EXPORT	Targets: inventory, eventrules, systemconfig, notificationrules, certificates, localizations, events. Format: export.json	NADCHSystem	automatic@nokia.com	an hour ago

29°C Partly sunny

Search

ENG US 4:01 PM 7/10/2026

21. Alarms from Server hosting NADCM

Steps: -

1. Login the NADCM GUI.
2. Generate the NADCM Hardware alarm

ipmitool -I lanplus -U <username> -P <Password> -H <10.ubm.mr.151> sensor thresh
'Volt_PVCCIN_CPU0' ucr 1.5

Result: - Alarms has been generated and test case passed.

```
[root@NAMBEL04 ~]#  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'K!Hw76K4puG53Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0  
Volt_PVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | 2.205 | na  
[root@NAMBEL04 ~]# ipmitool -U admin -P 'K!Hw76K4puG53Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0  
Volt_PVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | 2.205 | na  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'K!Hw76K4puG53Guk' -H 10.169.94.223 sensor thresh 'Volt_PVCCIN_CPU0' 1.5  
sensor thresh <id> <threshold> <settings>
```

```
[root@NAMBEL04 ~]# ipmitool -U admin -P 'K!Hw76K4puG53Guk' -H 10.169.94.223 sensor thresh 'Volt_PVCCIN_CPU0' ucr 1.5  
Locating sensor record 'Volt_PVCCIN_CPU0'.  
Setting sensor 'Volt_PVCCIN_CPU0' Upper Critical threshold to 1.500  
[root@NAMBEL04 ~]# ipmitool -U admin -P 'K!Hw76K4puG53Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0  
Volt_PVCCIN_CPU0 | 1.701 | Volts | cr | na | 1.431 | na | 1.503 | na  
[root@NAMBEL04 ~]#
```

← ↻ 🏠 🔒 Not secure https://10.169.94.153/agui/devices?filter.inventory_node_type=server&filter.obm_ip=10.169.94.223&selected=6a54a0bb8... 🔍 ⚙️ ☆ 👤 ⋮ Chat

T-Mobile Bookmarks Import favorites Pacto-NAMDAL04 Pacto-NAMELG04 Pacto-NAMNVL04 Privacy error Compass link Microsoft Entra ...

MantaRay HW Manager NAMBEL04

Type/Profile ▾ Name/ID ▾ Status ▾ Location ▾ Tags ▾ HW Labels ▾ FW versions ▾ Other ▾ 1 Items

Type	Name	Status	Hardware
📡	NAMBEL04	🟢 🚨	RP113

Active events

Severity	Device identification	Description	Sensor name	Value	Date (Read)
🚨	NAMBEL04	Voltage Abnormal	Volt_PVCCIN_CPU0	Upper Critical	2026/07/21 15:59:53
Management IP: 10.169.94.223 Location: / / / / / Device name: NAMBEL04 Description: Voltage Abnormal Device type: SERVER Origin: ipmbiosensor Read timestamp: 2026/07/21 15:59:53 GMT+05:30 Sensor type: Voltage Sensor name: Volt_PVCCIN_CPU0 Sensor number: 216 Sensor ID: 216 Status: Upper Critical Lower Critical: 1.431 Upper Critical: 1.503 Value: 1.710 (+/- 0) Volts					
⚠️	NAMBEL04	Predictive Failure - VIN_UV_WARNING - Asserted	PSU1_Status		2026/07/21 00:35:53
⚠️	NAMBEL04	Login failed from IP:10.147.179.96	Login Info		2026/07/21 00:35:53

All events

Severity	Device identification	Description	Sensor name	Value	Active	Date (Read)
🚨	NAMBEL04	Voltage Abnormal	Volt_PVCCIN_CPU0	Upper Critical	Yes	2026/07/21

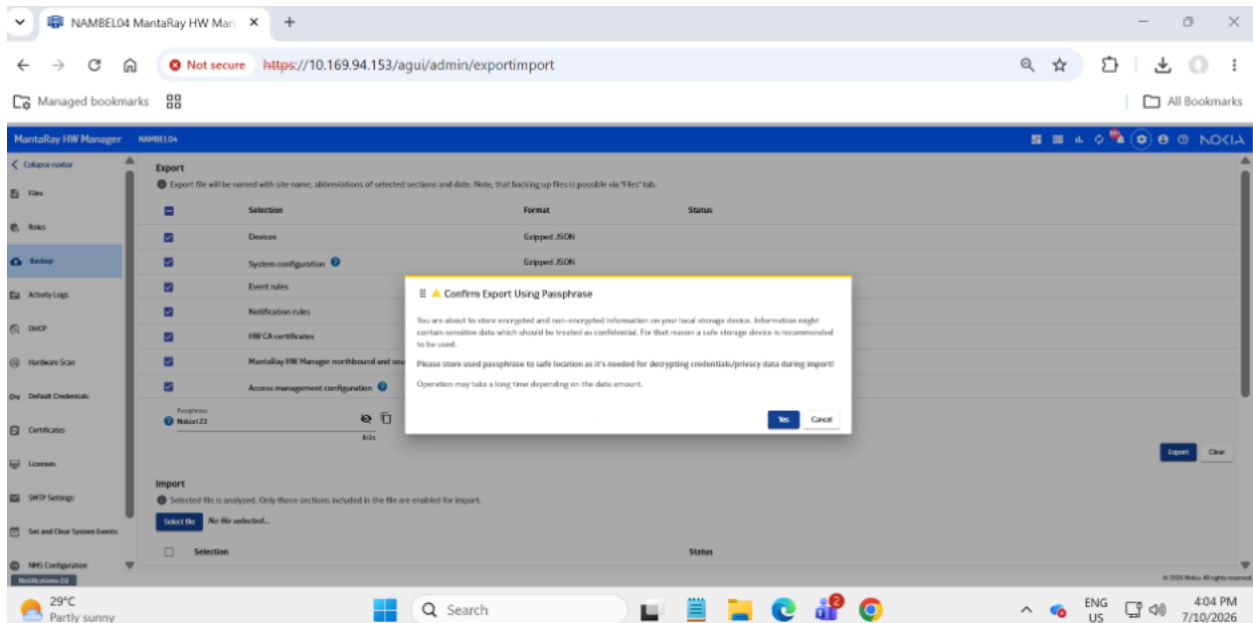
Notifications 0/0

22. NADCM Configuration export and import

Steps: -

1. Navigate to 'Administration'
2. Click on Backup
3. Export the backup with passphrase and it will be downloaded.
4. Import the backup with passphrase provided during export.

Result: -



NAMBEL04 MantaRay HW Man: x +

Not secure https://10.169.94.153/agui/admin/exportimport

Managed bookmarks

MantaRay HW Manager NAMBEL04

Export

Export file will be named with site name, abbreviations of selected sections and date. Note, that backing up files is possible via 'Files' tab.

Selection	Format	Status
☐ Devices	Gzipped JSON	✓ Devices successfully exported.
☐ System configuration	Gzipped JSON	✓ System configuration successfully exported.
☐ Event rules	Gzipped JSON	✓ Event rules successfully exported.
☐ Notification rules	Gzipped JSON	✓ Notification rules successfully exported.
☐ HW CA certificates	Gzipped JSON	✓ Certificates successfully exported.
☐ MantaRay HW Manager northbound and southbound certificates	Gzipped JSON	✓ MantaRay HW Manager northbound and southbound certificates successfully exported.
☐ Access management configuration	Gzipped JSON	✓ Access management configuration successfully exported.

Phrasephrase 00%

Import

Selected file is analyzed. Only those sections included in the file are enabled for import.

Select file No file selected...

Selection Status

29°C Partly sunny

Search

ENG US 4:05 PM 7/10/2026

NAMBEL04.dev_conf_everul_notifru...
t_access_nbsbcert_ExportData_2026-07-10T10_35_04.973Z.json.gz
377 KB • Done

HWM_Files_Export_2026-07-10T10_25_48Z.zip
3.2 KB • 9 minutes ago

NAMBEL04 MantaRay HW Man: x NAMBEL04 MantaRay HW Man: x +

Not secure https://10.169.94.153/agui/admin/exportimport

Managed bookmarks

MantaRay HW Manager NAMBEL04

Import

Selected file is analyzed. Only those sections included in the file are enabled for import.

Select file No file selected...

Selection	Status
☐ Devices	✓ Devices successfully imported.
☐ System configuration	✓ System configuration successfully imported.
☐ Event rules	✓ Event rules successfully imported.
☐ Notification rules	✓ Notification rules successfully imported.
☐ HW CA certificates	✓ Certificates successfully imported.
☐ MantaRay HW Manager northbound and southbound certificates	✓ MantaRay HW Manager northbound and southbound certificates successfully imported.

Set default user password? 0236

User federation password 0138

Phrasephrase 00% Ignore credentials

Import Clear

30°C Partly sunny

Search

ENG US 4:17 PM 7/10/2026

23. HW decommissioning to verify license capacity utilization

Steps: -

1. In NADCM GUI, navigate to 'Administration' → 'Licenses' and take note on the 'Used' and 'Free' capacity.
2. Go to 'Hardware Management' view, select device and then delete it from inventory.
3. Go back to 'Administration' → 'Licenses' and observe the 'Used' and 'Free' capacity

Result: - Below screenshot shows once a hardware is added in inventory, license utilization also increases.

The screenshot displays the MantaRay HW Manager interface. The left sidebar contains navigation options: Files, Rules, Backup, Activity Logs, DHCP, Hardware Scan, Default Credentials, Certificates, Licenses (selected), and SMTP Settings. The main content area is divided into two sections: Features and Licenses.

Features Section:

Feature code	Description	Capacity	Used	Free	Usage %
0000038242	Integration Express App SW	0	0	0	N/A %
0000045672	Base Capacity	720	292	428	41 %
0000038244	Integration Express NE Capacity	0	0	0	N/A %

Licenses Section:

System target ID: e8e5bf71000e0b3dbfac

License name	Serial number	Start time	End time	Capacity	Actions
MantaRay NH self monitor HW and ANB LK	2387171913767	a year ago	Expires in 4 years	0	
HW Management LK	7001024156113	a year ago	Expires in 4 years	700	

The interface also includes a 'Delete' button at the bottom right of the Licenses section. The bottom status bar shows the time as 7:39 AM on 7/21/2026.

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | +

Not secure https://10.169.94.153/agu/devices?filter.name=OPNAF004_Compute-U18R&selected=64217266c611fb82c74498d2

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas - TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile | Name/PP | Status | Location | Tags | HW Labels | FW versions | Other | -1 Items

Type Name Status Hardware

☒ OPNAF004_Compute-U18R OR18

Device status Selected devices: 1

Select a device

Name: OPNAF004_Compute-U18R IP: 10.145.217.90

OK UNDEF OK

Are you sure you want to continue?

Device Details

Name	OPNAF004_Compute-U18R
Management IPv4	10.145.217.90
Management MAC	00:30:64:3d:a3:34
Hardware profile	OR18
Product name	AO-SER2U-AS/NF2115.01
Product serial	CHATS1W1022014793
Device ID	64217266c611fb82c74498d2
Board product	AO-SER2U-AS/NF2115.01
Board part number	61-11101-1A30

Sensor Details

Notifications (1)

8:02 AM 7/21/2026

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | NAMBEL04 MantaRay HW Manag | +

Not secure https://10.169.94.153/agu/admin/licenses?filter.name=OPNAF004_Compute-U18R

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas - TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Files | Rules | Backup | Activity Logs | DHCP | Hardware Scan | Default Credentials | Certificates | Licenses | SMTP Settings

Features 3 features

Feature code	Description	Capacity	Used	Free	Usage %
0000038244	Integration Express NE Capacity	0	0	0	N/A %
0000038242	Integration Express App SW	0	0	0	N/A %
0000045672	Base Capacity	720	291	429	40 %

Licenses 2 licenses

+ Add Licenses System target ID: e8e5b71000e0b3dbfac

License name	Serial number	Start time	End time	Capacity	Actions
☐ MantaRay NH self monitor HW and ANB LK	2387171913767	a year ago	Expires in 4 years	0	
☐ HW Management LK	7001024156113	a year ago	Expires in 4 years	700	

8:02 AM 7/21/2026

ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW X NAMBEL04 MantaRay HW X NAMBEL04 MantaRay HW X NAMBEL04_AllDeviceData X

Not secure https://10.169.94.153/agu/devices?filter.obm_ip=10.145.217.90&selected=6a5f2cb2f67255baecd2f29b

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -

MantaRay HW Manager NAMBEL04

Type/Profile Name/Status Location Tags HW Labels FW versions Other 1 items

Type Name Status Hardware

☒ OPNAF004-Compute-U188

Device status Selected devices: 1

Select a device

Name: OPNAF004-Compute-U188 IP: 10.145.217.90

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

UNDEF
sensors

N/A
sensors

Device Details Sensor Details

Name	OPNAF004-Compute-U188
Management IPv4	10.145.217.90
Management MAC	00:30:64:3d:a3:34
Hardware profile	-
Product name	-
Product serial	-
Device ID	6a5f2cb2f67255baecd2f29b
Asset tag	Free form asset tag

Notifications (0)

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8:25 AM 7/21/2026

ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW X NAMBEL04 MantaRay HW X NAMBEL04 MantaRay HW X NAMBEL04_AllDeviceData X

Not secure https://10.169.94.153/agu/admin/licenses?filter.name=ATP%20TEST

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -

MantaRay HW Manager NAMBEL04

Collapse navbar

- Files
- Rules
- Backup
- Activity Logs
- DHCP
- Hardware Scan
- Default Credentials
- Certificates
- Licenses
- SMTP Settings

Features 3 features

Feature code	Description	Capacity	Used	Free	Usage %
0000038242	Integration Express App SW	0	0	0	N/A %
0000045672	Base Capacity	720	292	428	41 %
0000038244	Integration Express NE Capacity	0	0	0	N/A %

Licenses 2 licenses

+ Add Licenses System Target ID: e8e5bf71000e0b3dbfbc

License name	Serial number	Start time	End time	Capacity	Actions
☐ MantaRay NM self monitor HW and ANS LK	2387171913767	a year ago	Expires in 4 years	0	
☐ HW Management LK	7001024156113	a year ago	Expires in 4 years	700	

Delete

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8:25 AM 7/21/2026

You're sharing your screen Stop sharing

24. Export Inventory report

Steps: -

1. Select the Hardware Management tab.
2. Choose Export Devices.

Results: Successfully showing the hardware export from inventory.

The screenshot displays the NAMBEL04 MantaRay HW Manager web interface. The browser address bar shows the URL <https://10.169.94.153/agui/devices>. The interface includes a top navigation bar with the title "MantaRay HW Manager" and a sub-header "NAMBEL04". Below this, there are filter tabs for "Type/Profile", "Name/IP", "Status", "Location", "Tags", "HW Labels", "FW versions", and "Other". The main content area is a table listing various devices. Each row contains columns for "Type", "Name", "Status", "Hardware profile", "IPs", "MAC", "Location", and "Tags". The "Status" column shows green checkmarks for most devices, indicating they are online. The "Location" column lists various locations such as "100 Lab-Balcony-17Switch-Room00211025A01". The "Tags" column contains a button labeled "B3 MCS FW Update". A download notification bubble in the top right corner indicates that the file "NAMBEL04_AllDeviceData_2026-07-10T10_50_29.548Z.xlsx" (74.0 KB) has been successfully downloaded. The bottom of the screen shows a Windows taskbar with the date and time "4:20 PM 7/10/2026".

Type	Name	Status	Hardware profile	IPs	MAC	Location	Tags
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.100	00:30:64:14:07:2e	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.101	00:30:64:14:07:2f	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.102	00:30:64:14:07:30	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.103	00:30:64:14:07:31	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.104	00:30:64:14:07:32	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.105	00:30:64:14:07:33	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.106	00:30:64:14:07:34	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.107	00:30:64:14:07:35	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.108	00:30:64:14:07:36	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.109	00:30:64:14:07:37	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.110	00:30:64:14:07:38	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.111	00:30:64:14:07:39	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.112	00:30:64:14:07:3a	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.113	00:30:64:14:07:3b	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.114	00:30:64:14:07:3c	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.115	00:30:64:14:07:3d	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.116	00:30:64:14:07:3e	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.117	00:30:64:14:07:3f	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.118	00:30:64:14:07:40	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.119	00:30:64:14:07:41	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update
Computer	NAMBEL04-1000-Computer	Online	DE-0	10.145.8.120	00:30:64:14:07:42	100 Lab-Balcony-17Switch-Room00211025A01	B3 MCS FW Update

25. Rack Health check for HW Mgmt. switch

Steps: -

1. Login NADCM and select “Devices”
2. Select the specific switch
3. Select the Switch and you can see the rack topology view at right side.

Result: - Switch rack topology details can be verified in as below.

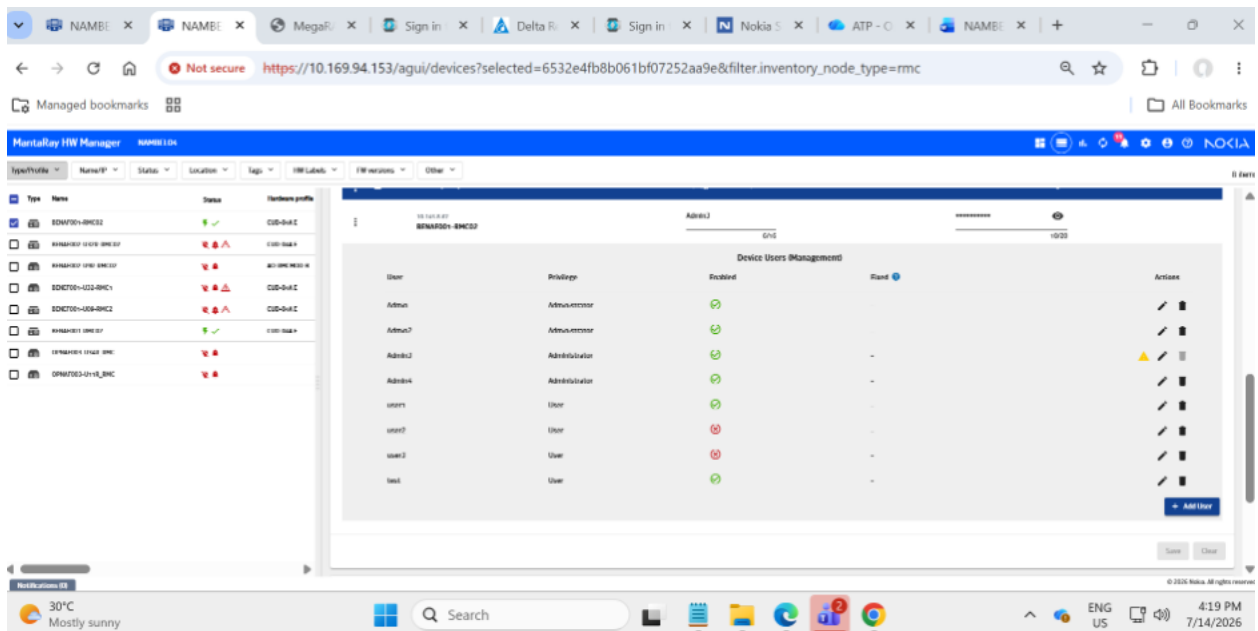
The screenshot displays the NADCM MantaRay HW Manager interface. On the left, a list of devices is shown, including various switches and line cards. The main panel on the right displays the 'Device status' for a selected switch, 'SABST004-020-MGMT-SW'. Below this, a 'Rack Topology Check' diagram is visible, showing a grid of ports (1-54) and their status (green for healthy, red for faulty). The diagram is divided into sections: 'PSU / Port Details', 'Other Interfaces', and 'Rack Topology Check'. The 'PSU / Port Details' section shows the switch's name, management IP, MAC, hardware profile, product name, product serial, and device ID. The 'Other Interfaces' section shows the switch's name, management IP, MAC, hardware profile, product name, product serial, and device ID. The 'Rack Topology Check' section shows the switch's name, management IP, MAC, hardware profile, product name, product serial, and device ID.

26. Device user management – RMC

Steps: -

1. In Devices → Select the RMC
2. Select Credential Management → Select MANAGEMENT IPMI/REST/SSH/gNMI.
3. Select the vertical ellipsis (:) Click Add user → create username and password.
4. Login to the RMC GUI using the new user and credentials.

Result: - New user creation has been completed successfully for OR18 server.



NAMEB: x

NAMEB: x

MegaR: x

Sign in: x

Delta P: x

Sign in: x

Nokia S: x

ATP - C: x

NAMEB: x

+

-

o

x

Not securehttps://10.145.8.67/html/index.htm

Managed bookmarks

All Bookmarks



Remote Monitor Controller

STATUS : ON

TEMP : 23.17 °C

▶ System

▶ Modules

▶ Configurations

Event Log

Power Management Status

System Input

Volt. Input 1st	Curr. Input 1st	Power Input 1st
206.50	10.56	2200.00
Volt. Input 2nd	Curr. Input 2nd	Power Input 2nd
208.00	11.00	2336.00
Volt. Input Total	Curr. Input Total	Power Input Total

Hardware Rev. :S0

Software Rev. :STD_x1.02B35

Date Time :2002-09-17 17:56:33

30°C

Mostly sunny

Search

ENG

US

4:34 PM

7/14/2026

27. Device user management – OR 18 servers

Steps: -

1. In Devices → Select the Server
2. Select Credential Management → Select MANAGEMENT IPMI/REST/SSH/gNMI
3. Select the vertical ellipsis (:), → Click Add user → create username and password.
4. Login to the BMC GUI of the server using the new user.

Result: - New user creation has been completed successfully for OR18 server.

The screenshot displays the MantaRay HW Manager web interface. The browser address bar shows the URL: https://10.169.94.153/agui/devices?filter.inventory_node_type=server&selected=5bfdca2d5d52700480c234c. The interface includes a sidebar with a list of devices, each with a checkbox, name, status, and hardware profile. The main panel shows the details for a selected device, including a 'Device status' section with a grid of icons for various components (Airflow, Current, Fan, Power, Temperature, Voltage, Other, Disk, CPU) and a 'Device Details' section with a table of attributes.

Attribute	Value
Name	BOM1001-0108-Compute
Management IPV4	10.145.8.103
Management MAC	00:30:84:76:07:ae
Hardware profile	OR18
Product name	AO-S08210-ASU2111-01
Product serial	C4072701-8002308
Device ID	5096a2f05d52700480c234c
Board product	AO-S08210-ASU2111-01
Board part number	51-33101-1A20
Asset tag	AT_5230d18633_0131_JC1
Product part number	MTV-S-NEAC3110108

Browser tabs: JIT NMNet, Sign in to your..., Sign in to Nokia..., NADCM cre.upd..., Consolidated N..., NAMBEL04 Man..., New Microsoft..., +

Address bar: https://10.169.94.153/agui/devices?filter.inventory_node_type=server&selected=5bfdca2d5d552700480c234c

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS..., login.microsoftonli..., NADCM cre.updat...

MantaRay I/W Manager NAMBEL04

Tab: Type/Profile - Name/ID

Warning: This Web UI For Securing Firmware versions of RAN Core devices only. If you wish to filter firmware versions of your RAN Core devices, please choose "Filter" (After add) tab on "Firmware version".

FW version: Other

1-100 of 236 Items

Bank selection: Selected devices: 1

Credential management: Selected devices: 1

USED PROTOCOLS: REGISTER, (PWS, PWS, PWS, PWS), DMMPV2/V3, TRAP RECEIVER, RAN-SECURITY

Device (Server): Management username: Management password

Device Users (Management)

User	Privilege	Enabled	Fixed	Actions
admin	administrator	Enabled	YES	
ncs	administrator	Enabled	YES	

Location management: TH0 Lab Belfame 1 / Switch Boxes / B0001 / 10 / 25 / 30

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23°C Cloudy Search

10:05 AM 7/10/2026

Browser tabs: JIT NMNet, Sign in to your..., Sign in to Nokia..., NADCM cre.upd..., Consolidated N..., NAMBEL04 Man..., New Microsoft..., +

Address bar: https://10.169.94.153/agui/devices?filter.inventory_node_type=server&selected=5bfdca2d5d552700480c234c

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS..., login.microsoftonli..., NADCM cre.updat...

MantaRay I/W Manager NAMBEL04

Tab: Type/Profile - Name/ID - Status - Location - Tags - HW Labels - FW version - Other

1-100 of 236 Items

Bank selection: Selected devices: 1

Credential management: Selected devices: 1

USED PROTOCOLS: DMMPV2/V3, TRAP RECEIVER, RAN-SECURITY

Device (Server): Management username: Management password

Device Users (Management)

User	Privilege	Enabled	Fixed	Actions
admin	administrator	Enabled	YES	
ncs	administrator	Enabled	YES	

Location management: TH0 Lab Belfame 1 / Switch Boxes / B0001 / 10 / 25 / 30

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23°C Cloudy Search

10:06 AM 7/10/2026

Add User

User name: Selector

If ENEC is in 16-character password mode, given longer password may be truncated automatically to 16 first characters. Please refer to device specific user manual for setting secure passwords.

User password: *****

Confirm user password: *****

Privilege: USER

Save Cancel

JIT NMNet

Sign in to you

Sign in to N

NADCM cre

Consolidated

NAMBEL04 I

New Micros

MegaRAC SI

Not securehttps://10.145.8.103/#dashboard

T-Mobile BookmarksCompasscyberarkSafeguardJIT NMNetMy AccessNTAC RAN OSS...login.microsoftNADCM cre_updat...

AirFrame BMC WebUItestuser - User | About | Log OutNOKIA

DashboardSensorSystem InventoryLogs & ReportsSettingsControl & Maintenance

DashboardControl Panel

1287 d 22 hrs

Up Time

More info

1780

Pending Disconnections

More info

71

Access Logs

More info

Today (0)

Details

No events for today..

30 days (0)

Details

No events for last 30 days..

Sensor Monitoring

All sensors are good now!

Currently recovered

23°C Cloudy

Search

ENG10:11 AM10/2026

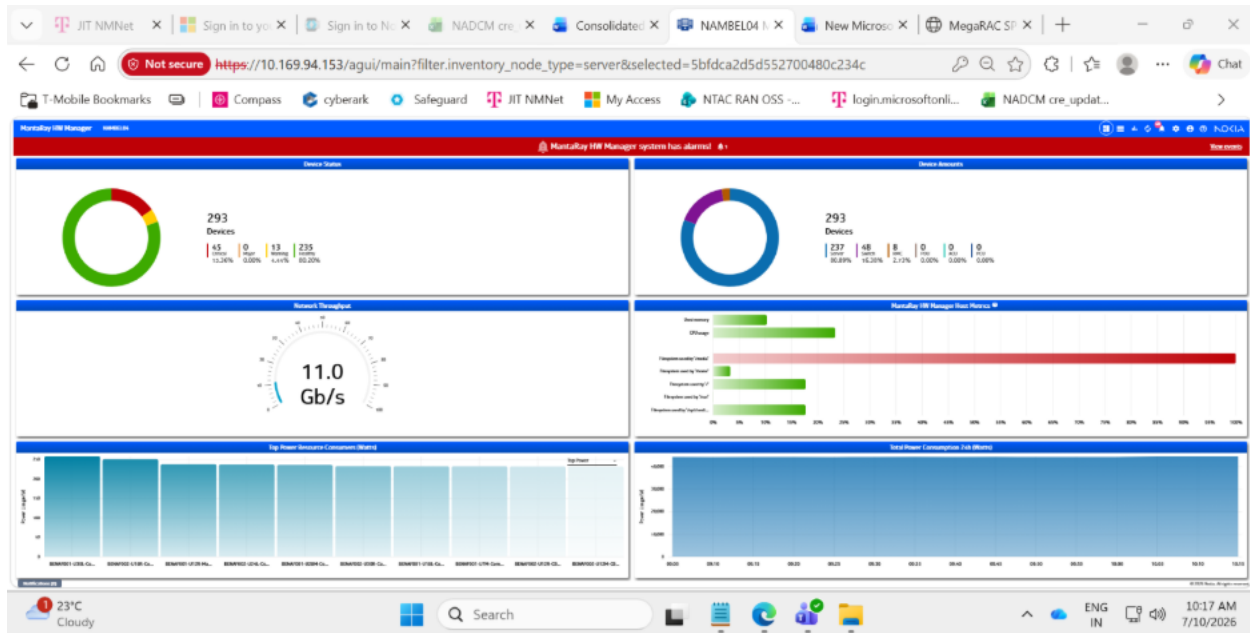
You're sharing your screenStop sharing

28. Verify System Name in NADCM Dashboard

Steps: -

1. Login to NADCM WEB GUI
2. The top banner should show the NADCM localized System name instead of IP address.

Result: - Successfully verified as shown below

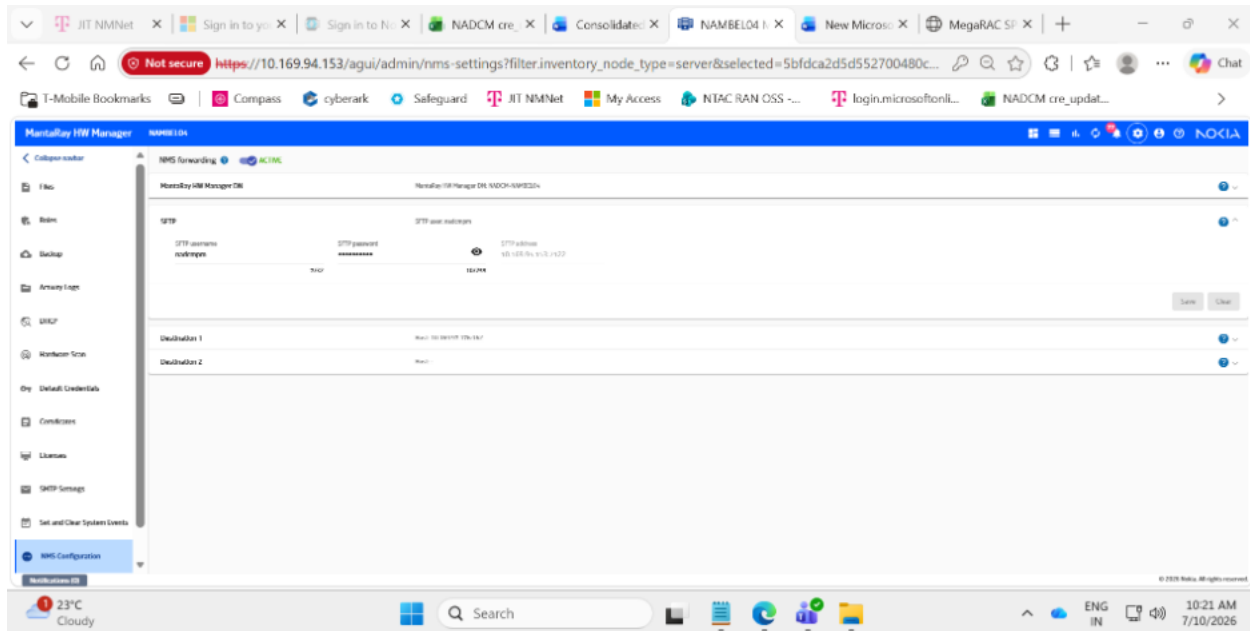


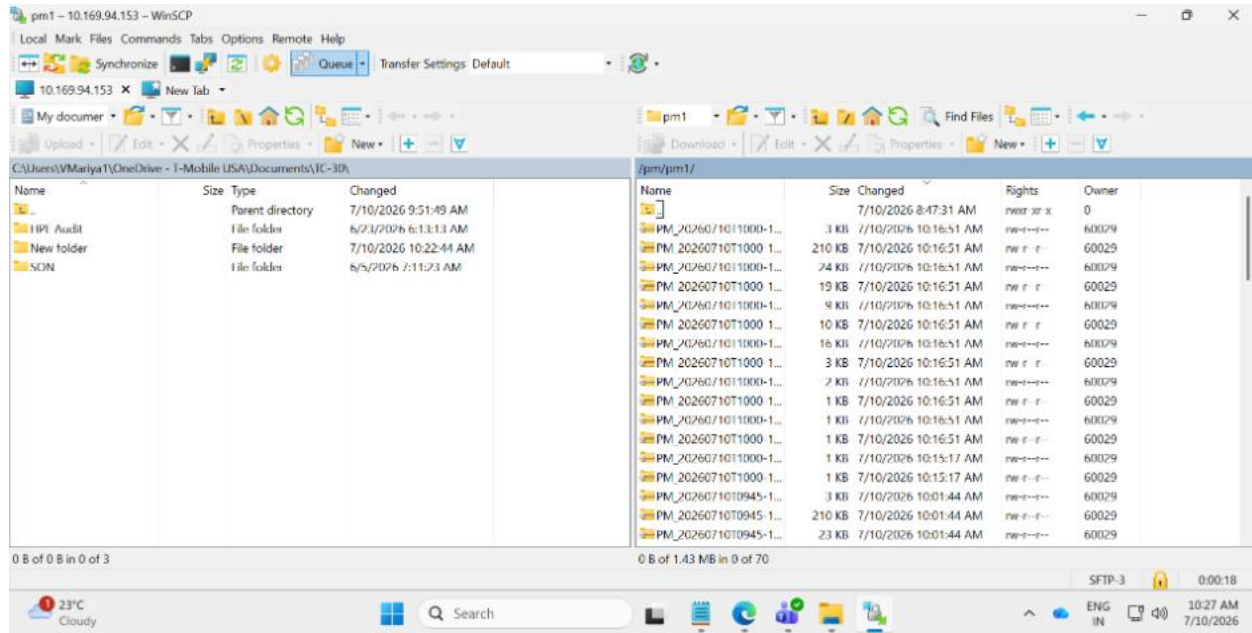
29. Verify NADCM PM Data Integration

Steps: -

1. Login to NADCM → Administration → NMS Interface.
2. Open WinSCP tool → Use SFTP address, SFTP username & Password
3. PM report will be available.

Result: -



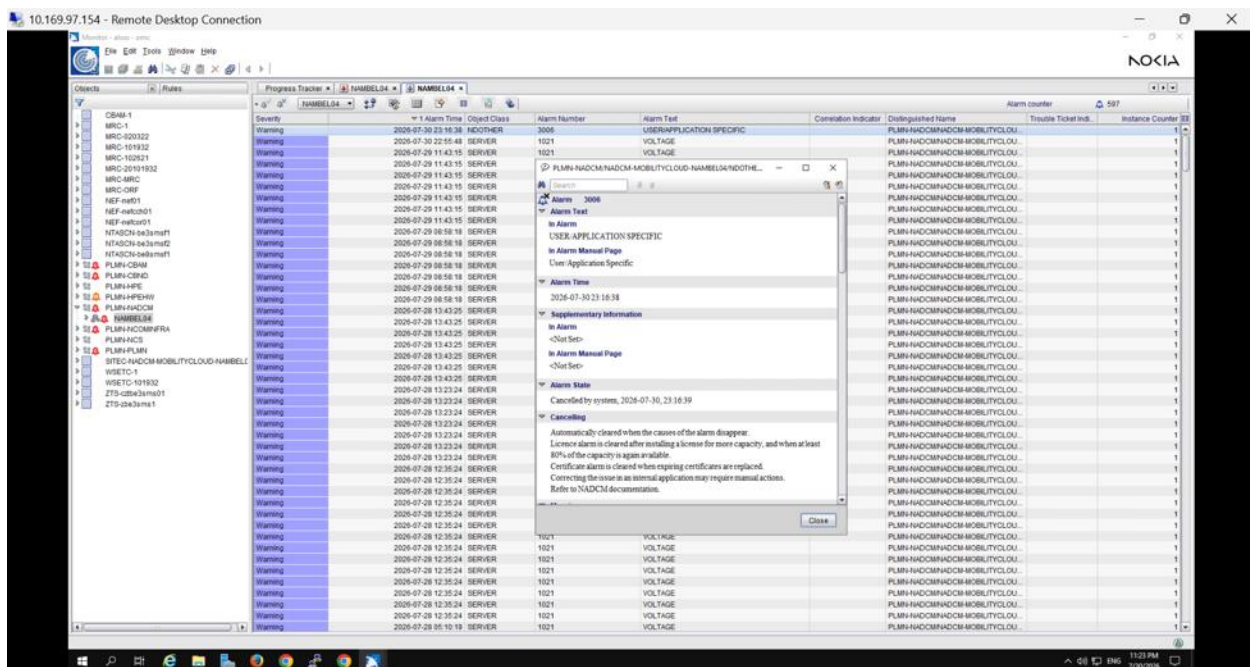
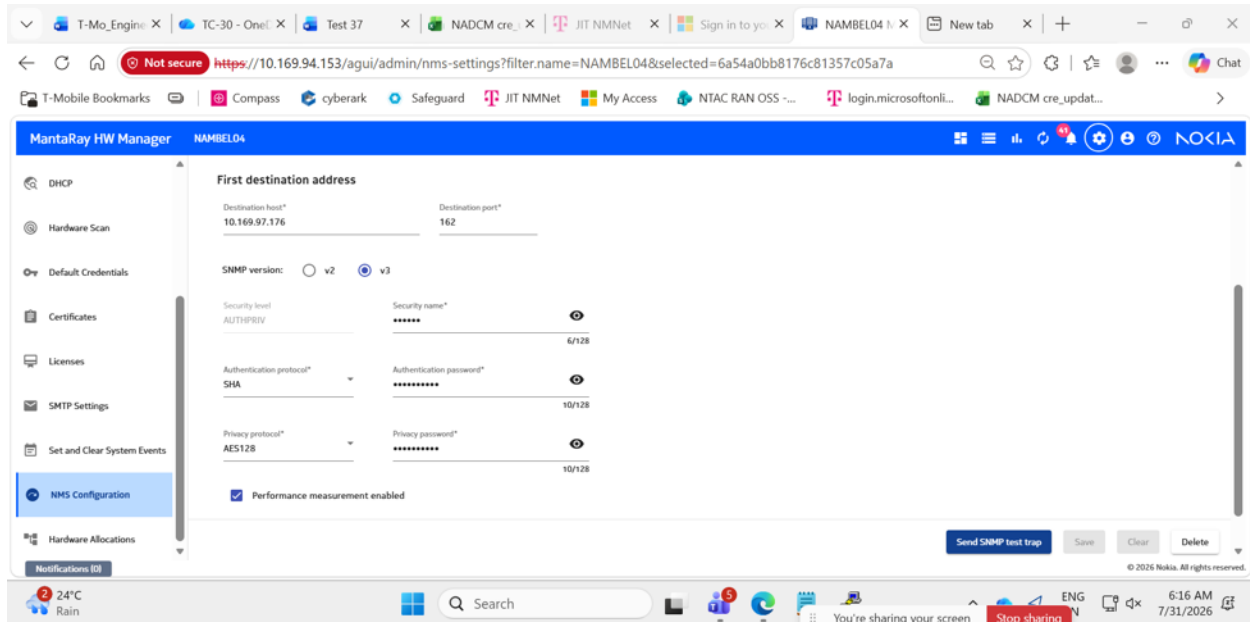


30. Test trap from NADCM WEB GUI

Steps: -

1. Login to NADCM WEB GUI.
2. Administration → NMS Configuration → Destination 1 → Click on “Send SNMP test trap”.
3. Go to the settings for the next Destination 2 → Click on “Send SNMP test trap”.
4. Login to NetAct
5. Verify the two trap messages are displayed under Warnings.

Result: - Test trap from NADCM GUI has been verified.



31. NADCM NTP SYNC

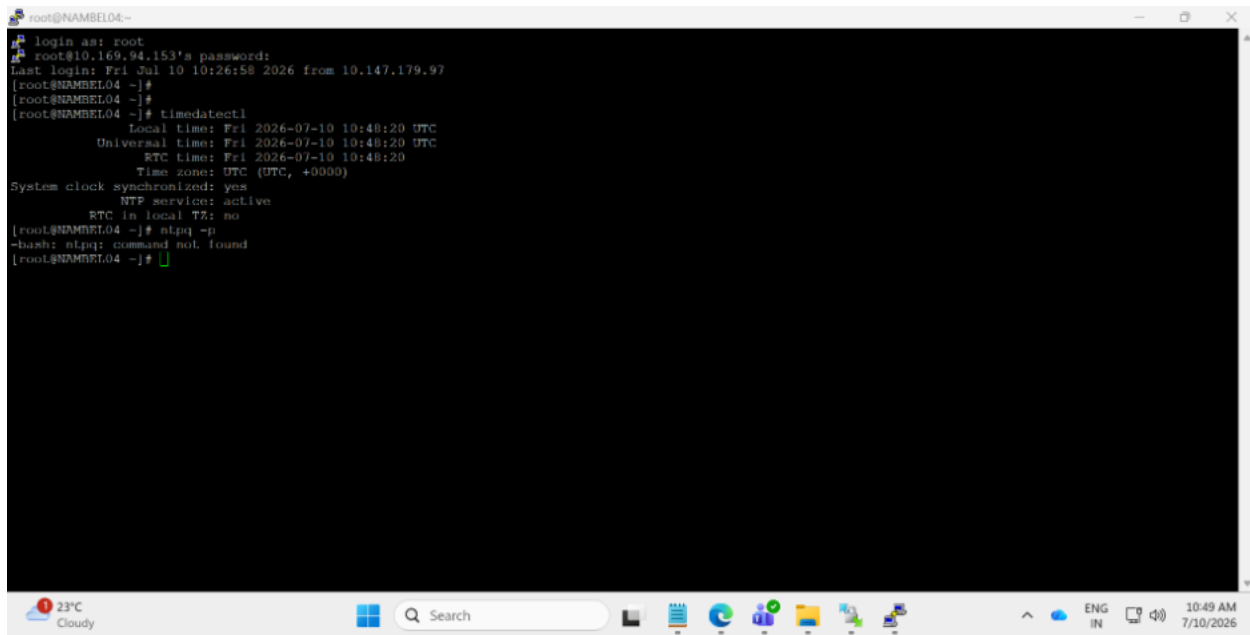
Steps: -

1. Login to NADCM as root user
2. Execute the following to check the NTP sync.

```
[root@NAMBEL02 ~]# timedatectl
```

```
[root@NAMBEL02 ~]# ntpq -p
```

Result: - Successfully showing the NTP sync as below.



```
root@NAMBEL04~  
login as: root  
root@10.169.94.153's password:  
Last login: Fri Jul 10 10:26:58 2026 from 10.147.179.97  
[root@NAMBEL04 ~]#  
[root@NAMBEL04 ~]#  
[root@NAMBEL04 ~]# timedatectl  
Local time: Fri 2026-07-10 10:48:20 UTC  
Universal time: Fri 2026-07-10 10:48:20 UTC  
RTC time: Fri 2026-07-10 10:48:20  
Time zone: UTC (UTC, +0000)  
System clock synchronized: yes  
NTP service: active  
RTC in local TZ: no  
[root@NAMBEL04 ~]# ntpq -p  
-bash: ntpq: command not found  
[root@NAMBEL04 ~]#
```

The screenshot shows a terminal window with a black background and white text. The text displays the output of several commands: a login prompt, a password prompt, a last login message, and the output of the 'timedatectl' command which shows the system clock is synchronized and NTP service is active. The 'ntpq -p' command results in a '-bash: ntpq: command not found' error. The terminal window is titled 'root@NAMBEL04~'. At the bottom of the image is a Windows taskbar showing the date and time as 10:49 AM on 7/10/2026, along with various system icons and a search bar.

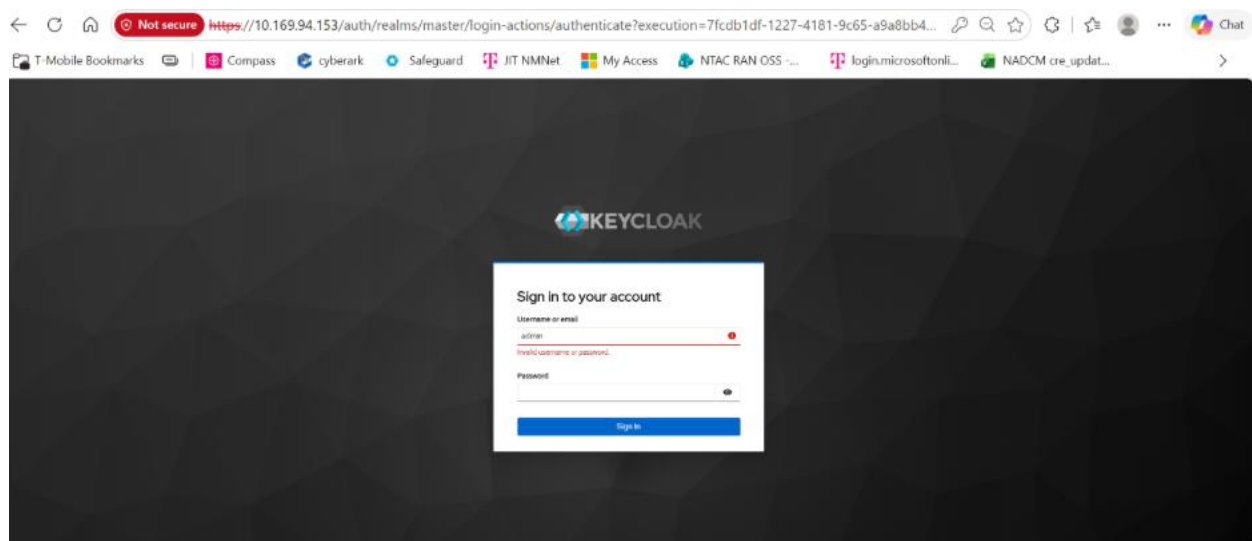
32. Keycloak Brute force Test

Steps: -

1. Verify Current settings for NADCM Keycloak to prevent brute force attacks below
2. login to Keycloak at https://NADCM_IP/auth/admin with wrong credentials

NOTE:

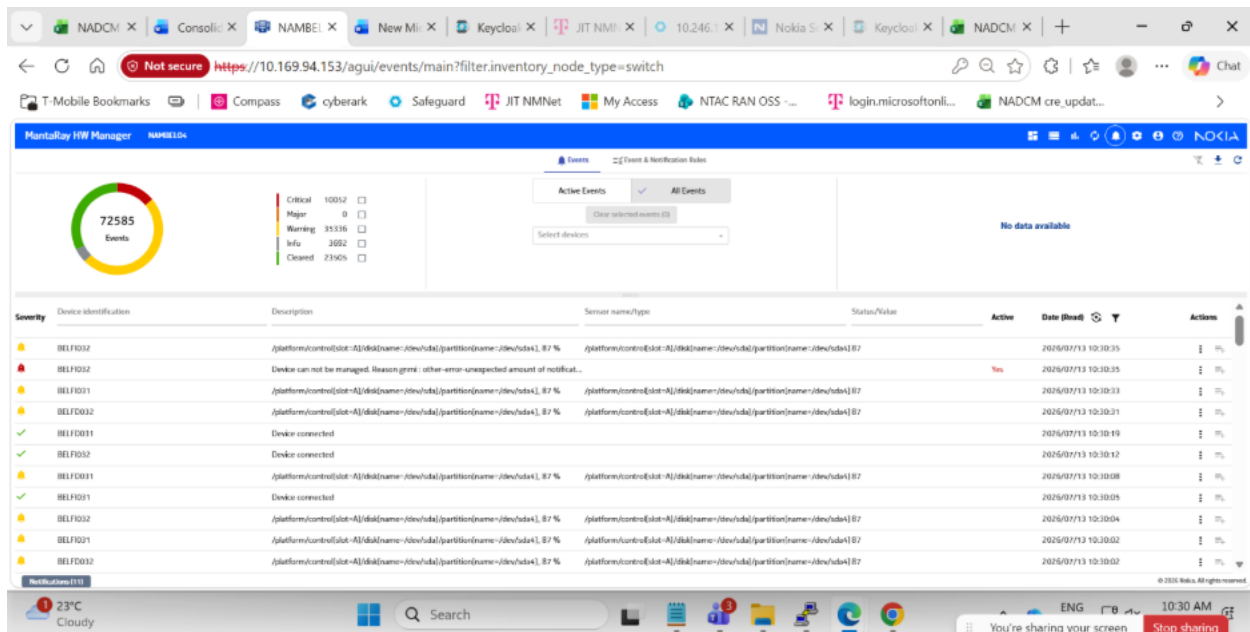
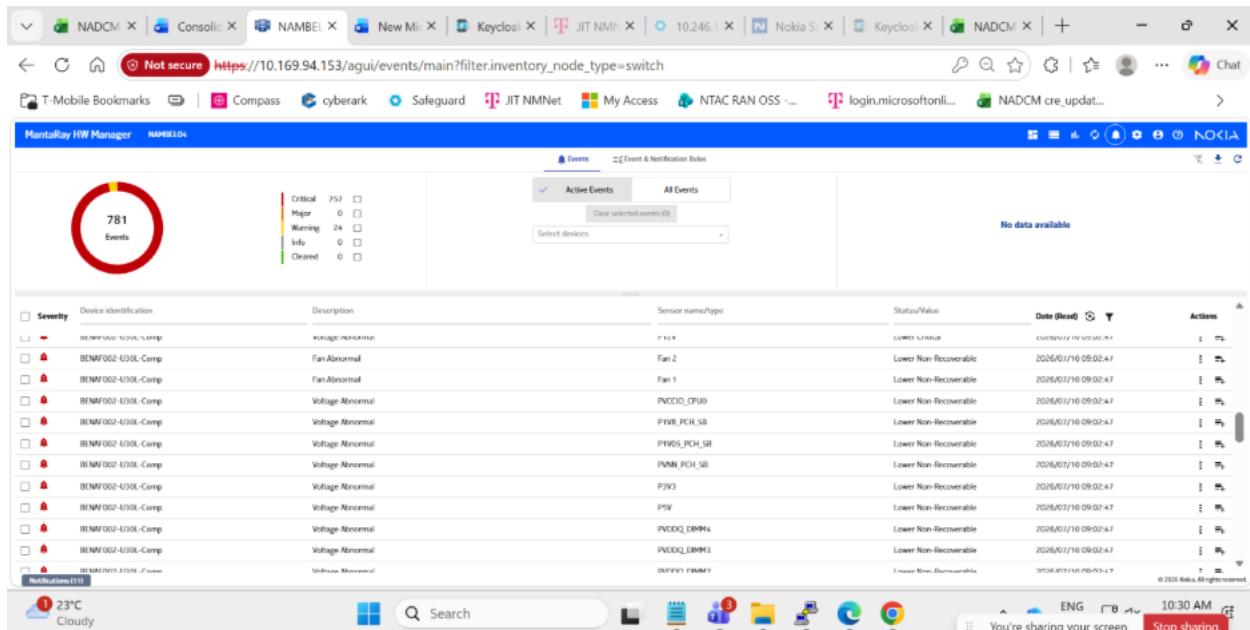
Failed attempt speed is factor, number of failed attempts is factor



33.Event Totals are properly updated; Filtering and display of Events Results managed properly

1. **Steps:** -Login to NADCM
2. Go to Events
3. List All Events
4. List All Active Events

Result: - Below screenshot shows the successful results.



34. SNMP AGENT WORKING

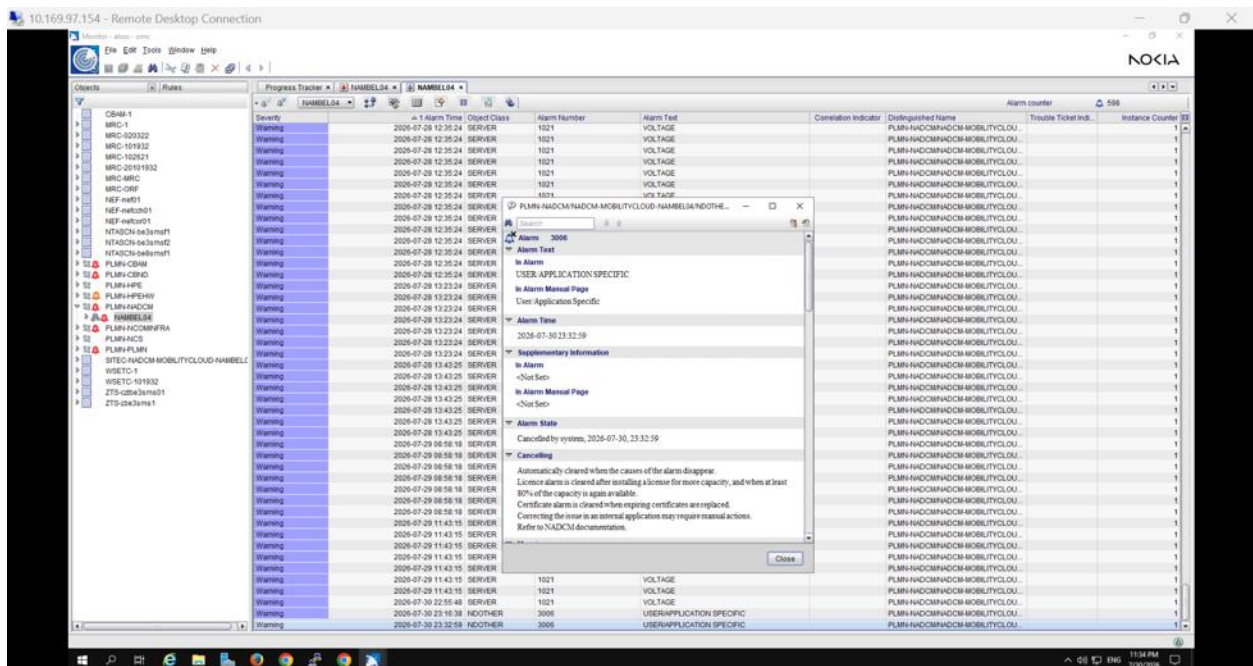
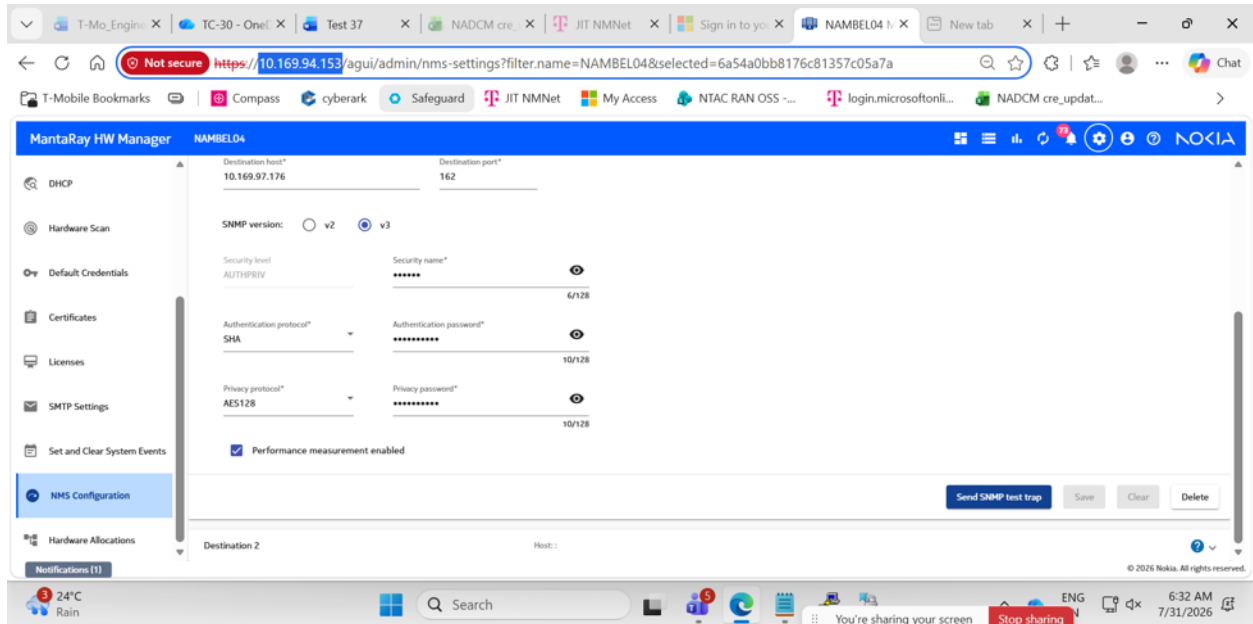
Steps: -

1. Login to NADCM as root
2. Setup tcpdump to capture SNMP

tcpdump -ni any port 162 -w /tmp/alarm.pcap

3. Generate test alarm by sending test trap by below procedures
4. Login the NADCM and go to Administration
5. Click on NMS Configuration
6. Send the test trap to the netact

Result: -



alarm.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.194.47.154	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
2	0.000002	10.194.47.154	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
3	0.000090	10.194.47.154	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
4	0.012364	10.194.47.158	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
5	0.012365	10.194.47.158	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
6	0.012414	10.194.47.158	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
7	0.036558	10.194.47.146	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
8	0.036560	10.194.47.146	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
9	0.036645	10.194.47.146	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
10	0.039951	10.194.47.150	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
11	0.039953	10.194.47.150	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
12	0.040023	10.194.47.150	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
13	0.097853	10.233.102.175	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
14	0.097923	10.169.94.153	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
15	0.097929	10.169.94.153	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
16	0.118171	10.233.102.175	10.169.97.176	SNMP	810	encryptedPDU: privKey Unknown

> Frame 1: Packet, 324 bytes on wire (2592 bits), 324 bytes captured (2592 bits)

Linux cooked capture v2

- Protocol: IPv4 (0x0800)
- Reserved: 0000
- Interface index: 4
- Link-layer address type: Ethernet (1)
- Packet type: Unicast to us (0)
- Link-layer address length: 6
- Source: Cisco_7a:d4:2f (18:59:f5:7a:d4:2f)
- Unused: 0000

> Internet Protocol Version 4, Src: 10.194.47.154, Dst: 10.169.94.153

> User Datagram Protocol, Src Port: 162, Dst Port: 162

> Simple Network Management Protocol

0000 08 00 00 00 00 00 04 00 01 00 06 18 59 f5 7aY z
 0010 d4 2f 00 00 45 00 01 30 c8 4e 00 00 35 11 18 d1 ./...E...N...S...
 0020 0a c2 2f 9a 0a 95 99 00 a2 00 a2 01 1c b3 5c/...^.....\...
 0030 30 82 01 10 02 01 03 30 0d 02 01 00 02 02 20 00 0.....0.....
 0040 04 01 03 02 01 03 04 39 30 37 04 0c 00 00 17 8b\$ 07.....
 0050 02 00 d8 9e f3 a4 24 20 02 01 0f 02 04 07 e8 7c|.....
 0060 c1 04 06 6e 65 74 61 63 74 04 0c 6f ef c5 9f 81netac t-o.....
 0070 8d 3d c5 82 c8 ca a0 04 08 dd 09 69 6c 78 ad 64-...l1x d.....
 0080 8a 04 81 c0 33 8f 8b 5a e8 cc ef c2 b1 c2 08 ed3 -Z.....
 0090 c4 34 2e 4c 41 40 12 22 66 11 cc 92 ca 95 ae 954.LAB...f.....
 00a0 a9 2e 79 a6 1c 48 a8 cc e2 37 8c 24 b6 8d fd 0dy...H...7...\$.....
 00b0 2d 34 9b fb 90 ff 9d 1d 9a 08 95 f9 bb 15 20 a84.....
 00c0 b3 df f1 f6 00 f9 90 60 f5 cb 85 3b 30 68 81 f3-...0h.....
 00d0 38 60 88 3d 98 36 db cf 2e b9 24 fb d0 30 98 0b 8'...6...\$-0.....
 00e0 6d 21 c2 a8 6e e0 d6 24 fb dc aa 3b aa f9 85 b9 m!...n...\$.....
 00f0 f2 14 e5 91 22 29 21 62 65 2d 83 6d 1e a7 2f 73")lb e-m.../s.....
 0100 83 e7 f9 b3 61 9c b3 09 f9 1a bf 34 f6 40 bd 54a...-4@T.....
 0110 d7 e8 90 96 57 24 6a b1 64 df 74 06 f8 4e 8a b6W\$J...d:t...N...
 0120 f6 57 44 57 0c 44 b5 ad b3 27 14 a2 61 f5 d2 79WDN D...a...y.....
 0130 a6 9a f0 16 84 88 d9 f6 72 25 99 2b f4 2e db 68r%+...h.....
 0140 cc 27 88 41A.....

Packets: 2970 Profile: Default

alarm.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.194.47.154	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
2	0.000002	10.194.47.154	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
3	0.000090	10.194.47.154	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
4	0.012364	10.194.47.158	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
5	0.012365	10.194.47.158	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
6	0.012414	10.194.47.158	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
7	0.036558	10.194.47.146	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
8	0.036560	10.194.47.146	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
9	0.036645	10.194.47.146	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
10	0.039951	10.194.47.150	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
11	0.039953	10.194.47.150	10.169.94.153	SNMP	324	encryptedPDU: privKey Unknown
12	0.040023	10.194.47.150	10.233.102.146	SNMP	324	encryptedPDU: privKey Unknown
13	0.097853	10.233.102.175	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
14	0.097923	10.169.94.153	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
15	0.097929	10.169.94.153	10.169.97.176	SNMP	964	encryptedPDU: privKey Unknown
16	0.118171	10.233.102.175	10.169.97.176	SNMP	810	encryptedPDU: privKey Unknown

> Internet Protocol Version 4, Src: 10.194.47.154, Dst: 10.169.94.153

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 304

Identification: 0xc84e (51278)

> 0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 53

Protocol: UDP (17)

Header Checksum: 0x18d1 [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.194.47.154

Destination Address: 10.169.94.153

[Stream index: 0]

> User Datagram Protocol, Src Port: 162, Dst Port: 162

> Simple Network Management Protocol

0000 08 00 00 00 00 00 04 00 01 00 06 18 59 f5 7aY z
 0010 d4 2f 00 00 45 00 01 30 c8 4e 00 00 35 11 18 d1 ./...E...N...S...
 0020 0a c2 2f 9a 0a 95 99 00 a2 00 a2 01 1c b3 5c/...^.....\...
 0030 30 82 01 10 02 01 03 30 0d 02 01 00 02 02 20 00 0.....0.....
 0040 04 01 03 02 01 03 04 39 30 37 04 0c 00 00 17 8b\$ 07.....
 0050 02 00 d8 9e f3 a4 24 20 02 01 0f 02 04 07 e8 7c|.....
 0060 c1 04 06 6e 65 74 61 63 74 04 0c 6f ef c5 9f 81netac t-o.....
 0070 8d 3d c5 82 c8 ca a0 04 08 dd 09 69 6c 78 ad 64-...l1x d.....
 0080 8a 04 81 c0 33 8f 8b 5a e8 cc ef c2 b1 c2 08 ed3 -Z.....
 0090 c4 34 2e 4c 41 40 12 22 66 11 cc 92 ca 95 ae 954.LAB...f.....
 00a0 a9 2e 79 a6 1c 48 a8 cc e2 37 8c 24 b6 8d fd 0dy...H...7...\$.....
 00b0 2d 34 9b fb 90 ff 9d 1d 9a 08 95 f9 bb 15 20 a84.....
 00c0 b3 df f1 f6 00 f9 90 60 f5 cb 85 3b 30 68 81 f3-...0h.....
 00d0 38 60 88 3d 98 36 db cf 2e b9 24 fb d0 30 98 0b 8'...6...\$-0.....
 00e0 6d 21 c2 a8 6e e0 d6 24 fb dc aa 3b aa f9 85 b9 m!...n...\$.....
 00f0 f2 14 e5 91 22 29 21 62 65 2d 83 6d 1e a7 2f 73")lb e-m.../s.....
 0100 83 e7 f9 b3 61 9c b3 09 f9 1a bf 34 f6 40 bd 54a...-4@T.....
 0110 d7 e8 90 96 57 24 6a b1 64 df 74 06 f8 4e 8a b6W\$J...d:t...N...
 0120 f6 57 44 57 0c 44 b5 ad b3 27 14 a2 61 f5 d2 79WDN D...a...y.....
 0130 a6 9a f0 16 84 88 d9 f6 72 25 99 2b f4 2e db 68r%+...h.....
 0140 cc 27 88 41A.....

Packets: 2970 Profile: Default

35. TLS Cert Verification

Steps: -

1. Execute below commands in NADCM.

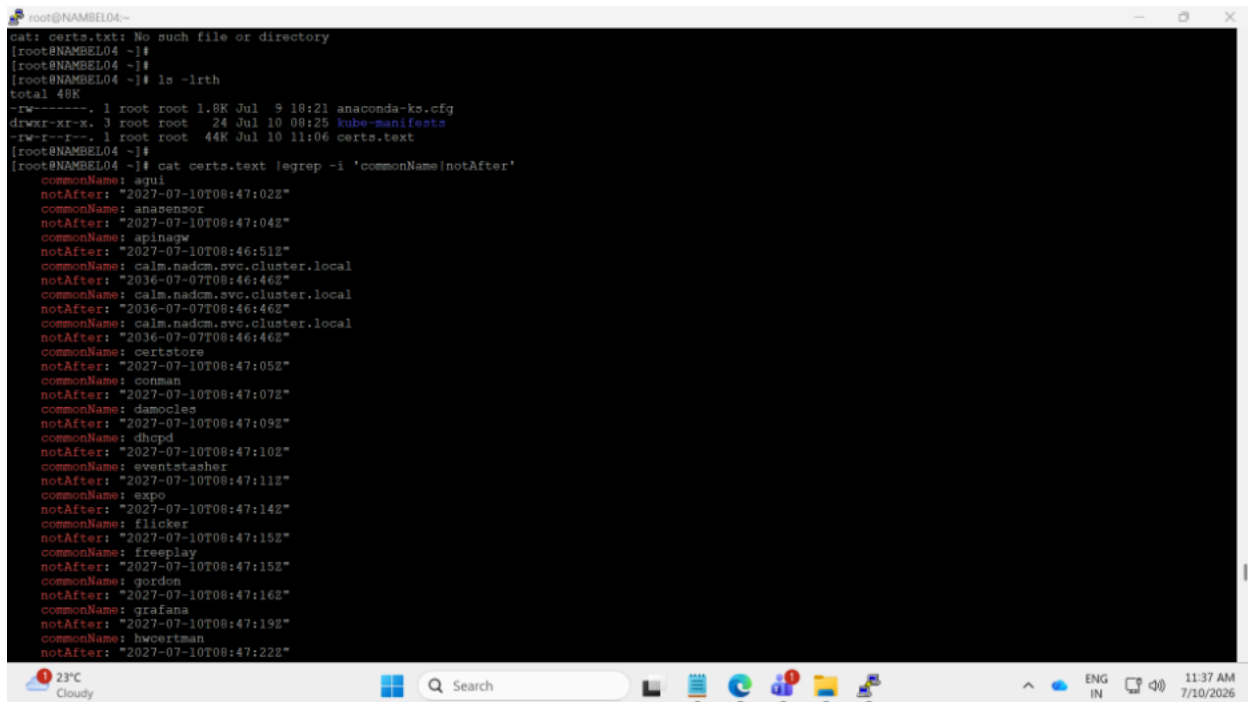
```
cat certs.txt | egrep -i 'commonName|notAfter'
```

```
cat certs.txt | egrep -i 'commonName|notAfter|notBefore|renewalTime'
```

2. Verify TLS cert is extended for 1 year.

Result: - certificate verification has been done by using the mentioned command.

```
cat certs.txt | egrep -i 'commonName|notAfter'
```



```
root@NAMBEL04:~
cat: certs.txt: No such file or directory
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ls -lrth
total 48K
-rw-r--r-- 1 root root 1.8K Jul  9 18:21 anaconda-ks.cfg
drwxr-xr-x 3 root root  24 Jul 10 08:25 kube-manifests
-rw-r--r-- 1 root root  44K Jul 10 11:06 certs.txt
[root@NAMBEL04 ~]# cat certs.txt | egrep -i 'commonName|notAfter'
commonName: agui
notAfter: "2027-07-10T08:47:02Z"
commonName: anasensor
notAfter: "2027-07-10T08:47:04Z"
commonName: apinagw
notAfter: "2027-07-10T08:46:51Z"
commonName: calm.nadcm.svc.cluster.local
notAfter: "2036-07-07T08:46:46Z"
commonName: calm.nadcm.svc.cluster.local
notAfter: "2036-07-07T08:46:46Z"
commonName: calm.nadcm.svc.cluster.local
notAfter: "2036-07-07T08:46:46Z"
commonName: certstore
notAfter: "2027-07-10T08:47:05Z"
commonName: conman
notAfter: "2027-07-10T08:47:07Z"
commonName: damocles
notAfter: "2027-07-10T08:47:09Z"
commonName: dhsdpd
notAfter: "2027-07-10T08:47:10Z"
commonName: eventstasher
notAfter: "2027-07-10T08:47:11Z"
commonName: expo
notAfter: "2027-07-10T08:47:14Z"
commonName: flicker
notAfter: "2027-07-10T08:47:15Z"
commonName: freeplay
notAfter: "2027-07-10T08:47:15Z"
commonName: gordon
notAfter: "2027-07-10T08:47:16Z"
commonName: grafana
notAfter: "2027-07-10T08:47:19Z"
commonName: hwcertman
notAfter: "2027-07-10T08:47:22Z"
```

```
cat certs.txt | egrep -i 'commonName|notAfter|notBefore|renewalTime'
```

```
root@NAMBEL04:~  
notAfter: "2027-07-10T08:47:00Z"  
commonName: rope  
notAfter: "2027-07-10T08:47:32Z"  
commonName: southgw  
notAfter: "2027-07-10T08:47:33Z"  
commonName: switch-inspector  
notAfter: "2027-07-10T08:47:34Z"  
commonName: tftpserver  
notAfter: "2027-07-10T08:47:35Z"  
commonName: tolkien  
notAfter: "2027-07-10T08:47:36Z"  
[root@NAMBEL04 ~]# cat certs.text | egrep -i 'commonName|notAfter|notBefore| renewalTime'  
commonName: agui  
notAfter: "2027-07-10T08:47:02Z"  
notBefore: "2026-07-10T08:47:02Z"  
renewalTime: "2027-06-10T08:47:02Z"  
commonName: anasensor  
notAfter: "2027-07-10T08:47:04Z"  
notBefore: "2026-07-10T08:47:04Z"  
renewalTime: "2027-06-10T08:47:04Z"  
commonName: apinagw  
notAfter: "2027-07-10T08:46:51Z"  
notBefore: "2026-07-10T08:46:51Z"  
renewalTime: "2027-06-10T08:46:51Z"  
commonName: calm.nadcm.svc.cluster.local  
notAfter: "2036-07-07T08:46:46Z"  
notBefore: "2026-07-10T08:46:46Z"  
renewalTime: "2036-06-22T08:46:46Z"  
commonName: calm.nadcm.svc.cluster.local  
notAfter: "2036-07-07T08:46:46Z"  
notBefore: "2026-07-10T08:46:46Z"  
renewalTime: "2036-06-22T08:46:46Z"  
commonName: calm.nadcm.svc.cluster.local  
notAfter: "2036-07-07T08:46:46Z"  
notBefore: "2026-07-10T08:46:46Z"  
renewalTime: "2036-06-22T08:46:46Z"  
commonName: certstore  
notAfter: "2027-07-10T08:47:05Z"  
notBefore: "2026-07-10T08:47:05Z"  
renewalTime: "2027-06-10T08:47:05Z"  
commonName: conman  
notAfter: "2027-07-10T08:47:07Z"  
notBefore: "2026-07-10T08:47:07Z"  
renewalTime: "2027-06-10T08:47:07Z"
```

23°C
Cloudy



Search



ENG

IN



11:38 AM
7/10/2026

36. NADCM Host Alarm

Steps: -

1. Login the NADCM GUI.
2. Generate any alarm which will reflect in NADCM.
3. Change the value for sensor threshold and check the alarm in NADCM Events.

Result: -

ipmitool -I lanplus -U "<Username>" -P <Password>" -H "<BMC IP>" sensor | grep 'Volt_PVCCIN_CPU0'

```
[root@NAMES04 ~]# ipmitool -I lanplus -U admin -P 'K&Hw76K4puGS3Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0
Volt_PVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | na | 2.205 | na
[root@NAMES04 ~]# ipmitool -U admin -P 'K&Hw76K4puGS3Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0
Volt_PVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | na | 2.205 | na
[root@NAMES04 ~]# ipmitool -I lanplus -U admin -P 'K&Hw76K4puGS3Guk' -H 10.169.94.223 sensor thresh 'Volt_PVCCIN_CPU0' 1.5
sensor thresh <id> <threshold> <setting>
```

```
[root@NAMES04 ~]# ipmitool -U admin -P 'K&Hw76K4puGS3Guk' -H 10.169.94.223 sensor thresh 'Volt_PVCCIN_CPU0' ucr 1.5
Locating sensor record 'Volt_PVCCIN_CPU0'...
Setting sensor 'Volt_PVCCIN_CPU0' Upper Critical threshold to 1.500
[root@NAMES04 ~]# ipmitool -U admin -P 'K&Hw76K4puGS3Guk' -H 10.169.94.223 sensor | grep Volt_PVCCIN_CPU0
Volt_PVCCIN_CPU0 | 1.701 | Volts | cr | na | 1.431 | na | na | 1.503 | na
[root@NAMES04 ~]#
```

Not securehttps://10.169.94.223/#dashboard

T-Mobile BookmarksImport favoritesPacto- NAMDAL04Pacto- NAMELG04Pacto- NAMNVL04Privacy errorCompass linkMicrosoft Entra ~...

AirFrame BMC WebUIadmin - Administrator | About | Log OutNOKIA

DashboardSensorSystem InfoSettingsLogs & ReportsControl & MaintainSyncRefresh

Home > Dashboard

Device Information
BMC Date&Time: 21 Jul 2026 10:32:03
More Info

0 d 12 hrs
Total System Uptime
Power Cycle

Today (51)Details
Login Info
38 events

30 days (51)Details
Login Info
38 events

Threshold Sensor Monitoring - 1 critical sensors

Volt_PVCCIN_CPU0	1.71 Volts		
Volt_PVCCIN_CPU0	1.71 Volts		
Volt_PVCCIN_CPU0	1.71 Volts		

WindowsTaskbar icons

26°C
Light rain4:02 PM
7/21/2026

Not securehttps://10.169.94.153/agui/devices?filter.inventory_node_type=server&filter.obm_ip=10.169.94.223&selected=6a54a0bb8...

T-Mobile BookmarksImport favoritesPacto- NAMDAL04Pacto- NAMELG04Pacto- NAMNVL04Privacy errorCompass linkMicrosoft Entra ~...

MantaRay HW ManagerNAMBEL04

Type/ProfileName/IPStatusLocationTagsHW LabelsFW versionsOther1 Items

Type	Name	Status	Hardware
Server	NAMBEL04	Online	SP110

Active events

Severity	Device Identification	Description	Sensor name	Value	Date (Read)
Critical	NAMBEL04	Voltage Abnormal	Volt_PVCCIN_CPU0	Upper Critical	2026/07/21 15:59:53
Management IP: 10.169.94.223Location: / / / / Device name: NAMBEL04Description: Voltage Abnormal Device type: SERVERSensor type: Voltage Origin: IpmbSensorSensor name: Volt_PVCCIN_CPU0 Read timestamp: 2026/07/21 15:59:53 GMT+05:30Sensor number: 216 Sensor ID: 216 Status: Upper Critical Lower Critical: 1.431 Upper Critical: 1.503 Value: 1.710 (+/- 0) Volts					
Warning	NAMBEL04	Predictive Failure - VIN_UV_WARNING - Asserted	PSU1_Status		2026/07/21 00:35:53
Warning	NAMBEL04	Login failed from IP:10.147.179.96	Login Info		2026/07/21 00:35:53

All events

Severity	Device Identification	Description	Sensor name	Value	Active	Date (Read)
Critical	NAMBEL04	Voltage Abnormal	Volt_PVCCIN_CPU0	Upper Critical	Yes	2026/07/21

WindowsTaskbar icons

26°C
Light rain4:03 PM
7/21/2026

37. NADCM Host Alarm-Netact

Execute **TC-48 NADCM Host Alarm** and check the alarm in netact.

Result: -

The screenshot shows the MantaRay HW Manager web interface. At the top, a red circle highlights '782 Events'. Below this, a table lists event severities: Critical (741), Major (0), Warning (41), Info (0), and Cleared (0). The main table displays a list of events. The first event is for device NAMBEL04, with a description 'Voltage Abnormal' and sensor 'Volt_FVCCIN_CPU0'. The status is 'Upper Critical' and the date is '2026/07/31 05:55:54'. A detailed view of this event is shown below the table, including management IP, device name, device type, origin, read timestamp, location, description, sensor name/type, status/value, and date. The status is 'Upper Critical' and the value is '1.710 (+/- 0) Volts'. Other events for BELFI032 and OPARF002_Compute-U09 are also visible.

Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Read)	Actions
Critical	NAMBEL04	Voltage Abnormal	Volt_FVCCIN_CPU0	Upper Critical	2026/07/31 05:55:54	
Major						
Warning						
Info						
Cleared						

Management IP: 10.169.94.223
Device name: NAMBEL04
Device type: SERVER
Origin: ipmisensor
Read timestamp: 2026/07/31 5:55:54 GMT+00:00

Location: Voltage Abnormal
Description: Voltage Abnormal
Sensor type: Voltage
Sensor name: Volt_FVCCIN_CPU0
Sensor number: 216
Sensor ID: 216
Status: Upper Critical
Lower Critical: 1.431
Upper Critical: 1.503
Value: 1.710 (+/- 0) Volts

```
Volt_CPU0 DEF | 1.225 | Volts | ok | na | 1.078 | na | na | 1.323 | na
Volt_P12V | 12.033 | Volts | ok | na | 10.773 | na | na | 13.230 | na
Volt_P12V_AUX | 11.970 | Volts | ok | na | 11.214 | na | na | 13.041 | na
Volt_P1V05_PCH | 1.043 | Volts | ok | na | 0.945 | na | na | 1.155 | na
Volt_P1V9_PCH | 1.728 | Volts | ok | na | 1.620 | na | na | 1.989 | na
Volt_P3V3 | 3.264 | Volts | ok | na | 2.975 | na | na | 3.621 | na
Volt_P5V | 4.982 | Volts | ok | na | 4.498 | na | na | 5.538 | na
Volt_FVCCIN_CPU0 | 1.701 | Volts | ok | na | 1.431 | na | na | 2.205 | na
Volt_FVCCIN_CPU1 | 1.701 | Volts | ok | na | 1.431 | na | na | 2.205 | na
Volt_FVCCIO_CPU0 | 0.980 | Volts | ok | na | 0.882 | na | na | 1.057 | na
Volt_FVCCIO_CPU1 | 0.994 | Volts | ok | na | 0.882 | na | na | 1.057 | na
Volt_FVNN_PCH | 0.987 | Volts | ok | na | 0.763 | na | na | 1.106 | na
Volt_P3V_BAT | 2.877 | Volts | ok | na | 2.205 | na | na | 3.591 | na
Watchdog | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
PSU_FAN_0 | 5700.000 | RPM | ok | na | na | na | na | na | na
PSU_FAN_1 | 5700.000 | RPM | ok | na | na | na | na | na | na
SYS_CPU_Util | 6.000 | percent | ok | na | na | na | na | na | na
SYS_Mem_Util | 2.000 | percent | ok | na | na | na | na | na | na
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor | grep Volt_FVCCIN_CPU0
Volt_FVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | na | 2.205 | na
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor thresh 'Volt_FVCCIN_CPU0' ucr 1.5
Locating sensor record 'Volt_FVCCIN_CPU0'...
Setting sensor "Volt_FVCCIN_CPU0" Upper Critical threshold to 1.500
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor sensor thresh 'Volt_FVCCIN_CPU0' ucr 2.205°C
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor thresh 'Volt_FVCCIN_CPU0' ucr 2.205
Locating sensor record 'Volt_FVCCIN_CPU0'...
Setting sensor "Volt_FVCCIN_CPU0" Upper Critical threshold to 2.205
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor | grep Volt_FVCCIN_CPU0
Volt_FVCCIN_CPU0 | 1.710 | Volts | ok | na | 1.431 | na | na | 2.205 | na
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# ipmitool sensor | grep Volt_FVCCIN_CPU0
```

A screenshot of a Windows taskbar. The taskbar is dark grey and contains several application icons: the Start button (Windows logo), Search (magnifying glass), Task View (three horizontal bars), Microsoft Edge (blue 'e'), File Explorer (yellow folder), OneDrive (blue cloud), Google Chrome (red, yellow, and green circles), and a small icon of a person. On the right side of the taskbar, there are icons for network, volume, and battery, followed by the system clock showing '10:55 PM' and the date '7/30/2026'.

38. Server Alarm FM

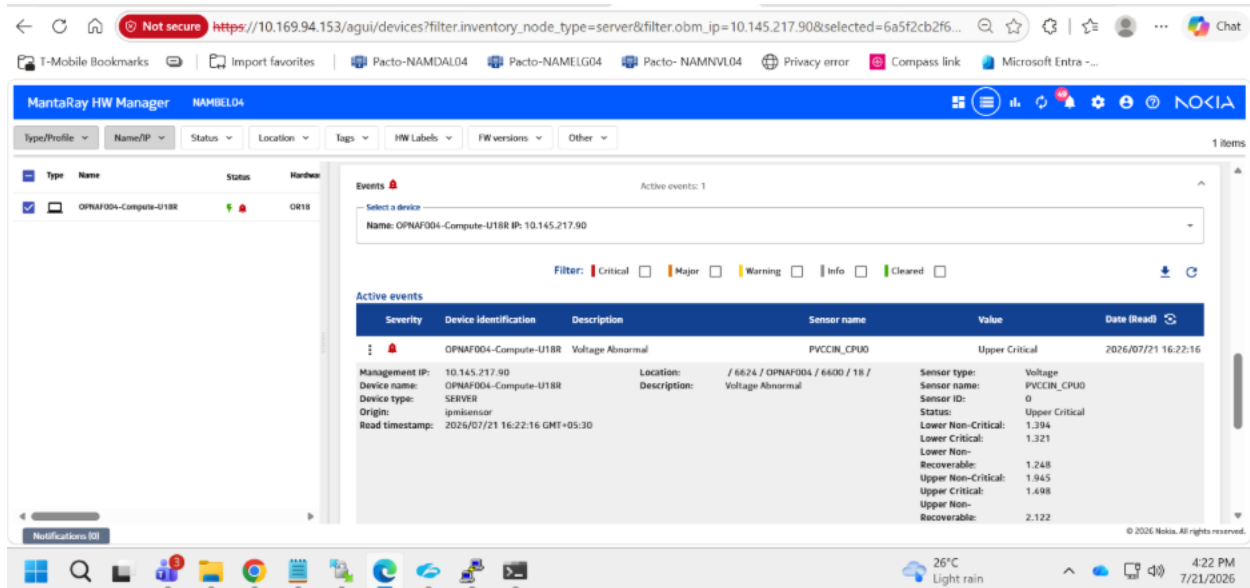
Steps: -

1. Consider any server and SSH into the NADCM
2. Using the `ipmitool lanplus` command, Check sensor readings of particular server.
2. Update the Outlet temperature sensor threshold to generate alarm on the compute.
3. Verify Critical alarm for the temperature event on NADCM.

Result: - Alarm has been generated manually to verify the test case.

```
root@NAMBEL04 ~#  
[nokia@NAMBEL04 ~]$  
[nokia@NAMBEL04 ~]$  
[nokia@NAMBEL04 ~]$  
[nokia@NAMBEL04 ~]$ sudo -i  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'Q%$^f1Qa$w#edTler2' -H 10.145.217.90 sensor  
PVCCIN_CPU0 | 1.778 | Volts | ok | 1.248 | 1.321 | 1.394 | 1.945 | 2.028 | 2.122  
PVCCIN_CPU1 | 1.789 | Volts | ok | 1.248 | 1.321 | 1.394 | 1.945 | 2.028 | 2.122  
PVDDQ_DIMM1 | 1.211 | Volts | ok | 1.022 | 1.078 | 1.134 | 1.260 | 1.316 | 1.372  
PVDDQ_DIMM2 | 1.211 | Volts | ok | 1.022 | 1.078 | 1.134 | 1.260 | 1.316 | 1.372  
PVDDQ_DIMM3 | 1.211 | Volts | ok | 1.022 | 1.078 | 1.134 | 1.260 | 1.316 | 1.372  
PVDDQ_DIMM4 | 1.211 | Volts | ok | 1.022 | 1.078 | 1.134 | 1.260 | 1.316 | 1.372  
P5V | 5.076 | Volts | ok | 4.222 | 4.481 | 4.740 | 5.232 | 5.491 | 5.750  
P3V3 | 3.317 | Volts | ok | 2.787 | 2.958 | 3.129 | 3.454 | 3.625 | 3.796  
PVNN_PCH_SB | 0.994 | Volts | ok | 0.714 | 0.763 | 0.805 | 1.050 | 1.099 | 1.148  
PIV05_PCH_SB | 1.064 | Volts | ok | 0.889 | 0.945 | 0.994 | 1.099 | 1.155 | 1.204  
PIV8_PCH_SB | 1.824 | Volts | ok | 1.560 | 1.654 | 1.748 | 1.927 | 2.021 | 2.115  
PVCCIO_CPU0 | 1.001 | Volts | ok | 0.805 | 0.854 | 0.903 | 1.050 | 1.099 | 1.148  
PVCCIO_CPU1 | 1.001 | Volts | ok | 0.805 | 0.854 | 0.903 | 1.050 | 1.099 | 1.148  
VDRP_IN | 3.038 | Volts | ok | 2.216 | 2.279 | 2.384 | na | 3.545 | na  
Inlet Temp | 27.000 | degrees C | ok | 0.000 | 0.000 | 0.000 | 55.000 | 60.000 | na  
Outlet Temp | 39.000 | degrees C | ok | 0.000 | 0.000 | 0.000 | 62.000 | 67.000 | na  
P0 Temp | 42.000 | degrees C | ok | 0.000 | 0.000 | 0.000 | 85.000 | 95.000 | 105.000  
P1 Temp | 46.000 | degrees C | ok | 0.000 | 0.000 | 0.000 | 85.000 | 95.000 | 105.000  
Fan 1 | 4788.000 | RPM | ok | 0.000 | 1008.000 | 1323.000 | 16065.000 | 16065.000 | 16065.000  
Fan 2 | 4788.000 | RPM | ok | 0.000 | 1008.000 | 1323.000 | 16065.000 | 16065.000 | 16065.000  
PI2V | 12.480 | Volts | ok | na | 10.800 | 11.700 | na | 13.200 | na  
Power | 175.000 | Watts | ok | na | na | na | na | na | na  
Current | 14.000 | Amps | ok | na | na | na | na | na | na  
PowerStatus | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
Pwr_Sts | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
ChassisButton | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
FreeBmcMemory | 253.000 | megabyte | ok | 3.000 | 9.000 | 27.000 | na | na | na  
CPU1ERR | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
PCHTHERMTRIP | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
CPU0PROCROT | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
CPU1PROCROT | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
BIOS_POST_CMPLT | 0x0 | discrete | 0x0080 | na | na | na | na | na | na  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'Q%$^f1Qa$w#edTler2' -H 10.145.217.90 sensor | grep Volt_PVCCIN_CPU0
```

```
[root@NAMBEL04 ~]#  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'Q%$^f1Qa$w#edTler2' -H 10.145.217.90 sensor | grep PVCCIN_CPU0  
PVCCIN_CPU0 | 1.778 | Volts | ok | 1.248 | 1.321 | 1.394 | 1.945 | 2.028 | 2.122  
[root@NAMBEL04 ~]# ipmitool -I lanplus -U admin -P 'Q%$^f1Qa$w#edTler2' -H 10.145.217.90 sensor | grep PVCCIN_CPU0  
PVCCIN_CPU0 | 1.778 | Volts | ok | 1.248 | 1.321 | 1.394 | 1.945 | 2.028 | 2.122  
[root@NAMBEL04 ~]#
```



39. Server Alarm FM -Netact

Steps: -

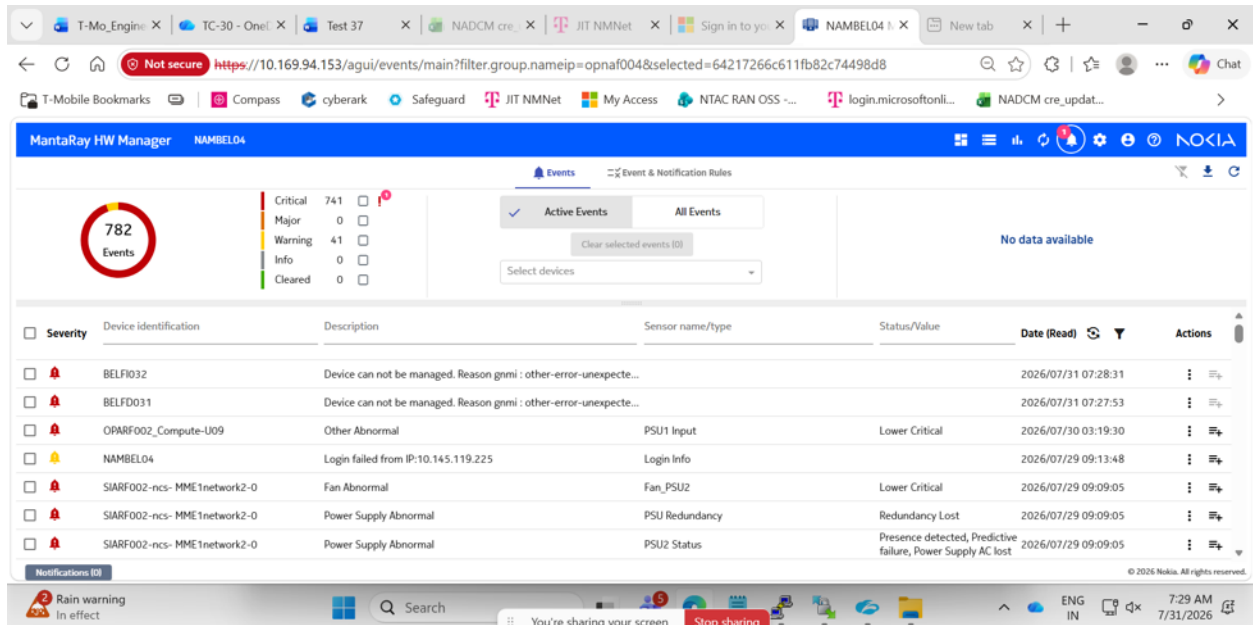
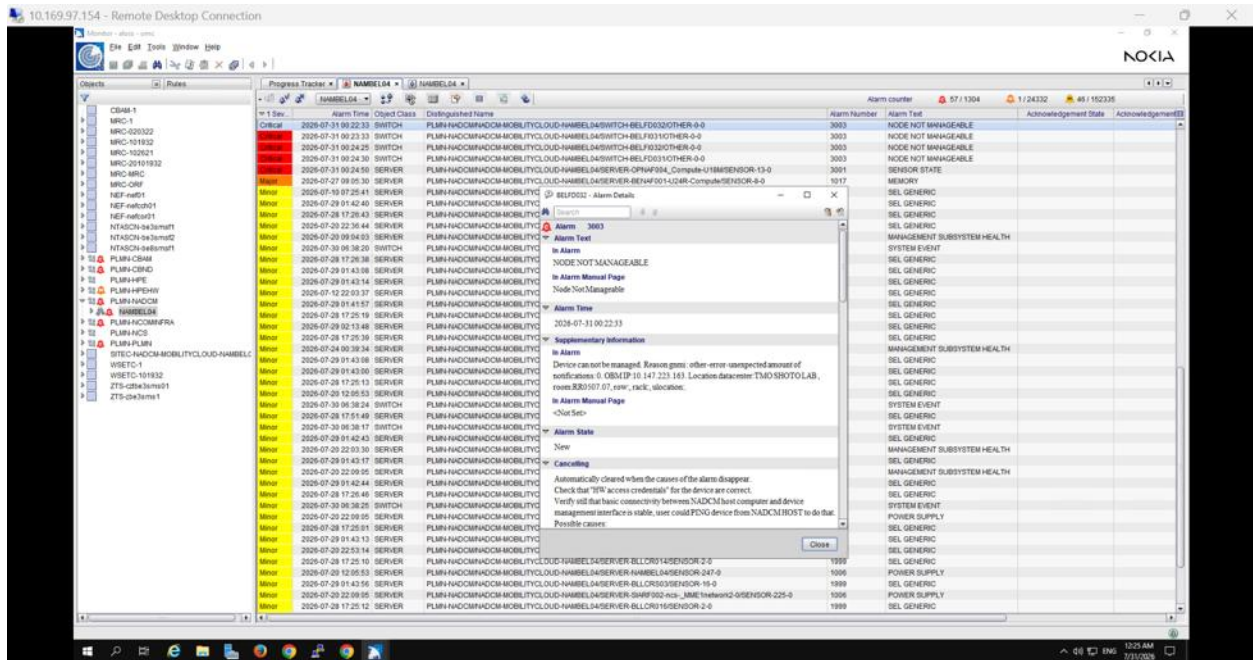
Execute **TC-56 Server Alarm FM** and confirm that alarm is reflecting in netact.

Result: - Below screenshot shows server alarm in NADCM that can be seen in NetAct.

```

root@NMBEL04:~#
CPU1ERR          | 0x0 | discrete | 0x0080 | na | na | na | na | na | na | na
PCHTHERMTRIP     | 0x0 | discrete | 0x0080 | na | na | na | na | na | na | na
CPU0PROCHOT      | 0x0 | discrete | 0x0080 | na | na | na | na | na | na | na
CPU1PROCHOT      | 0x0 | discrete | 0x0080 | na | na | na | na | na | na | na
BIOS_POST_CMPLT  | 0x0 | discrete | 0x0080 | na | na | na | na | na | na | na
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep Volt_FVCCIN_CPU0
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep PVCCIN_CPU0
PVCCIN_CPU0      | 1.768 | Volts | ok | 1.248 | 1.321 | 1.394 | 1.945 | 2.028 | 2.122
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor thresh 'PVCCIN_CPU0' ucr 2.205
Locating sensor record 'PVCCIN_CPU0'...
Invalid Threshold data values. Cannot Set Threshold Data.
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor thresh 'PVCCIN_CPU0' ucr 1.248
Locating sensor record 'PVCCIN_CPU0'...
Invalid Threshold data values. Cannot Set Threshold Data.
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep Outlet Temp
grep: Temp: No such file or directory
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep Outlet Temp
grep: Temp: No such file or directory
^C[root@NMBEL04 ~]#
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep Outlet_Temp
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep VBAT_HM
VBAT_HM          | 3.017 | Volts | ok | 2.216 | 2.279 | 2.384 | na | 3.545 | na
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor thresh 'VBAT_HM' ucr 2.505
Locating sensor record 'VBAT_HM'...
Setting sensor "VBAT_HM" Upper Critical threshold to 2.505
[root@NMBEL04 ~]#
[root@NMBEL04 ~]# ipmitool -I lanplus -U admin -P "O%s*FIQa@ws#edTier2" -H 10.145.217.89 sensor | grep VBAT_HM
VBAT_HM          | 3.017 | Volts | cr | 2.216 | 2.279 | 2.384 | na | 2.511 | na
[root@NMBEL04 ~]# █

```

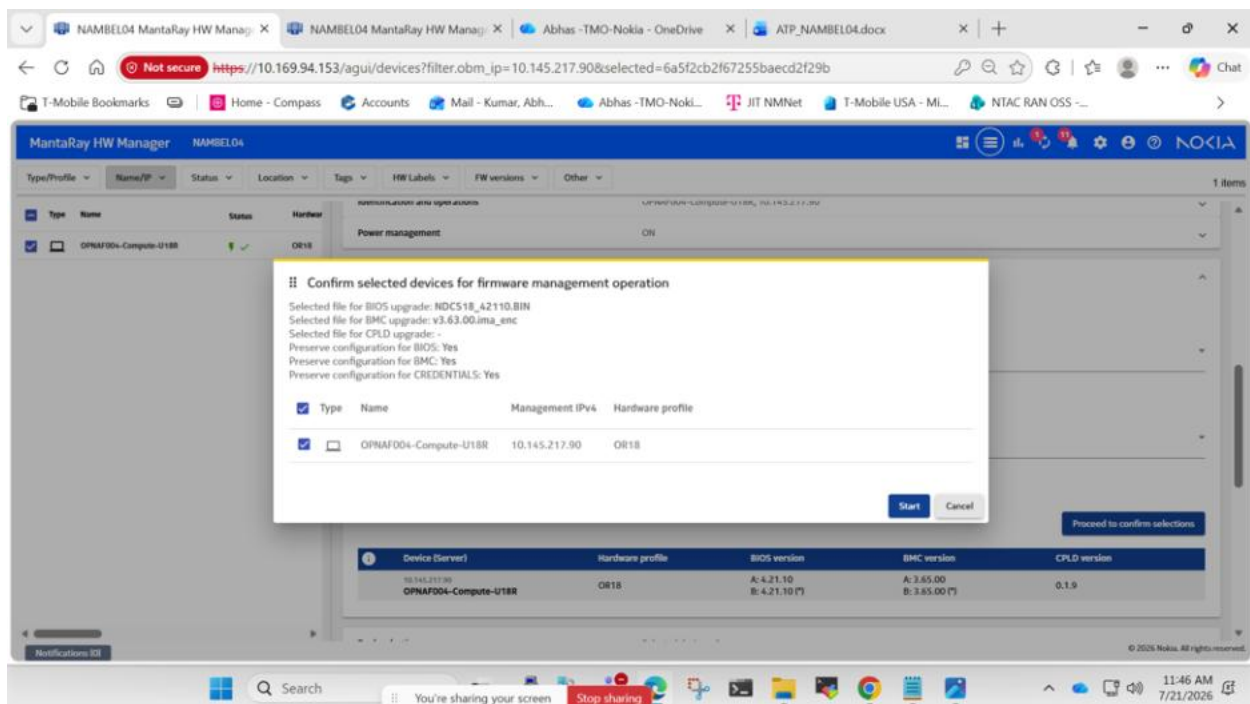


40. BMC/BIOS Downgrade and Upgrade in OR18

Steps: -

1. In Devices → Select the server → Firmware Management
2. Choose BMC & BIOS image files.
3. Confirm the devices and image files → Start the activity.
4. Once completed, perform the same steps again to start the upgrade.

Result: - Successfully updated BMC/BIOS of server using NADCM.



NAMBEL04 MantaRay HW Manag... X NAMBEL04 MantaRay HW Manag... X Abhas - TMO-Nokia - OneDrive X ATP_NAMBEL04.docx X

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.217.90&selected=6a5f2cb2f67255baecd2f29b

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MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18B	✔	OR18

Files selection:

binrc_fw
File: NDCS18_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update model(s) might be powered off!

binrc_fw
File: v3.63.00.ima_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

[Proceed to confirm selections](#)

Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18B	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.63.00 (*) B: 3.65.00	0.1.9

Bank selection Selected devices: 1

Credential management Selected devices: 1

USED PROTOCOLS REDFISH MANAGEMENT (PMU/REST/SSH/JPMH) SNOOPv2/v3 TRAP RECEIVER KVM/VT/VMWARE SSH

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MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18B	✔	OR18

Power management ON

Firmware management Selected devices: 1

Files selection:

binrc_fw
File: NDCS18_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update model(s) might be powered off!

binrc_fw
File: v3.63.00.ima_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

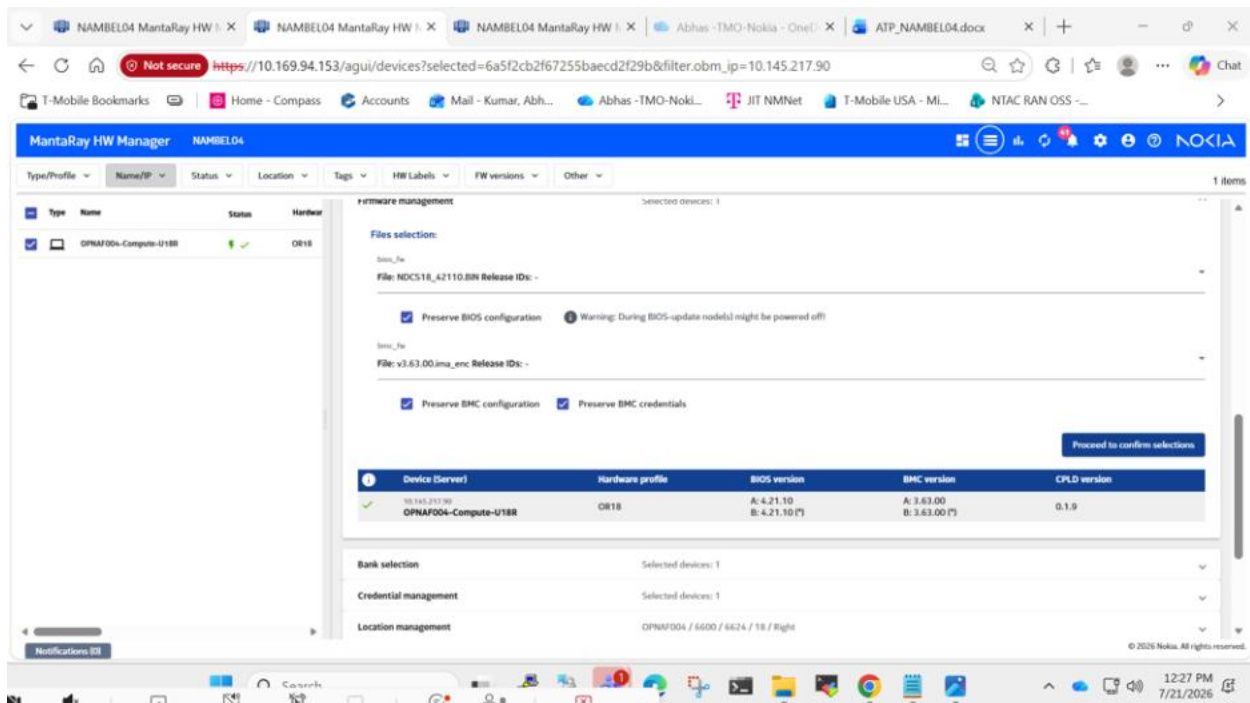
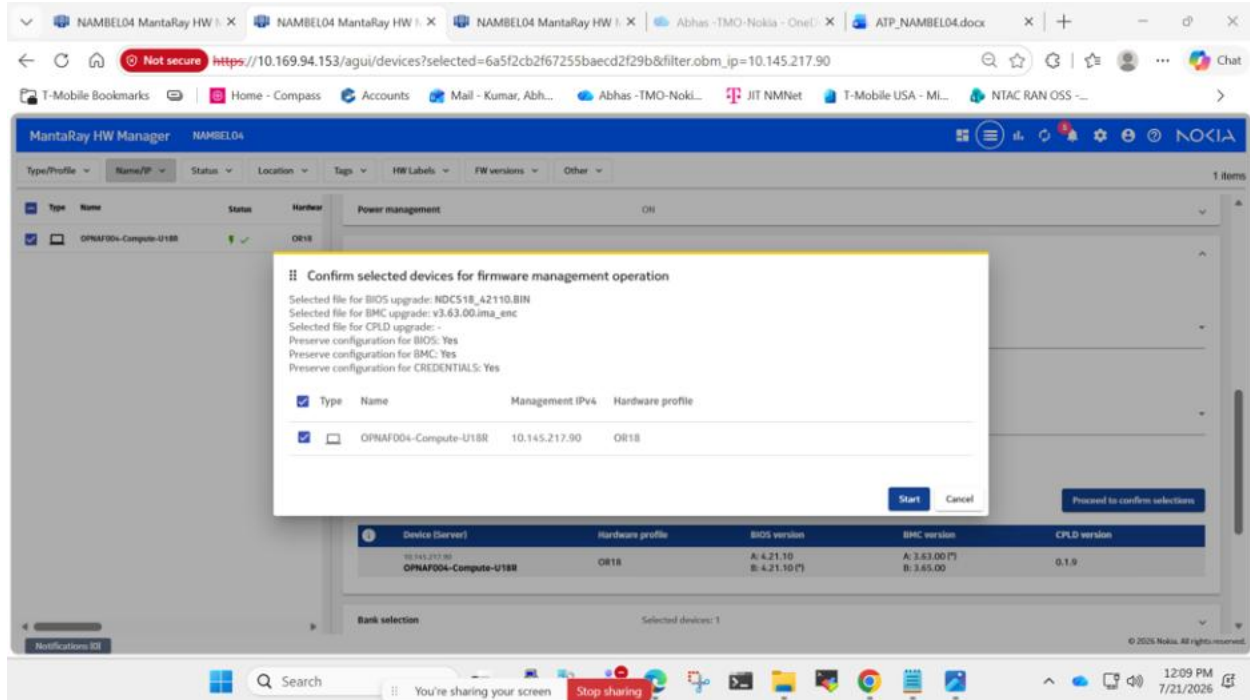
[Proceed to confirm selections](#)

Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18B	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.63.00 (*) B: 3.65.00	0.1.9

Bank selection Selected devices: 1

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MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18R	✓	OR18

Power management ON

Firmware management Selected devices: 1

Files selection:

File: NDC518_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update nodeid might be powered off!

File: v3.65.00.ima_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

Proceed to confirm selections

Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18R	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.63.00 B: 3.63.00 (*)	0.1.9

Bank selection Selected devices: 1

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NAMBEL04 MantaRay X NAMBEL04 MantaRay X NAMBEL04 MantaRay X NAMBEL04 MantaRay X Abhas -TMO-Nokia X ATP_NAMBEL04.docx X

Not secure https://10.169.94.153/aguui/devices?selected=6a5f2cb2f67255baecd2f29b&filter.obm_ip=10.145.217.90

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MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18R	✓	OR18

Power management ON

Firmware management Selected devices: 1

Files selection:

File: NDC518_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update nodeid might be powered off!

File: v3.65.00.ima_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

Proceed to confirm selections

Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18R	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.63.00 B: 3.63.00 (*)	0.1.9

Bank selection Selected devices: 1

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Confirm selected devices for firmware management operation

Selected file for BIOS upgrade: NDC518_42110.BIN
Selected file for BMC upgrade: v3.65.00.ima_enc
Selected file for CPLD upgrade: -
Preserve configuration for BIOS: Yes
Preserve configuration for BMC: Yes
Preserve configuration for CREDENTIALS: Yes

Type	Name	Management IPv4	Hardware profile
☒	OPNAF004-Compute-U18R	10.145.217.90	OR18

Start Cancel Proceed to confirm selections

NAMBELO4 MantaRay X NAMBELO4 MantaRay X NAMBELO4 MantaRay X NAMBELO4 MantaRay X Abhas -TMO-Nokia X ATP_NAMBELO4.docx X

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MantaRay HW Manager NAMBELO4

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18R		OR18

bin_cfe
File: NDCS18_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update modeld might be powered off!

bin_bmc
File: v3.65.00.bmc_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

Proceed to confirm selections

Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18R	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.65.00 (*) B: 3.63.00	0.1.9

Bank selection Selected devices: 1

Credential management Selected devices: 1

Location management OPNAF004 / 6600 / 6624 / 18 / Right

Interface management Interface count: 1

Notifications 0/3

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12:49 PM 7/21/2026

NAMBELO4 Manti X NAMBELO4 Manti X NAMBELO4 Manti X NAMBELO4 Manti X NAMBELO4 Manti X Abhas -TMO-Ni X ATP_NAMBELO4 X

Not secure https://10.169.94.153/aguui/devices?selected=6a5f2cb2f67255baecd2f29b8&filter.obm_ip=10.145.217.90

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBELO4

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNAF004-Compute-U18R		OR18

Power management ON

Firmware management Selected devices: 1

Files selection:

bin_cfe
File: NDCS18_42110.BIN Release ID: -

☒ Preserve BIOS configuration Warning: During BIOS-update modeld might be powered off!

bin_bmc
File: v3.65.00.bmc_enc Release ID: -

☒ Preserve BMC configuration ☒ Preserve BMC credentials

Proceed to confirm selections

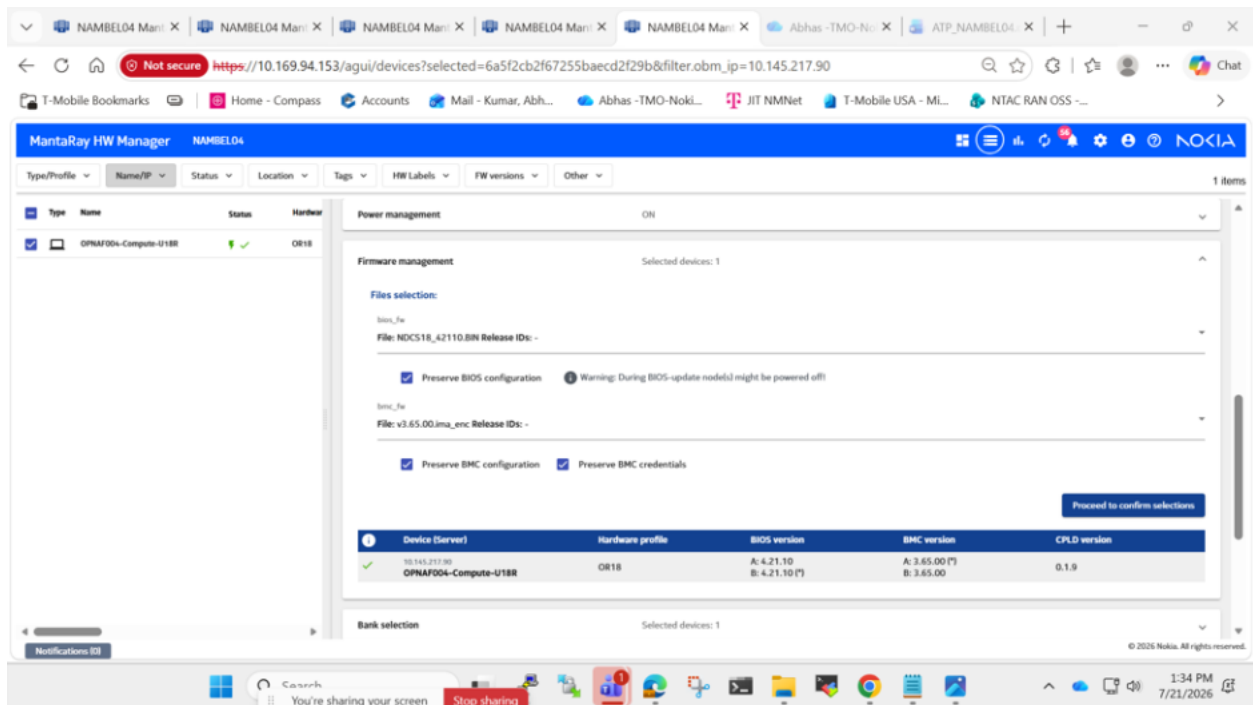
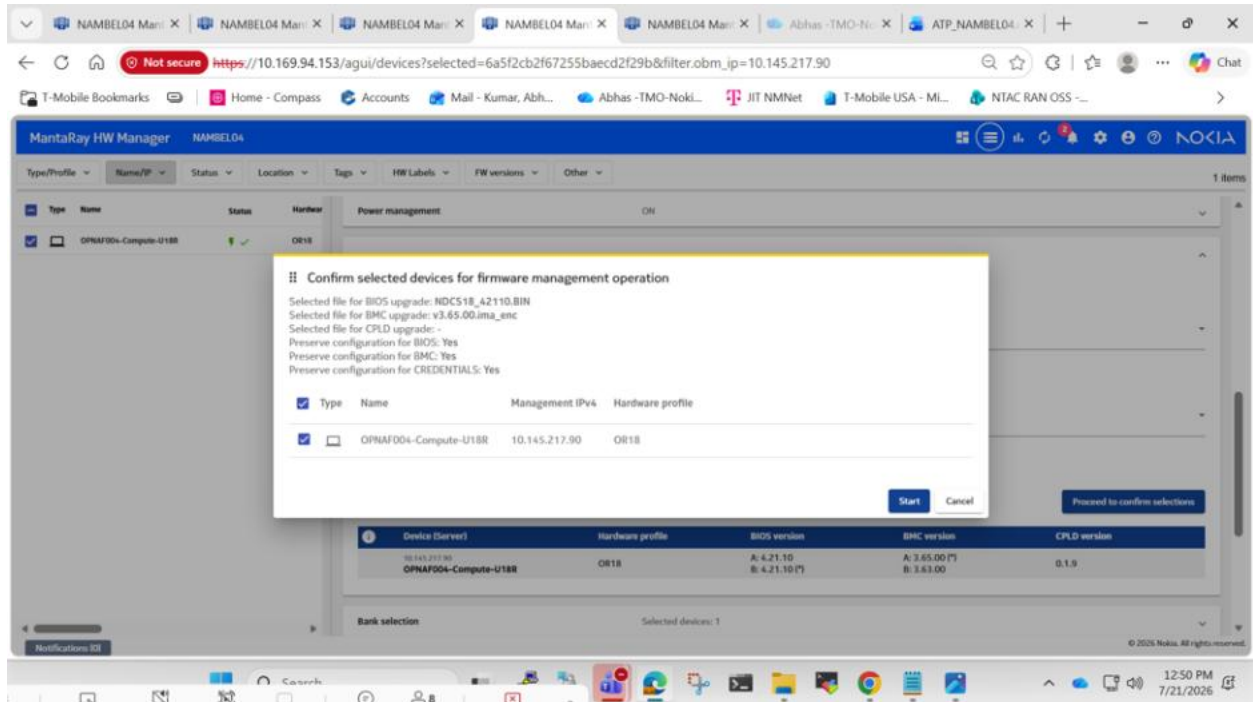
Device (Server)	Hardware profile	BIOS version	BMC version	CPLD version
10.145.217.90 OPNAF004-Compute-U18R	OR18	A: 4.21.10 B: 4.21.10 (*)	A: 3.65.00 (*) B: 3.63.00	0.1.9

Bank selection Selected devices: 1

Notifications 0/3

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12:49 PM 7/21/2026



Windows taskbar showing the Start button, search bar, and several pinned applications including Edge, Word, PowerPoint, and Teams. The system tray on the right shows the time as 1:35 PM on 7/21/2026.

41. Kubernetes cert verification

Steps: -

1. Execute kubeadm alpha certs check-expiration and verify the validity of all certificates.

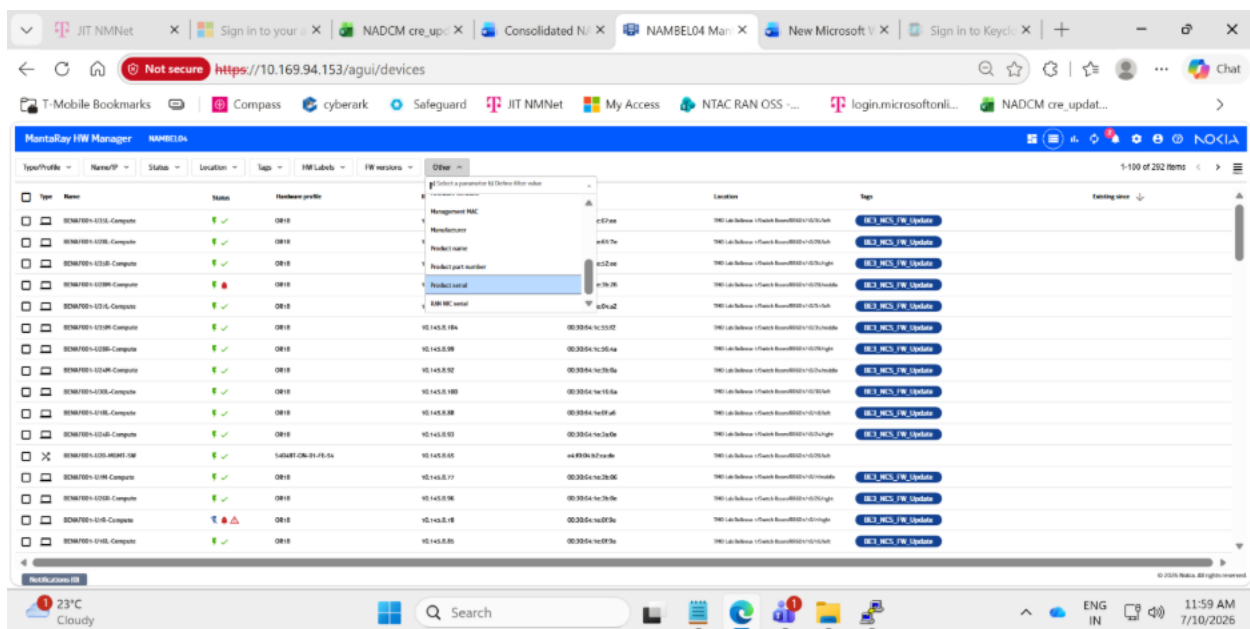
[root@NAMBEL01 ~]# kubeadm alpha certs check-expiration

Result: - Successfully showing the Kubernetes certificate.

```
root@NAMBEL04~
Steps: -
Execute kubeadm alpha certs check-expiration and verify the validity of all certificates.
[
[root@NAMBEL01 ~]# kubeadm alpha certs check-expiration
[
Result: - Successfully showing the Kubernetes certificate.
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]#
[root@NAMBEL04 ~]# kubeadm alpha certs check-expiration
Kubeadm experimental sub-commands
[root@NAMBEL04 ~]# kubeadm certs check-expiration
[check-expiration] Reading configuration from the "kubeadm-config" ConfigMap in namespace "kube-system"...
[check-expiration] Use 'kubeadm init phase upload-config --config your-config-file' to re-upload it.
W0710 11:54:40.434414 308545 utils.go:69] The recommended value for "clusterDNS" in "KubeletConfiguration" is: [10.233.0.10]; the provided value is: [169.254.25.10]

CERTIFICATE          EXPIRES              RESIDUAL TIME    CERTIFICATE AUTHORITY  EXTERNALLY MANAGED
admin.conf            Jul 10, 2027 08:52 UTC 364d             ca                      no
apiserver             Jul 10, 2027 08:52 UTC 364d             ca                      no
apiserver-kubelet-client Jul 10, 2027 08:52 UTC 364d             ca                      no
controller-manager.conf Jul 10, 2027 08:52 UTC 364d             ca                      no
front-proxy-client    Jul 10, 2027 08:52 UTC 364d             front-proxy-ca         no
scheduler.conf        Jul 10, 2027 08:52 UTC 364d             ca                      no
super-admin.conf      Jul 10, 2027 08:52 UTC 364d             ca                      no

CERTIFICATE AUTHORITY EXPIRES              RESIDUAL TIME    EXTERNALLY MANAGED
ca                    Jul 07, 2036 08:23 UTC 9y               no
front-proxy-ca       Jul 07, 2036 08:23 UTC 9y               no
[root@NAMBEL04 ~]#
```



43. Events can be filtered with (1) Device Identification (name or IP) and/or (2) with a given date or date range

Steps: -

1. In NADCM go to “Events”
2. Click on select devices (either Name or IP)
3. Click on date column to filter events from date or date range.

Result: - Successfully showing Event filtration can be done by name or IP or by a given date or date range.

The screenshot shows the NADCM Events page in a web browser. The browser's address bar displays the URL: https://10.169.94.153/agu/events/main?filter.serial_number=50LQPK2. The page features a sidebar with a circular gauge showing 776 events and a list of event categories: Critical (752), Major (0), Warning (24), Info (0), and Cleared (0). The main content area displays a table of events with columns for Severity, Device Identification, Description, Status/Value, Date/Time, and Actions. A modal dialog titled "Time Range Filter (for read timestamp)" is open, showing a calendar for July 2024. The calendar has the 1st selected. Below the calendar, there are fields for Start Time (hour, minute, second) and End Time (hour, minute, second). The Start Time is currently 0:0:0 and the End Time is 23:0:0. The dialog has "Save" and "Cancel" buttons.

Severity	Device Identification	Description	Status/Value	Date/Time	Actions
☐	RELF001	Device cannot be managed. Reason: green - other error		2024/07/10 12:02:35	
☐	RELF002	Device cannot be managed. Reason: green - other error		2024/07/10 12:02:13	
☐	RELF003	Device cannot be managed. Reason: green - other error		2024/07/10 12:02:07	
☐	BENAF002-0230-EMC02	Device cannot be managed. Reason: green - other error		2024/07/10 10:47:48	
☐	NADCMtest	Login failed from IP: 10.142.179.2		2024/07/10 10:47:00	
☐	BENAF001-0141-Comp	Device cannot be managed. Reason: redfish - other error		2024/07/10 10:44:19	
☐	OPNF001-Master-1-023	Device cannot be managed. Reason: green - other error		2024/07/10 10:44:15	
☐	SANAF004-014	Device cannot be managed. Reason: green - other error		2024/07/10 10:44:10	
☐	BENAF001-0148-Compute	Device cannot be managed. Reason: redfish - other error		2024/07/10 10:43:46	
☐	BENAF001-0148-Unload	Device cannot be managed. Reason: redfish - other error		2024/07/10 10:43:42	
☐	BENAF001-0144-Compute	Device cannot be managed. Reason: redfish - other error		2024/07/10 10:43:42	

44. NADCM interface certificates' tab, NADCM North Bound certificate can be viewed

Steps: -

1. In NADCM go to "Administration".
2. Click on select Certificate.
3. Click on NADCM interface certificates.

Result: - Successfully showing north bound certificates

The screenshot shows the NADCM interface with the 'Certificates' tab selected. The interface is divided into two main sections: 'MantaRay HW Manager Northbound Certificate (Web UI/REST)' and 'MantaRay HW Manager Southbound Certificate'. Each section contains a table with the following columns: Cert type, Start time, End time, Upload time, Automatic renewal, and Actions.

Cert type	Start time	End time	Upload time	Automatic renewal	Actions
default-self-signed	4 hours ago	Expires in 2 years	an hour ago	ON	[Edit] [Delete]

Cert type	Start time	End time	Upload time	Automatic renewal	Actions
default-self-signed	a day ago	Expires in 3 years	an hour ago	ON	[Edit] [Delete]

The interface also includes a sidebar with various navigation options such as Files, Roles, Backup, Activity Logs, DHCP, Hardware Scan, Default Credentials, Certificates (selected), Licenses, SNMP Settings, Set and Clear System Events, and NMS Configuration. The bottom of the screen shows a Windows taskbar with the date and time as 12:05 PM on 7/10/2026.

46. Verify alarms are seen in NADCM Event Manager

Steps: -

1. Login NADCM
2. Go to Events
3. Verify the alarms.

Result: - Alarms has been verified in NADCM.

The screenshot displays the NADCM Event Manager interface. At the top, there's a navigation bar with 'Events' and 'Event & Notification Rules' tabs. A summary section on the left shows a donut chart for '772 Events' and a legend for event severity: Critical (745), Major (0), Warning (27), Info (0), and Cleared (0). The main area shows a table of active events. The table has columns for Severity, Device identification, Description, Sensor name/type, Status/Value, Date (with a refresh icon), and Actions. The events listed include 'Device can not be managed', 'Voltage Abnormal', and several 'Login failed' events from IP 10.145.119.225.

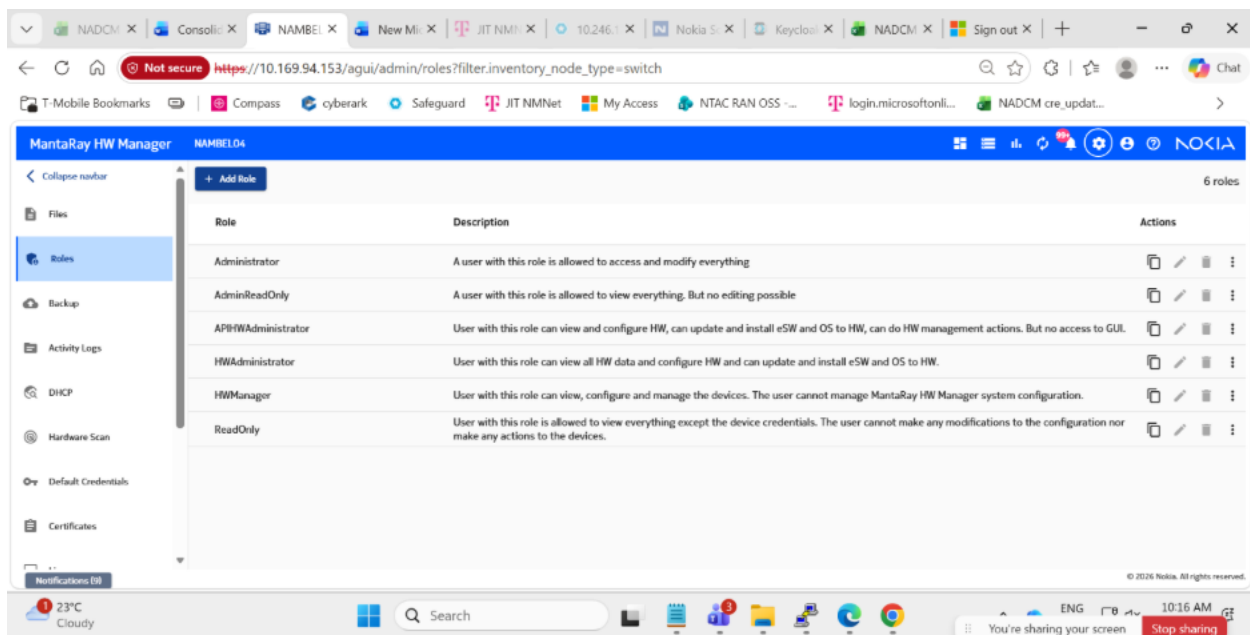
Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Refresh)	Actions
Critical	BELFD002	Device can not be managed. Reason given: other error-unexpected amount of notifications: 0			2025/01/24 10:31:49	Info
Critical	BELFD001	Device can not be managed. Reason given: other error-unexpected amount of notifications: 0			2025/01/24 10:29:30	Info
Warning	BENW002-039M-Comp	Voltage Abnormal	VBAU_HIM	Lower Critical	2025/01/24 10:28:21	Info
Warning	NAMBEL04	Login failed from IP:10.145.119.225	Login Info		2025/01/24 08:05:48	Info
Warning	BELCR002	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:10:50	Info
Warning	BELCR013	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:10:48	Info
Warning	BELCR003	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:10:01	Info
Warning	BELCR002	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:55	Info
Warning	BELCR001	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:40	Info
Warning	BELCR011	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:20	Info
Warning	BELCR001	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:14	Info
Warning	BELCR001	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:12	Info
Warning	BELCR016	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:11	Info
Warning	BELCR014	Login failed from IP:10.145.119.225	Login Info		2025/01/24 07:09:10	Info

47. Role management & default credentials set up

Steps: -

1. Go to Administration
2. Click on Roles → Add role and save.
3. Newly added roles will be reflected in keycloak.
4. Login keycloak. Check on Realm roles.
5. In NADCM, Go to Administration
6. Click on Default Credentials
7. Set the username and password.
8. Add the new device and set the default credentials.

Result: - Role management and default credentials set up has been verified successfully.



NADCM x Consolix x NAMBEL x New Mi x JIT NMNet x 10.246.1 x Nokia S x Keycloak x NADCM x Sign out x

Not secure https://10.169.94.153/agu/admin/roles?filter.inventory_node_type=switch

T-Mobile Bookmarks Compass cyberark Safeguard JIT NMNet My Access NTAC RAN OSS login.microsoftonline NADCM cre, updat...

MantaRay HW Manager NAMBEL04

Files Rules Backup Activity Logs DHCP Hardware Scan Default Credentials Certificates

Role Administrator AdminReadOnly APB-WAdministrator HWAdministrator HWManager ReadOnly

Add role

Role name* ATPTest

Role description Testing

Device Credentials ☒ ☐ ?

DHCP server administration ☒ ☐ ?

Export and Import ☒ ☐ ?

File management ☒ ☐ ?

Firmware Update ☒ ☐ ?

Hardware Allocations ☒ ☐ ?

Hardware Scan ☒ ☐ ?

Install OS ☒ ☐ ?

Save Cancel

6 roles

Actions

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23°C Cloudy Search 10:17 AM

NADCM x Consolix x NAMBEL x New Mi x Keycloak x JIT NMNet x 10.246.1 x Nokia S x Keycloak x NADCM x

Not secure https://10.169.94.153/auth/admin/master/console/#/NADCM/roles

T-Mobile Bookmarks Compass cyberark Safeguard JIT NMNet My Access NTAC RAN OSS login.microsoftonline NADCM cre, updat...

KEYCLOAK admin

MantaRay HW Manager Current realm Manage realms Manage Clients Client scopes Realm roles Users Groups Sessions Events Configure Realm settings Authentication

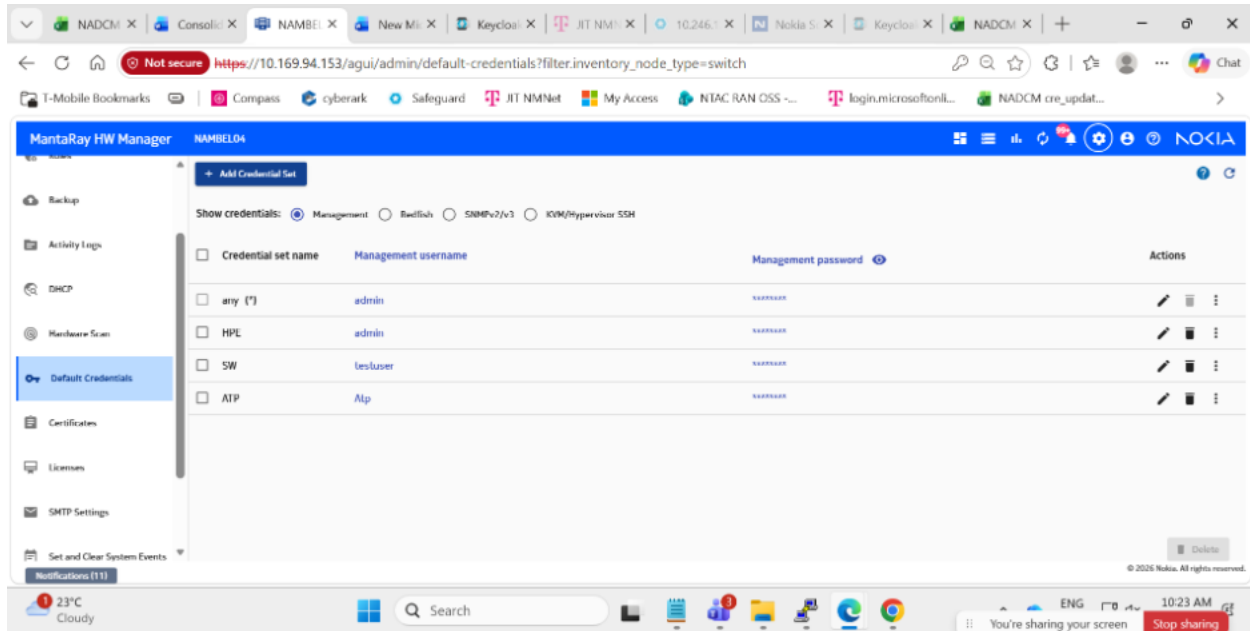
Realm roles

Realm roles are the roles that you define for use in the current realm. Learn more

Search role by name Create role Refresh 1-10

Role name	Compo...	Description
Administrator	False	A user with this role is allowed to access and modify everything
AdminReadOnly	False	A user with this role is allowed to view everything. But no editing possible
APB-WAdministrator	False	User with this role can view and configure HW, can update and install eSW and OS to HW, can do HW management actions. But no access to GUI
ATPTest	False	Testing
default-roles-nadcm	True	role_default-roles
HWAdministrator	False	User with this role can view all HW data and configure HW and can update and install eSW and OS to HW
HWManager	False	User with this role can view, configure and manage the devices. The user cannot manage MantaRay HW Manager system configuration
offline_access	False	role_offline-access
ReadOnly	False	User with this role is allowed to view everything except the device credentials. The user cannot make any modifications to the configuration nor make any actions to the devices
uma_authorization	False	-

23°C Cloudy Search 10:21 AM



48. Performance management reporting to NMS

Steps: -

1. Login to NADCM then Administration → NMS Configuration
2. Open WinSCP tool → Use SFTP address, SFTP username & Password
3. PM report will be available.

Result: - Successfully shows below the performance management report.

pm1 - nadcmpm@10.169.94.153 - WinSCP

Local Mark Files Commands Tabs Options Remote Help

Synchronize Queue Transfer Settings Default

nadcmpm@10.169.94.153 x New Tab

My documenter Upload Edit Properties New Download Edit Properties New Find Files

C:\Users\VMaria1\OneDrive - T-Mobile USA\Documents\TC-30\ /pm/pm1/

Name	Size	Type	Changed	Name	Type	Size	Changed	Rights	Owner
ATP Test case		File folder	7/31/2026 6:43:41 AM	PM_20260731T0715-0...		3 KB	7/31/2026 7:31:46 AM	rw-r--r-x	60029
HPE Audit		File folder	6/23/2026 6:13:13 AM	PM_20260731T0715-0...		211 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
NADCM Upgrade		File folder	7/23/2026 3:53:46 AM	PM_20260731T0715-0...		23 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
alarm.pcap	1,076 KB	PCAP File	7/31/2026 6:34:49 AM	PM_20260731T0715-0...		20 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		9 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		10 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		16 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		3 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		2 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		1 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		1 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		1 KB	7/31/2026 7:31:46 AM	rw-r--r--	60029
				PM_20260731T0715-0...		1 KB	7/31/2026 7:30:18 AM	rw-r--r--	60029
				PM_20260731T0715-0...		1 KB	7/31/2026 7:30:18 AM	rw-r--r--	60029
				PM_20260731T0700-0...		3 KB	7/31/2026 7:16:45 AM	rw-r--r--	60029
				PM_20260731T0700-0...		211 KB	7/31/2026 7:16:45 AM	rw-r--r--	60029
				PM_20260731T0700-0...		24 KB	7/31/2026 7:16:45 AM	rw-r--r--	60029

0 B of 1.05 MB in 0 of 4 0 B of 14.1 MB in 0 of 686

SFTP-3 0:00:25

25°C Light rain Search You're sharing your screen Stop sharing ENG IN 7:32 AM 7/31/2026

10.169.97.154 - Remote Desktop Connection

MantaRay NM Start Page - albi Performance Manager - albi

Not secure https://alocswb1cal.netact.mgmt/pm/

NOKIA MantaRay NM Performance Manager+ cnc

Reporting

Filter... * Board 1 Last refresh: 07/31/2026 12:34:20 AM

Reporting Suite

- ALL-nokia may collect scores
- LB-ASN Lawful Interception G...
- L/C-ASN Lawful Interception C...
- L/QAPPS-Nokia Multi-access E...
- LNBS-LTE Base Transceiv...
- MSMS-Movius Maseon 5000 M...
- MF-NetAct Mediation Framew...
- MGB-ASN Multimedia Gateway
- MSF-Radiays Multimedia Reso...
- MSLITE-OragonWave Microwa...
- MSLT-Nokia Microwave Radio T...
- NADCM-Nokia AirFrame Data...
- Device metrics
- Host CPU
- Host Relaysystem
- Host memory
- FDU sensor, banks
- FDU sensor, devices

My Content Shared with Me

REFRESH

Period start time T	NADCM name	CPU_AVE (CPU_AVE) [%] (CPU...	CPU_CURR (CPU_CURR) [%] (CPU...	CPU_MAX (CPU_MAX) [%] (CPU...	CPU_MIN (CPU_MIN) [%] (CPU...	period_duration (period_du...
07/31/2026 12:00:00 AM	NAMBELO4	4.77	6.41	12.28	2.18	15
07/30/2026 11:45:00 PM	NAMBELO4	4.57	2.88	13.61	2.22	15
07/30/2026 11:30:00 PM	NAMBELO4	4.59	11.41	11.41	2.62	15
07/30/2026 11:15:00 PM	NAMBELO4	4.26	3.82	8.42	1.56	15
07/30/2026 11:00:00 PM	NAMBELO4	4.49	1.28	16.12	1.28	15
07/30/2026 10:45:00 PM	NAMBELO4	5.48	5.72	14.99	2.29	15
07/30/2026 10:30:00 PM	NAMBELO4	4.31	2.97	13.05	1.57	15
07/30/2026 10:15:00 PM	NAMBELO4	4.99	6.22	16.07	1.39	15
07/30/2026 10:00:00 PM	NAMBELO4	6.51	2.77	18.21	2.77	15
07/30/2026 9:45:00 PM	NAMBELO4	5.21	2.89	11.37	2.15	15
07/30/2026 9:30:00 PM	NAMBELO4	5.37	9.07	10.93	2.01	15
07/30/2026 9:15:00 PM	NAMBELO4	7.3	4.39	14.34	3.46	15
07/30/2026 9:00:00 PM	NAMBELO4	6.15	3.59	11.07	3.59	15
07/30/2026 8:45:00 PM	NAMBELO4	5.22	2.8	14.14	2.79	15
07/30/2026 8:30:00 PM	NAMBELO4	9.16	1.8	14.14	1.8	15

Count: 34

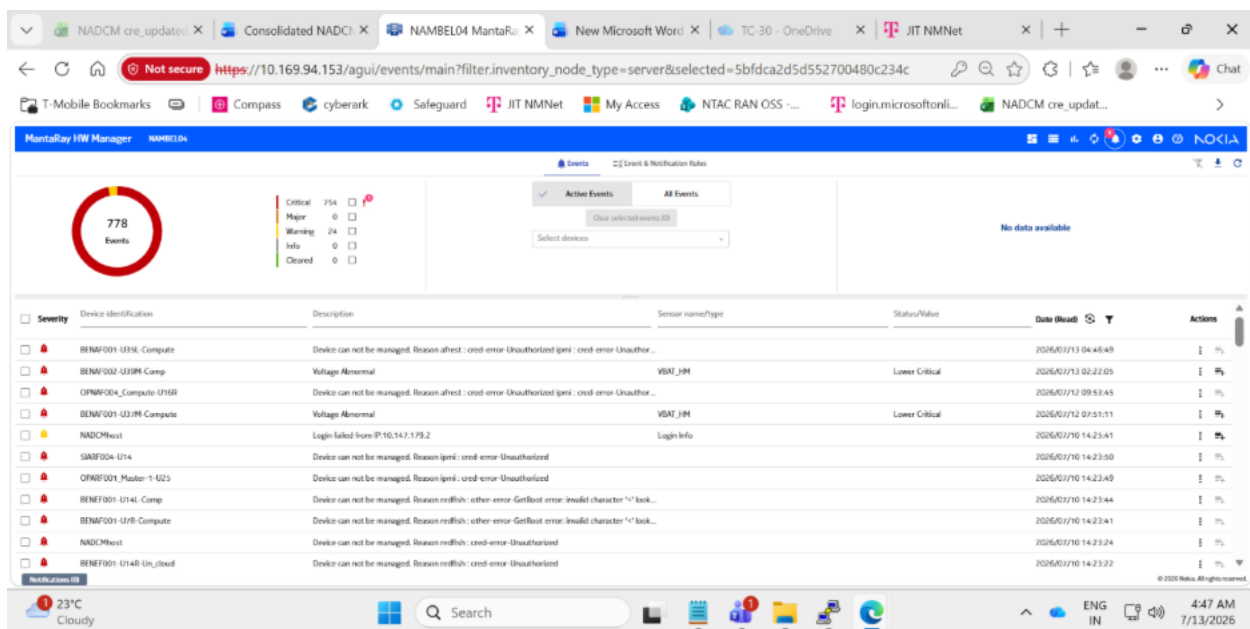
12:34 AM 31/3/2026

49. Ability to alarm about wider variety of management protocol related issues

Steps: -

1. In NADCM, Select the server.
2. Go to Credential management.
3. Change the password and check the alarm in Events.

Result: -



The screenshot displays the NADCM Events page. At the top, there is a summary section with a donut chart showing 778 Events. Below this, a table lists various events with columns for Severity, Device Identification, Description, Sensor name/Type, Status/Value, Date (Read), and Actions. The events include warnings for voltage abnormalities and critical errors for device management and login failures.

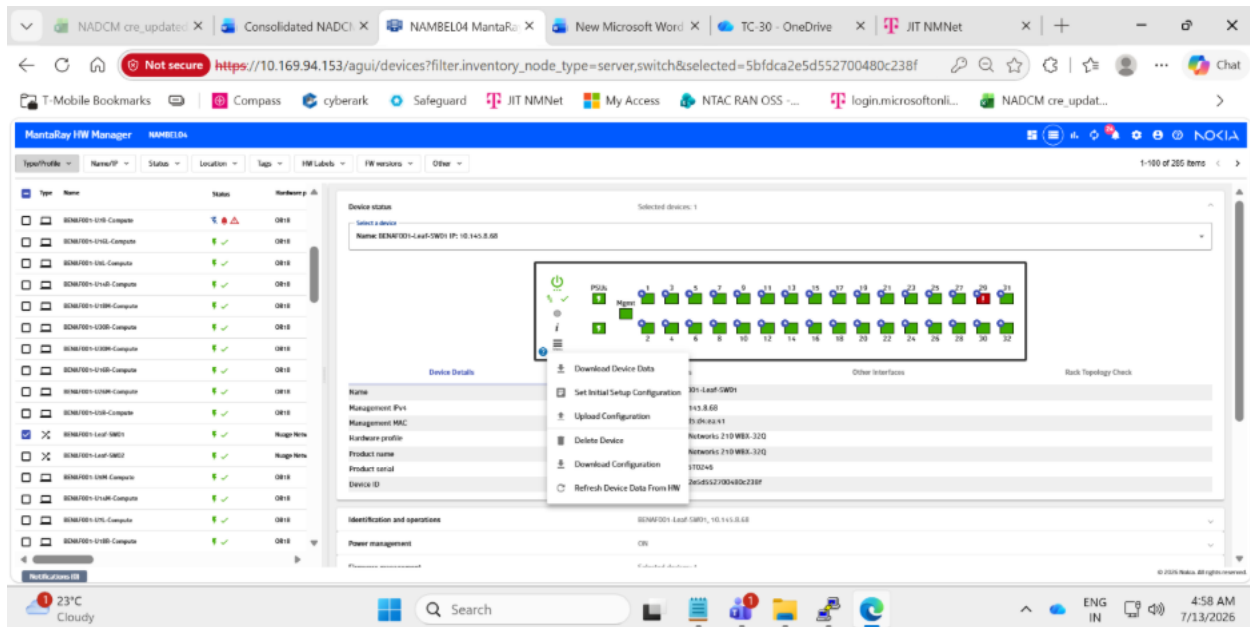
Severity	Device Identification	Description	Sensor name/Type	Status/Value	Date (Read)	Actions
Critical	BENAF001-U336-Compute	Device can not be managed. Reason:afrect : cred error Unauthorized ipmi : cred error Unauthorized			2026/07/13 04:46:49	
Warning	BENAF002-U339M-Comp	Voltage Abnormal	VBAT_JPM	Lower Critical	2026/07/13 02:22:05	
Critical	OPNAP004-Compute-U16R	Device can not be managed. Reason:afrect : cred error Unauthorized ipmi : cred error Unauthorized			2026/07/12 09:53:45	
Warning	BENAF001-U33JM-Compute	Voltage Abnormal	VBAT_JPM	Lower Critical	2026/07/12 07:51:51	
Info	NADCMtest	Login failed from IP:10.147.179.2	Login Info		2026/07/10 14:23:41	
Critical	SBRF004-U14	Device can not be managed. Reason:ipmi : cred error Unauthorized			2026/07/10 14:23:50	
Critical	OPNAP001-Master-1-U25	Device can not be managed. Reason:ipmi : cred error Unauthorized			2026/07/10 14:23:49	
Critical	BENEF001-U14L-Comp	Device can not be managed. Reason:redfish : other error GetBoot error: invalid character '\0' look...			2026/07/10 14:23:44	
Critical	BENAF001-U18-Compute	Device can not be managed. Reason:redfish : other error GetBoot error: invalid character '\0' look...			2026/07/10 14:23:41	
Critical	NADCMtest	Device can not be managed. Reason:redfish : cred error Unauthorized			2026/07/10 14:23:24	
Critical	BENEF001-U14R-Unc_cloud	Device can not be managed. Reason:redfish : cred error Unauthorized			2026/07/10 14:23:22	

50. Download IXR Switch Configuration using NADCM

Steps: -

1. Login NADCM and select **“Devices”** Select the Switch
2. Goto  icon
2. Select Download Configuration
4. Once get success verify the same in Administration → Files.

Result: - IXR Leaf Switch configuration has been downloaded successfully.



[illegible]

51. Firmware version check in IXR Switch

Steps: -

1. Login NADCM → select “**Devices**”.
2. Select the Switch and you can see the devices view at right side.

Result: - Firmware version of IXR switch can be verified from below screenshot.

The screenshot displays the NADCM web interface. The browser address bar shows the URL: https://10.169.94.153/aguil/devices?filter.inventory_node_type=server,switch&selected=5bfdca2e5d52700480c2390. The interface is titled "MantaRay ITR Manager" and "NAMBEL04".

On the left, a table lists various devices. The selected device is "BOM7001-Leaf-SW02". The right pane shows the details for this device, including:

- Basic information:** Name: BOM7001-Leaf-SW02, Management IPV4: 10.142.8.50, Management MAC: 00:99:00:07:56:A1, Hardware profile: Nage Networks 710-WBX-320, Product name: Nage Networks 710-WBX-320, Product serial: N518v1T0171, Device ID: vbfba2e5d52700480c2390.
- Identification and operations:** BOM7001-Leaf-SW02, 10.142.8.50.
- Power management:** ON.
- Firmware management:** Selected devices: 1. A table shows the firmware details:

Device checked	Hardware profile	Current version	Link
BOM7001-Leaf-SW02	Nage Networks 710-WBX-320	Switch, Ver 20.10.14-1019	-

The bottom of the interface shows a Windows taskbar with the date and time: 5:02 AM, 7/13/2026.

52. Device view on IXR switch

Steps: -

1. Login NADCM → select “**Devices**”.
1. Select the Switch and you can see the devices view at right side.

Result: - Device details can be verified in as below.

The screenshot displays the NADCM (Network Administration and Configuration Manager) web interface. The browser address bar shows the URL: https://10.169.94.153/aguil/devices?filter.inventory_node_type=switch&selected=6414aa22c611fb82c7448643. The interface is titled "MantaRay HW Manager" and shows a list of devices on the left. The selected device, "OPNAT004-SW01", is highlighted. The right pane displays the "Device status" and "Device Details" for this switch. The "Device status" section shows a diagram of the switch's ports, with a legend indicating "Ports" and "Status". The "Device Details" section provides information about the switch, including its name, management IP, management MAC, hardware profile, product name, product serial, and device ID. The "Identification and operations" section shows the switch's name and IP address. The "Power management" section shows the switch's power status. The "Device Details" section also includes a "Back Topology Check" button.

Type	Name	Status	Hardware
Switch	OPNAT003-SW02	OK	21484-016
Switch	OPNAT001-023-MGMT-SW	OK	1x4x4T-OK
Switch	OPNAT002-023-MGMT-SW	OK	1x4x4T-OK
Switch	OPNAT001-023-MGMT-SW	OK	1x4x4T-OK
Switch	OPNAT003-SW02	OK	21484-016
Switch	OPNAT004-SW01	OK	21484-016
Switch	OPNAT004-SW02	OK	21484-016
Switch	OPNAT003-SW01	OK	21484-016
Switch	OPNAT001-MGMT-SW-023	OK	1x4x4T-OK
Switch	OPNAT001-MGMT-SW-023	OK	1x4x4T-OK
Switch	OPNAT002-023-Leaf-SW01	OK	21484-016
Switch	OPNAT002-023-Leaf-SW02	OK	21484-016
Switch	OPNAT001-023-Leaf-SW01	OK	21484-016
Switch	OPNAT002-023-Leaf-SW02	OK	21484-016
Switch	OPNAT001-023-Leaf-SW01	OK	21484-016
Switch	OPNAT002-023-Leaf-SW02	OK	21484-016
Switch	OPNAT001-023-Leaf-SW01	OK	21484-016
Switch	OPNAT002-023-Leaf-SW02	OK	21484-016

Device status

Selected device: 1

Select a device

Name: OPNAT004-SW01 IP: 10.143.217.68

Device Details

PSU / Port Details

Other interfaces

Back Topology Check

Name: OPNAT004-SW01

Management IP: 10.143.217.68

Management MAC: 3c:8f:f9:2d:f0:28

Hardware profile: NADCM Networks 210-WBS-32Q

Product name: NADCM Networks 210-WBS-32Q

Product serial: NS223711087

Device ID: 6414aa22c611fb82c7448643

Identification and operations

OPNAT004-SW01, 10.143.217.68

Power management

OK

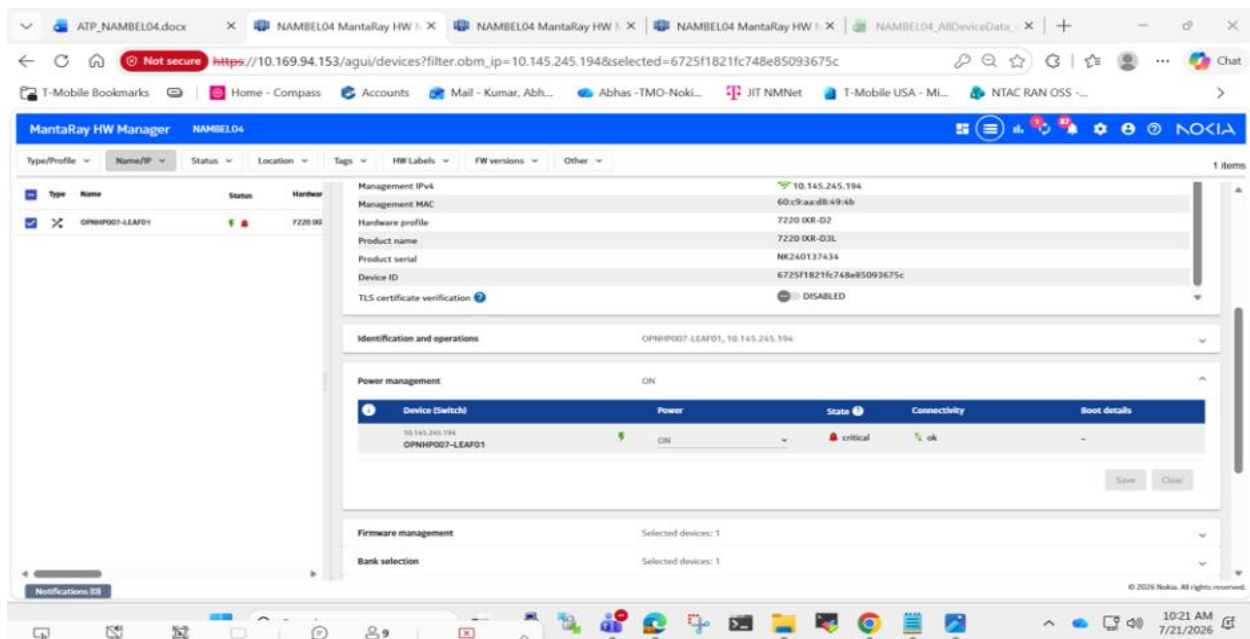
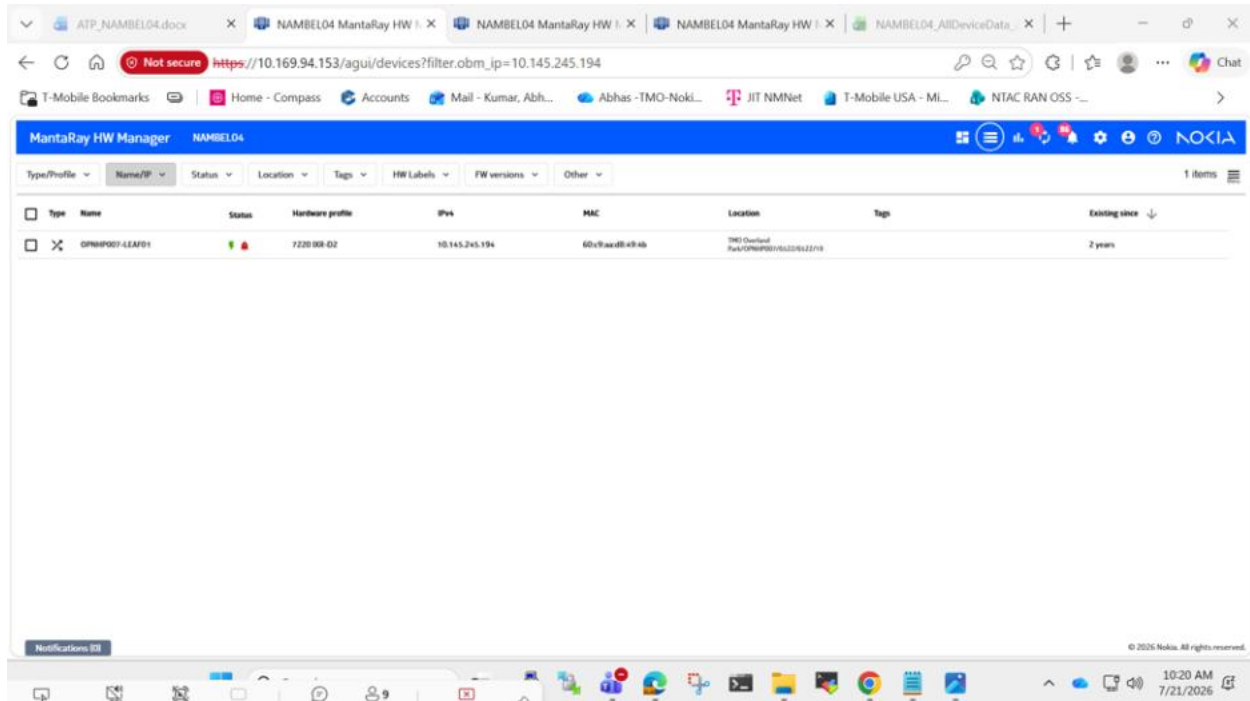
Power management

Selected devices: 1

53. Power OFF/ON for IXR Leaf Switch

Steps: -

1. In NADCM, go to Devices.
2. Select the required Leaf Switch → Power management.
3. Go to power management and select reload. Then save it.



ATP_NAMBEL04.docx x NAMBEL04 MantaRay HW : x NAMBEL04 MantaRay HW : x NAMBEL04 MantaRay HW : x NAMBEL04_AllDeviceData : x + -

Not secure https://10.169.94.153/agu/devices?filter.obm_ip=10.145.245.194&selected=6725f1821fc748e85093675c

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
OPNBP007-LEAF01		● ● ●	7220 95

Select a device
Name: OPNBP007-LEAF01 IP: 10.145.245.194

Filter: ☒ Critical ☐ Major ☐ Warning ☐ Info ☐ Cleared

Active events

Severity	Device Identification	Description	Sensor name	Value	Date (Read)
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=METRO-E_VRF)/ip...	/bfd/network-instance(name=METRO-E_VRF)/peer(local-discriminator=16392)	down	2026/07/21 09:14:41
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer...	/bfd/network-instance(name=NMNET_VRF)/peer(local-discriminator=16391)	down	2026/07/21 09:14:31
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer...	/bfd/network-instance(name=NMNET_VRF)/peer(local-discriminator=16390)	down	2026/07/21 09:14:29
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer...	/bfd/network-instance(name=NMNET_VRF)/peer(local-discriminator=16389)	down	2026/07/21 09:14:28
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=METRO-E_VRF)/ip...	/bfd/network-instance(name=METRO-E_VRF)/peer(local-discriminator=16388)	down	2026/07/21 09:14:28
Critical	OPNBP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer...	/bfd/network-instance(name=NMNET_VRF)/peer(local-discriminator=16387)	down	2026/07/21 09:14:28

All events

Severity	Device Identification	Description	Sensor name	Value	Active	Date (Read)
Info	OPNBP007-LEAF01	Device connected				2026/07/21 09:15:55

Notifications (0)

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ATP_NAMBEL04.docx x NAMBEL04 MantaRay HW Mana... x NAMBEL04 MantaRay HW Mana... x NAMBEL04_AllDeviceData_2026-... x + -

Not secure https://10.169.94.153/agu/devices?selected=6725f1821fc748e85093675c&filter.obm_ip=10.145.245.194

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
OPNBP007-LEAF01		● ● ●	7220 95

Select a device
Name: OPNBP007-LEAF01 IP: 10.145.245.194

Selected devices: 1

Device Details

Device Details	PSU / Port Details	Other Interfaces	BGP Details	Other Sensors	Rack Topology Check
Name	OPNBP007-LEAF01				
Management IPv4	10.145.245.194				
Management MAC	60:c9:ad:b8:49:4b				
Hardware profile	7220 DR-02				
Product name	7220 DR-03L				
Product serial	NK240137434				
Device ID	6725f1821fc748e85093675c				
TLS certificate verification	DISABLED				

Notifications (1)

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ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW Mana X NAMBEL04 MantaRay HW Mana X NAMBEL04_AllDeviceData_2026- X +

Not secure https://10.169.94.153/aguil/devices?selected=6725f1821fc748e85093675c&filter.obm_ip=10.145.245.194

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas - TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNHP007-LEAF01		7220 95

Management IPv4 10.145.245.194
Management MAC 60:c9:ad:b8:49:4b
Hardware profile 7220 OKR-02
Product name 7220 OKR-03L
Product serial NK240137434
Device ID 6725f1821fc748e85093675c
TLS certificate verification DISABLED

Identification and operations OPNHP007-LEAF01, 10.145.245.194

Power management Not Available Updating 1/1

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.194 OPNHP007-LEAF01		RELOAD	critical	ok

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Notifications (1)

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10:28 AM 7/21/2026

ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW Mana X NAMBEL04 MantaRay HW Mana X NAMBEL04_AllDeviceData_2026- X +

Not secure https://10.169.94.153/aguil/devices?selected=6725f1821fc748e85093675c&filter.obm_ip=10.145.245.194

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas - TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile Name Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
☒	OPNHP007-LEAF01		7220 95

Management IPv4 10.145.245.194
Management MAC 60:c9:ad:b8:49:4b
Hardware profile 7220 OKR-02
Product name 7220 OKR-03L
Product serial NK240137434
Device ID 6725f1821fc748e85093675c
TLS certificate verification DISABLED

Identification and operations OPNHP007-LEAF01, 10.145.245.194

Power management ON

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.194 OPNHP007-LEAF01		RELOAD	critical	ok

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Notifications (1)

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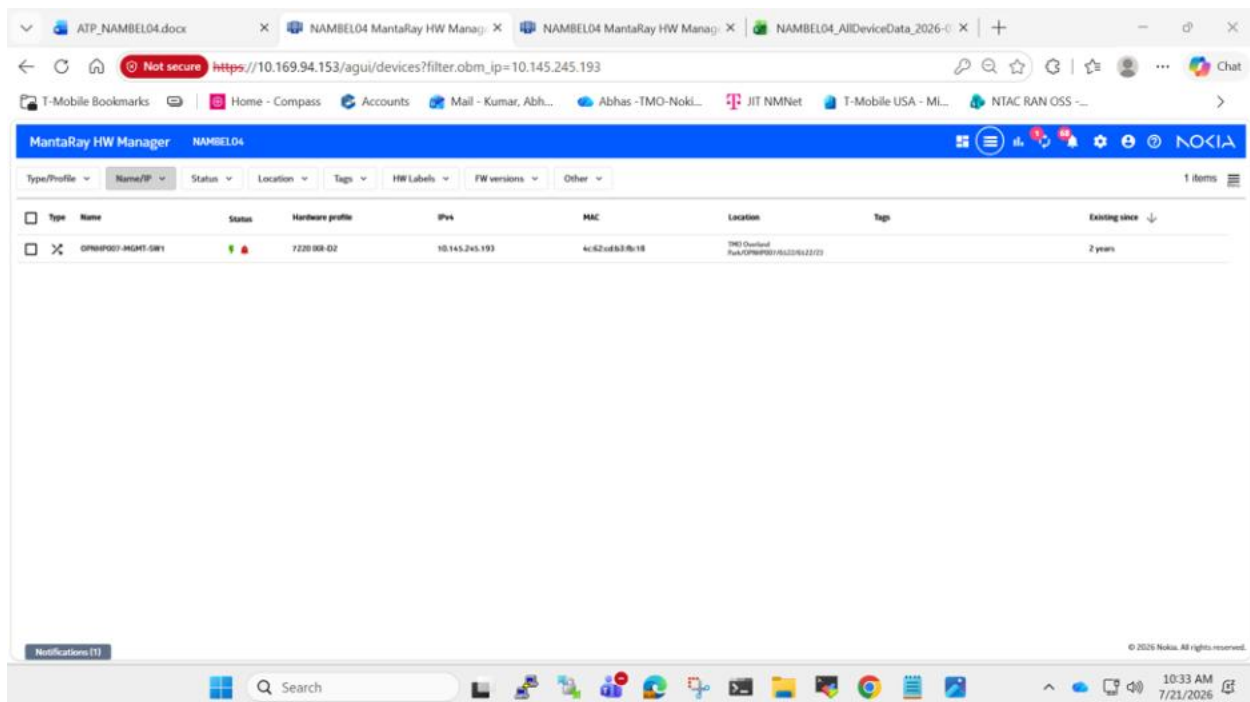
10:32 AM 7/21/2026

54. Power reset for IXR Hardware Management Switch

Steps: -

1. In NADCM, go to Devices.
2. Select HW mgmt. Switch
3. Go to Power Management
4. Perform Power reset.
5. Verify the switch was reset.

Result: - Successfully shows power restart hardware mgmt. switch from NADCM.



The screenshot displays the MantaRay HW Manager interface. The browser address bar shows the URL https://10.169.94.153/agui/devices?filter.obm_ip=10.145.245.193. The interface includes a navigation bar with the title "MantaRay HW Manager" and the identifier "NAMBEL04". Below the navigation bar, there are filter tabs for "Type/Profile", "Name/IP", "Status", "Location", "Tags", "HW Labels", "FW versions", and "Other". A table lists the hardware devices, with one item visible:

Type	Name	Status	Hardware profile	IPs	MAC	Location	Tags	Existing since
✕	OPNWP007-MSGMT-08Y1		7220-988-02	10.145.245.193	4c:62:cd:b3:7b:18	TMO Overland Pala/CPNMB011/0320/0421/03		2 years

The bottom of the screen shows a Windows taskbar with the search bar, task icons, and system tray information indicating the time is 10:33 AM on 7/21/2026.

ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW Mana X NAMBEL04 MantaRay HW Mana X NAMBEL04_AllDeviceData_2026- X

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093... Summarize Chat

T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -

MantaRay HW Manager NAMBEL04

Type/Profile Name/Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
OPNIP007-MGMT-SW1		● ● ●	7220 95

Events ▲ Active events: 3

Select a device
Name: OPNIP007-MGMT-SW1 IP: 10.145.245.193

Filter: ☒ Critical ☐ Major ☐ Warning ☐ Info ☐ Cleared

Active events

Severity	Device Identification	Description	Sensor name	Value	Date (Read)
●	OPNIP007-MGMT-SW1	/bfd/network-instance(name=NMMET_IP_VRF3/pe...	/bfd/network-instance(name=NMMET_IP_VRF3/peer(local- discriminator=16385)	down	2026/07/10 09:01:23
●	OPNIP007-MGMT-SW1	/bfd/network-instance(name=NMMET_IP_VRF3/pe...	/bfd/network-instance(name=NMMET_IP_VRF3/peer(local- discriminator=16385)	down	2026/07/10 09:01:23
●	OPNIP007-MGMT-SW1	/interface(name=mgmt0), admin enabled (still do...	/interface(name=mgmt0)	admin enabled (still down)	2026/07/10 09:01:22

All events

Severity	Device Identification	Description	Sensor name	Value	Active	Date (Read)
----------	-----------------------	-------------	-------------	-------	--------	-------------

Notifications (1)

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10:33 AM 7/21/2026

ATP_NAMBEL04.docx X NAMBEL04 MantaRay HW Mana X NAMBEL04 MantaRay HW Mana X NAMBEL04_AllDeviceData_2026- X

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093675a Chat

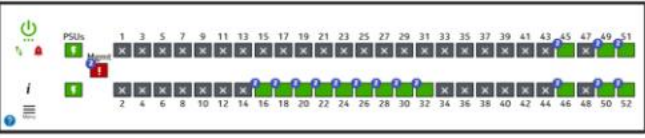
T-Mobile Bookmarks Home - Compass Accounts Mail - Kumar, Abh... Abhas -TMO-Noki... JIT NMNet T-Mobile USA - ML... NTAC RAN OSS -

MantaRay HW Manager NAMBEL04

Type/Profile Name/Status Location Tags HW Labels FW versions Other 1 items

Type	Name	Status	Hardware
OPNIP007-MGMT-SW1		● ● ●	7220 95

Select a device
Name: OPNIP007-MGMT-SW1 IP: 10.145.245.193



Device Details PSU / Port Details Other Interfaces BGP Details Other Sensors Rack Topology Check

Name	OPNIP007-MGMT-SW1
Management IPv4	10.145.245.193
Management MAC	Ac62c6b3fb18
Hardware profile	7220 IXR-02
Product name	7220 IXR-01
Product serial	NS1435T0001
Device ID	6725f1821fc748e85093675a
TLS certificate verification	DISABLED

Identification and operations OPNIP007-MGMT-SW1, 10.145.245.193

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10:34 AM 7/21/2026

You're sharing your screen Stop sharing

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag... | NAMBEL04 MantaRay HW Manag... | NAMBEL04_AllDeviceData_2026-0... | +

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093675a

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas -TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile | Name/IP | Status | Location | Tags | HW Labels | FW versions | Other | 1 items

Type	Name	Status	Hardware
☒	OPNH007-MGMT-SW1	● ● ●	7220 05

Hardware profile

Product name: 7220 IXR-01
Product serial: NS2435T0001
Device ID: 6725f1821fc748e85093675a
TLS certificate verification: ● DISABLED

Identification and operations

OPNH007-MGMT-SW1, 10.145.245.193

Power management

ON

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.193 OPNH007-MGMT-SW1	●	ON	● critical	● ok

Save Clear

Firmware management

Selected devices: 1

Bank selection

Selected devices: 1

Credential management

Selected devices: 1

Duration management

TMO Overload Park / OPH007 / 6522 / 6522 / 21

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Notifications (1)

Search

10:34 AM 7/21/2026

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag... | NAMBEL04 MantaRay HW Manag... | NAMBEL04_AllDeviceData_2026-0... | +

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093675a

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas -TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile | Name/IP | Status | Location | Tags | HW Labels | FW versions | Other | 1 items

Type	Name	Status	Hardware
☒	OPNH007-MGMT-SW1	● ● ●	7220 05

Hardware profile

Product name: 7220 IXR-01
Product serial: NS2435T0001
Device ID: 6725f1821fc748e85093675a
TLS certificate verification: ●

Identification and operations

OPNH007-MGMT-SW1, 10.145.245.193

Power management

ON

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.193 OPNH007-MGMT-SW1	●	RELOAD	● critical	● ok

Save Clear

Firmware management

Selected devices: 1

Bank selection

Selected devices: 1

Credential management

Selected devices: 1

Duration management

TMO Overload Park / OPH007 / 6522 / 6522 / 21

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Notifications (1)

Search

10:34 AM 7/21/2026

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag... | NAMBEL04 MantaRay HW Manag... | NAMBEL04_AllDeviceData_2026-0... | +

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093675a

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas -TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile | Name/IP | Status | Location | Tags | HW Labels | FW versions | Other | 1 items

Type	Name	Status	Hardware
☒	OPNHP007-MGMT-SW1		7220 05

Hardware profile

Product name: 7220 IXR-01
Product serial: NS2435T0001
Device ID: 6725f1821fc748e85093675a
TLS certificate verification: DISABLED

Identification and operations OPNHP007-MGMT-SW1, 10.145.245.193

Power management Not Available Updating 1/1

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.193 2/2 OPNHP007-MGMT-SW1	RELOAD	critical	ok	-

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Credential management Selected devices: 1

Duration management TMO Overload Park / OPNHP007 / 6522 / 6522 / 21

Notifications (1)

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10:35 AM 7/21/2026

ATP_NAMBEL04.docx | NAMBEL04 MantaRay HW Manag... | NAMBEL04 MantaRay HW Manag... | NAMBEL04_AllDeviceData_2026-0... | +

Not secure https://10.169.94.153/aguui/devices?filter.obm_ip=10.145.245.193&selected=6725f1821fc748e85093675a

T-Mobile Bookmarks | Home - Compass | Accounts | Mail - Kumar, Abh... | Abhas -TMO-Noki... | JIT NMNet | T-Mobile USA - ML... | NTAC RAN OSS -...

MantaRay HW Manager NAMBEL04

Type/Profile | Name/IP | Status | Location | Tags | HW Labels | FW versions | Other | 1 items

Type	Name	Status	Hardware
☒	OPNHP007-MGMT-SW1		7220 05

Hardware profile

Product name: 7220 IXR-01
Product serial: NS2435T0001
Device ID: 6725f1821fc748e85093675a
TLS certificate verification: DISABLED

Identification and operations OPNHP007-MGMT-SW1, 10.145.245.193

Power management Not Available Updating 1/1

Device (Switch)	Power	State	Connectivity	Boot details
10.145.245.193 2/2 OPNHP007-MGMT-SW1	RELOAD	critical	unreachable	-

Save Clear

Firmware management Selected devices: 1

Bank selection Selected devices: 1

Credential management Selected devices: 1

Duration management TMO Overload Park / OPNHP007 / 6522 / 6522 / 21

Notifications (1)

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10:36 AM 7/21/2026

The screenshot displays the MantaRay HW Manager interface. The top navigation bar is blue with the title 'MantaRay HW Manager' and 'NAMEBL04'. Below the navigation bar, there are tabs for 'Type/Profile', 'Name/IP', 'Status', 'Location', 'Tags', 'HW Labels', 'FW versions', and 'Other'. The main content area shows details for a device named 'OPNHP007-MGMT-SW1'. The details are organized into sections: 'Hardware profile' (Product name: 7220 IXX-01, Product serial: NS2435T0001, Device ID: 6725f1821fc748e85093675a, TLS certificate verification: DISABLED), 'Identification and operations' (IP: 10.145.245.193), 'Power management' (ON), and 'Device (Switch)' table. The table has columns for Device (Switch), Power, State, Connectivity, and Boot details. The device is listed as 'OPNHP007-MGMT-SW1' with a 'RELOAD' button and a 'critical' state. The bottom of the interface shows a 'Notifications (1)' section.

55. IXR PM Data

Steps: -

1. Login NADCM and select “Devices”.
2. Select the specific switch
3. Click on “BGP Details” right side

Result: - Switch BGP details can be verified in as below.

The screenshot displays the NADCM web interface. The top navigation bar shows the 'MantaRay I/W Manager' and 'NAMBEL04'. The left sidebar contains a list of devices with columns for Type, Name, Status, Location, Tags, HW Labels, FW versions, and Other. The main panel is divided into three sections: 'Device Details' (showing a port status diagram), 'PSU / Port Details' (a table of port information), and 'Other Interfaces' (a table of interface information). The bottom status bar shows the system temperature (23°C), search bar, and system time (5:11 AM, 7/13/2026).

Description	Admin/Operative status	Speed (Gbit/s)	In/Out Data	In/Out Errors
NULL 0	up / up	0	0 / 0	0 / 0
Port channel 1	up / up	200	2.4 GB / 2.05 GB	0 / 0
Vlan 1	down / down	200	0 / 0	0 / 0
Vlan 10	up / up	200	3.04 GB / 1.4 GB	0 / 0
Vlan 100	up / up	10	2.54 GB / 135 MB	0 / 0
Vlan 11	up / up	200	1.0 GB / 3.36 GB	0 / 0
Vlan 12	up / up	200	2.12 GB / 1.31 GB	0 / 0
Vlan 2000	up / up	10	431.48 MB / 1.17 GB	0 / 0
Vlan 300	up / up	10	2.07 GB / 1.07 GB	0 / 0
Vlan 501	up / up	10	15.58 MB / 3.06 GB	0 / 0
Vlan 502	up / up	10	1.07 GB / 113.21 MB	0 / 0
Vlan 503	up / up	10	3.71 GB / 1.07 GB	0 / 0
Vlan 504	up / up	10	3.71 GB / 10.64 MB	0 / 0
Vlan 505	up / up	10	2.34 GB / 1.36 GB	0 / 0
Vlan 506	up / up	10	2.64 GB / 1.56 GB	0 / 0
Vlan 507	up / up	10	1.73 GB / 954.04 MB	0 / 0

56. Alarm for IXR HW Mgmt Switch

Steps: -

1. Login to NADCM
2. Select the specific HW Mgmt IXR switch
3. We may try any operations like power reset.

Result: - Below is the Alarm shown in the NADCM.

The screenshot displays the MantaRay HW Manager interface. At the top, there's a navigation bar with the title 'MantaRay HW Manager' and a user profile 'NAMBELO4'. Below this, a summary section shows '10 Events' in a red circle, with a breakdown of event counts: Critical (10), Major (0), Warning (0), Info (0), and Cleared (0). To the right, there are tabs for 'Active Events' and 'All Events', and a 'Clear selected events' button. Below this, a table lists events with columns for Severity, Device identification, Description, Sensor name/type, Status/Value, Date (Read), and Actions. The table contains several entries, including 'OPNH007-MGMT-SW1' and 'BENHP002-U21-Leaf-SW01'. At the bottom, there's a 'Notifications' section and a system status bar showing '24°C' and 'Light rain'.

Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Read)	Actions
	2/39		discriminator=16386			
	OPNH007-MGMT-SW1	/bfd/network-instance(name=NMNET_IP_VRF)/peer(local-discrimi...	/bfd/network-instance(name=NMNET_IP_VRF)/peer(local-discriminator=16385)	down	2026/07/10 09:01:23	
	BENHP002-U21-Leaf-SW01	Device can not be managed, snmpv2 used			2026/07/10 09:01:23	
	OPNH007-MGMT-SW1	/interface(name=mgmt0), admin enabled (still down)	/interface(name=mgmt0)	admin enabled (still down)	2026/07/10 09:01:22	
	BENAF002-U21-Leaf-SW01	Device can not be managed, snmpv2 used			2026/07/10 09:01:22	
	BENHP002-U23-Leaf-SW02	Device can not be managed, snmpv2 used			2026/07/10 09:01:17	
	BENAF002-U23-Leaf-SW02	Device can not be managed, snmpv2 used			2026/07/10 09:01:13	

57. Alarm for IXR Leaf Switch

Steps: -

1. Login to NADCM
2. Select the specific Leaf IXR switch
3. Down the interface and check the Events.

Result: - Below is the Alarm shown in the NADCM

The screenshot displays the NADCM (Network Alarm Detection and Control Module) interface. The top section shows 'Hardware label management' and 'Selected devices: 1'. Below this, 'CPU, disk and memory' status is shown as 'CPUs: -, Disks: -, Memory modules: -'. The 'Events' section indicates 'Active events: 18'. A dropdown menu shows the selected device: 'Name: OPNHP007-LEAF01 IP: 10.145.245.194'. A filter bar allows selection of event severity: Critical (red), Major (orange), Warning (yellow), Info (blue), and Cleared (green). The 'Active events' table lists six events, all with a severity of 'Critical' and a value of 'down'. The events are related to BFD (Bidirectional Forwarding Detection) for various network instances (CORENET_VRF, NMNET_VRF, METRO-E_VRF) and a default instance. The table columns are: Severity, Device Identification, Description, Sensor name, Value, and Date (Read). The bottom of the interface shows a Windows taskbar with the search bar, task icons, and system tray information including language (ENG IN) and time (2:19 PM 7/31/2026).

Severity	Device Identification	Description	Sensor name	Value	Date (Read)
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=CORENET_VRF)/peer[local-discriminator=16392]	/bfd/network-instance(name=CORENET_VRF)/peer[local-discriminator=16392]	down	2026/07/22 16:36:36
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer[local-discriminator=16391]	/bfd/network-instance(name=NMNET_VRF)/peer[local-discriminator=16391]	down	2026/07/22 16:36:34
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=METRO-E_VRF)/peer[local-discriminator=16390]	/bfd/network-instance(name=METRO-E_VRF)/peer[local-discriminator=16390]	down	2026/07/22 16:36:34
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=METRO-E_VRF)/peer[local-discriminator=16389]	/bfd/network-instance(name=METRO-E_VRF)/peer[local-discriminator=16389]	down	2026/07/22 16:36:33
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=NMNET_VRF)/peer[local-discriminator=16388]	/bfd/network-instance(name=NMNET_VRF)/peer[local-discriminator=16388]	down	2026/07/22 16:36:33
Critical	OPNHP007-LEAF01	/bfd/network-instance(name=default)/peer[local-discriminator=16400]	/bfd/network-instance(name=default)/peer[local-discriminator=16400]	down	2026/07/22 16:35:52

58. IXR switch Rack Topology Check

Steps: -

1. Login NADCM and select “Devices”
2. Select the specific switch
3. Select the Switch and you can see the rack topology view at right side.

Result: - Switch rack topology details can be verified in as below.

The screenshot displays the NADCM (Network Assurance and Diagnostic Center Manager) web interface. The browser address bar shows the URL: https://10.169.94.153/agui/devices?filter.inventory_node_type=switch&selected=5bfdca2e5d52700480c238f5bfdca2.... The interface is titled "MantaRay HW Manager" and shows a list of devices on the left. The selected device is "SABT001-1001-1001-001". The right side of the interface displays the "Rack Topology Check" for this device. The topology is visualized as a grid of ports, with each port having a status icon (green for OK, yellow for warning, red for error). Below the topology, a table lists the interface details for each port.

Interface	Switch	MAC	Inventory MAC	Location	Port	Status	Interface description	Accept	Remove
8/1	S0a2	b0a0:67:1a				Unconfirmed new MAC	UD1-MME1-1-8-P1	☐	☐
11/1	S0a2	b0a0:67:57				Unconfirmed new MAC	UD1-MME1-1-8-P3	☐	☐
13/1	a0:1a:84:b6:a3:0c					Unconfirmed new MAC	UD1-MME1-1-8-P5	☐	☐
15/1	S0a2	b0a0:70:4a				Unconfirmed new MAC	UD1-MME1-1-8-P1	☐	☐
15/2	S0a2	b0a0:a6:7a				Unconfirmed new MAC	UD1-MME1-1-1-P1	☐	☐
15/3	S0a2	b0a0:66:16				Unconfirmed new MAC	UD1-MME1-1-2-P1	☐	☐
15/4	S0a2	b0a0:6a:22				Unconfirmed new MAC	UD1-MME1-1-3-P1	☐	☐
15/5	S0a2	b0a0:6a:af				Unconfirmed new MAC	UD1-MME1-1-4-P1	☐	☐
15/6	S0a2	b0a0:6a:4b				Unconfirmed new MAC	UD1-MME1-1-1-P3	☐	☐
15/7	S0a2	b0a0:7a:4b				Unconfirmed new MAC	UD1-MME1-1-2-P3	☐	☐
15/8	S0a2	b0a0:70:2b				Unconfirmed new MAC	UD1-MME1-1-3-P3	☐	☐

59. Overload Prevention

Steps: -

Keycloak concurrent sessions are limited to 50 at any given time

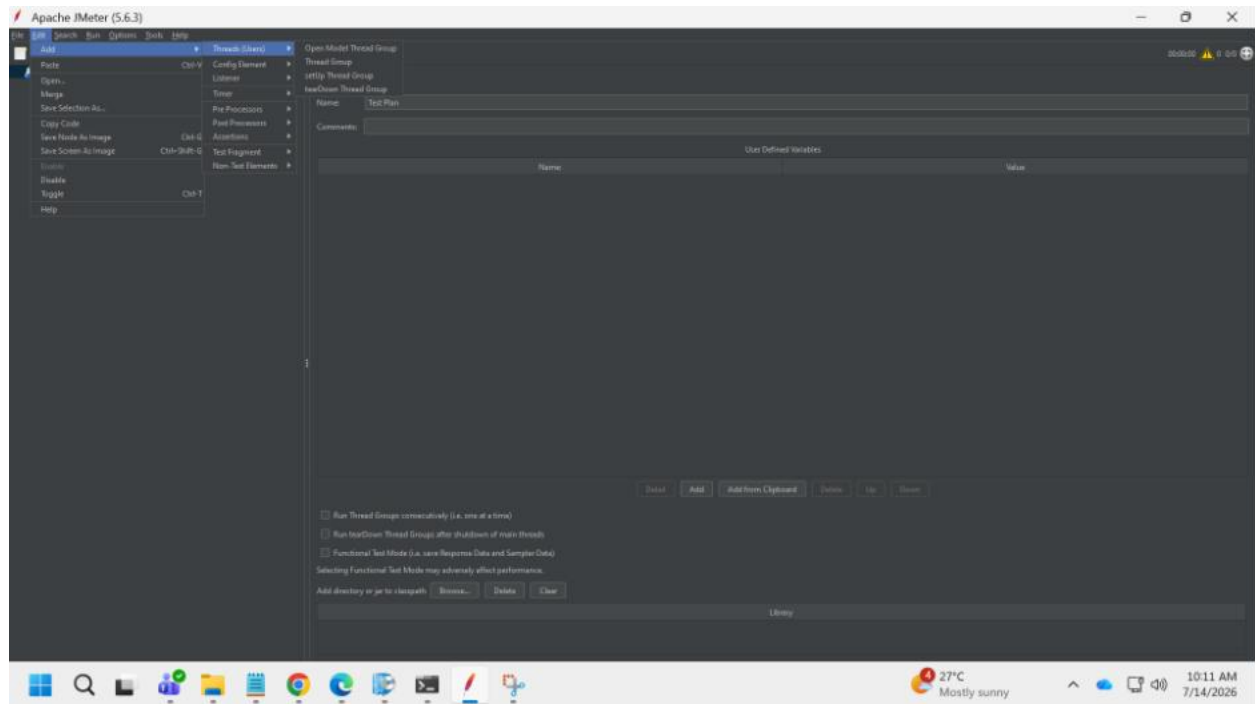
Prerequisites/conditions:

1. MantaRay HW Manager is installed (as a fresh installation, or as an upgrade)
2. A new test user exists in Keycloak, Follow the procedure from official customer documentation. a. Create a user “Configuring Security in MantaRay HW Manager” → “User accounts security” → “Managing users in Keycloak”. b. Make sure that password is set using “Configuring Security in MantaRay HW Manager” → “User accounts security” → “Managing user in Password”
3. Apache JMeter (5.6.3) GUI is running Execution: Create a new test plan & add a thread group, Add HTTP request sampler and Add listener to view the results
4. Configure protocol, server name or ip where HWM is installed and use path as “/auth/realms/NADCM/protocol/openid-connect/token” and make sure to use parameters as shown in the screenshot. Use values for “username” and “password” which is created as a pre-condition.

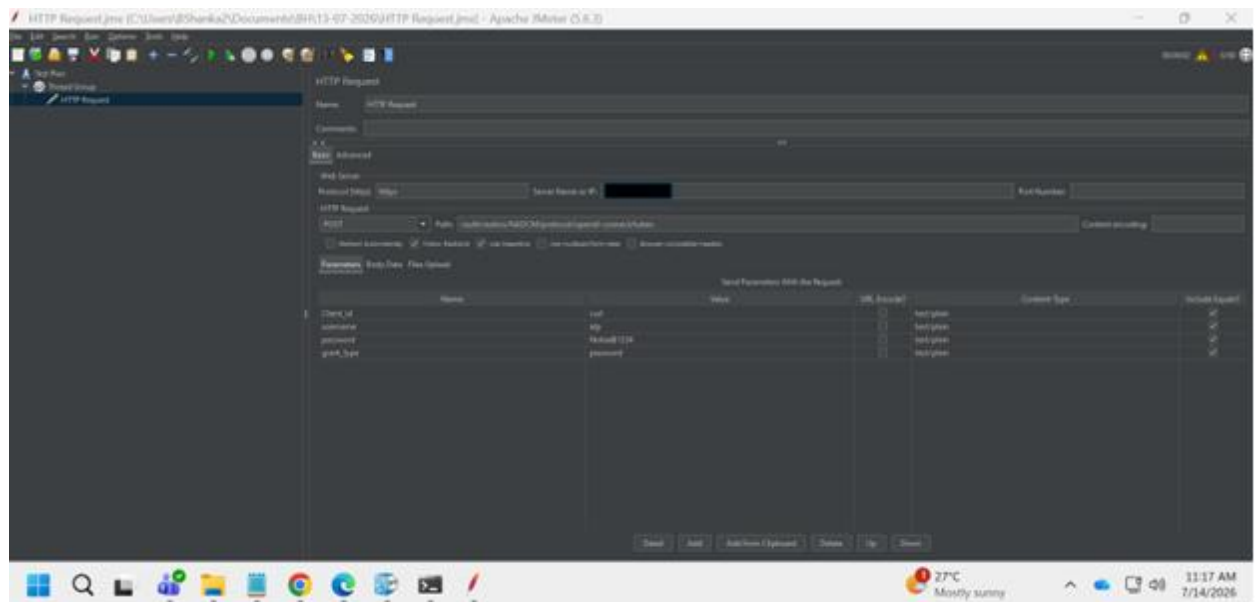
Expected results: There should be only 50 successful requests to get token for the user and all other requests should be failed.

Result: -

1. Create a new test plan
2. Add a thread group



3. Add HTTP request sampler



HTTP Request.jmx (C:\Users\BShanka2\Documents\BH\HTTP Request.jmx) - Apache JMeter (5.6.3)

FileEditViewOptionsHelp

Test Plan

Thread Group

HTTP Request

View Results in Table

View Results in Table

Name:View Results in Table

Comments:

Write results to file / Read from file

Filename:Log Log/Display Only:ErrorsSuccessesConfigure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
25	12/01/11:207	Thread Group 1-6	HTTP Request	1145	Success	2503	320	1146	708
26	12/01/11:409	Thread Group 1-7	HTTP Request	1151	Success	2503	320	1150	703
27	12/01/11:500	Thread Group 1-8	HTTP Request	1163	Success	2503	320	1163	706
28	12/01/12:027	Thread Group 1-9	HTTP Request	1149	Success	2503	320	1148	704
29	12/01/12:107	Thread Group 1-10	HTTP Request	1152	Success	2503	320	1152	703
40	12/01/15:752	Thread Group 1-1	HTTP Request	1181	Success	2503	320	1181	800
41	12/01/15:827	Thread Group 1-2	HTTP Request	1157	Success	2503	320	1157	708
42	12/01/15:957	Thread Group 1-3	HTTP Request	1154	Success	2503	320	1154	701
43	12/01/16:037	Thread Group 1-4	HTTP Request	1168	Success	2503	320	1168	701
44	12/01/16:157	Thread Group 1-5	HTTP Request	1144	Success	2503	320	1148	701
45	12/01/16:236	Thread Group 1-6	HTTP Request	1166	Success	2503	320	1167	800
46	12/01/16:336	Thread Group 1-7	HTTP Request	1157	Success	2503	320	1157	701
47	12/01/16:437	Thread Group 1-8	HTTP Request	1175	Success	2503	320	1174	701
48	12/01/16:537	Thread Group 1-9	HTTP Request	1158	Success	2503	320	1157	702
49	12/01/16:639	Thread Group 1-10	HTTP Request	1181	Success	2516	320	1181	701
50	12/01/42:102	Thread Group 1-1	HTTP Request	1185	Success	2516	320	1185	824
51	12/01/42:205	Thread Group 1-2	HTTP Request	1151	Success	2516	320	1151	752
52	12/01/42:300	Thread Group 1-3	HTTP Request	1137	Success	579	320	1137	701
53	12/01/42:400	Thread Group 1-4	HTTP Request	1138	Success	579	320	1138	800
54	12/01/42:507	Thread Group 1-5	HTTP Request	1125	Success	579	320	1125	760
55	12/01/43:007	Thread Group 1-6	HTTP Request	1134	Success	579	320	1133	791
56	12/01/43:108	Thread Group 1-7	HTTP Request	1143	Success	579	320	1140	800
57	12/01/43:208	Thread Group 1-8	HTTP Request	1148	Success	579	320	1140	799
58	12/01/43:308	Thread Group 1-9	HTTP Request	1128	Success	579	320	1128	701
59	12/01/43:408	Thread Group 1-10	HTTP Request	1136	Success	579	320	1136	701

Sort automatically:Child samples?

No. of Samples:59

Latest Sample:1136

Success:100%

Errors:0%

Windows Taskbar

29°C Mostly sunny

12:03 PM 7/14/2026

60. Optimized hardware discovery

Steps: -

- 1.Go to Administration
- 2.Click on DHCP
- 3.Choose an IP from the IP pool
- 4.Click edit option and enable the Instant HW scan option for the chosen IP
- 5.Go to hardware Scan and Perform hardware scan for the IP by providing the IP in the add scan range.

The screenshot displays the MantaRay HW Manager web interface. The left sidebar contains navigation options: Files, Roles, Backup, Activity Logs, DHCP (selected), Hardware Scan, Default Credentials, and Certificates. The main content area shows the DHCP configuration for a specific IP pool. At the top, the DHCP server is listed as 'bond0' with MAC address '50:5b:4b:b4:aa:fe' and IP address '10.169.94.153/27'. Below this, the 'DHCP global option mappings for IPv4' section shows 0 mappings. The 'DHCP subnets for IPv4' section lists 15 subnets. The table below shows the configuration for three subnets:

Subnet enabled	Instant HW scan enabled	Subnet	IP address pools	Gateway	TFTP server	Domain	DNS	Lease default	Lease maximum	Unknown clients	Unknown host ports	Actions
⊘	⊘	10.145.8.64/26	10.145.8.68 - 10.145.8.114	10.145.8.65				-	-	✓	✓	✎ 🗑 ⋮
✓	⊘	10.145.224.128/26	10.145.224.134 - 10.145.224.178	10.145.224.129				-	-	⊘	✓	✎ 🗑 ⋮
✓	⊘	10.145.52.0/26	10.145.52.7 - 10.145.52.45	10.145.52.1				-	-	✓	✓	✎ 🗑 ⋮

The bottom of the interface shows a Windows taskbar with the date and time as 8:42 AM on 7/13/2026.

Browser tabs: NADCM, Consolix, NAMBEL, New Mi, JIT NM, 10.246, Nokia S, Keycloak, NADCM, TC-30

Address bar: https://10.169.94.153/agu/admin/dhcpd?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a5bfdc...

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoft, NADCM cre, updat...

MantaRay HW Manager

Subnet parameters

Subnet: **ENABLED** Instant hardware scan: **ENABLED**

Options

☐ Deny unknown clients ☐ Deny unknown host ports

Subnet *

Subnet: 10.145.52.0 Network: 255.255.255.192

IP address pools (Hint: Allowed IP range is 10.145.52.0 - 10.145.52.63)

First address: 10.145.52.7 Last address: 10.145.52.45

DHCP options

Gateway IP: 10.145.52.1 TFTP server

Buttons: Next, Save, Cancel

System tray: 23°C Cloudy, Search, ENG IN, 8:43 AM 7/13/2026

Browser tabs: NADCM, Consolix, NAMBEL, New Mi, JIT NM, 10.246, Nokia S, Keycloak, NADCM, TC-30

Address bar: https://10.169.94.153/agu/admin/dhcpd?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a5bfdc...

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoft, NADCM cre, updat...

MantaRay HW Manager

DHCP global option mappings for IPv4

0 global option mappings

DHCP subnets for IPv4

15 subnets

Subnet enabled	Instant HW scan enabled	Subnet	IP address pools	Gateway	TFTP server	Domain	DNS	Lease default	Lease maximum	Unknown clients	Unknown host ports	Actions
		10.145.8.64/26	10.145.8.68 - 10.145.8.114	10.145.8.65				-	-			
		10.145.224.128/26	10.145.224.134 - 10.145.224.178	10.145.224.129				-	-			
		10.145.52.0/26	10.145.52.7 - 10.145.52.45	10.145.52.1				-	-			
		10.147.6.64/27	10.147.6.71 - 10.147.6.92	10.147.6.65				a minute	a minute			
		10.169.127.96/27	10.169.127.99 - 10.169.127.101 10.169.127.105 - 10.169.127.105 10.169.127.108 - 10.169.127.109 10.169.127.111 - 10.169.127.113 10.169.127.118 - 10.169.127.119 10.169.127.120 - 10.169.127.124	10.169.127.97				-	-			
		10.147.91.64/26	10.147.91.67 - 10.147.91.77	10.147.91.65				-	-			
		10.147.91.192/26	10.147.91.195 - 10.147.91.206	10.147.91.193				-	-			

Notification: Subnet "10.145.52.0" successfully modified a few seconds ago. [Show more](#)

System tray: 23°C Cloudy, Search, ENG IN, 9:05 AM 7/13/2026

NADCM x Consolix x NAMBEL x New Mil x JIT NM x 10.246.1 x Nokia S x Keycloak x NADCM x TC 30 x

Not secure https://10.169.94.153/agu/admin/hwscan?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a,5bfd...

T-Mobile Bookmarks Compass cyberark Safeguard JIT NMNet My Access NTAC RAN OSS login.microsoft... NADCM cre, updat...

MantaRay HW Manager NAMBEL04

Wait time between scans? 60 sec

Enable deep inspection once for new server node

Enable IPv4 DHCP automatically for the management port of a new device

Add new scan range

☒ Auto scan

First address* 10.145.52.7 Last address* 10.145.52.45

Credential set

Notes ATP

Max 10 characters 3/10

Save Cancel

Auto scan	First ↓	Last ↓	Credential set ↓	Notes	Actions
☐	10.147.62.67				
☐	10.147.255.35				
☐	10.147.223.163				

Notifications (0)

23°C Cloudy Search ENG IN 9:06 AM 7/13/2026

NADCM x Consolix x NAMBEL x New Mil x JIT NM x 10.246.1 x Nokia S x Keycloak x NADCM x TC 30 x

Not secure https://10.169.94.153/agu/admin/hwscan?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a,5bfd...

T-Mobile Bookmarks Compass cyberark Safeguard JIT NMNet My Access NTAC RAN OSS login.microsoft... NADCM cre, updat...

MantaRay HW Manager NAMBEL04

Wait time between scans? 60 sec

Enable deep inspection once for new server node

Enable IPv4 DHCP automatically for the management port of a new device

Auto scan	First ↓	Last ↓	Credential set ↓	Notes	Actions
☐	10.145.52.7	10.145.52.45	any	ATP	
☐	10.147.62.67	10.147.62.67			
☐	10.147.255.35	10.147.255.35	SW		
☐	10.147.223.163	10.147.223.163	SW		

Notifications (0)

23°C Cloudy Search ENG IN 9:07 AM 7/13/2026

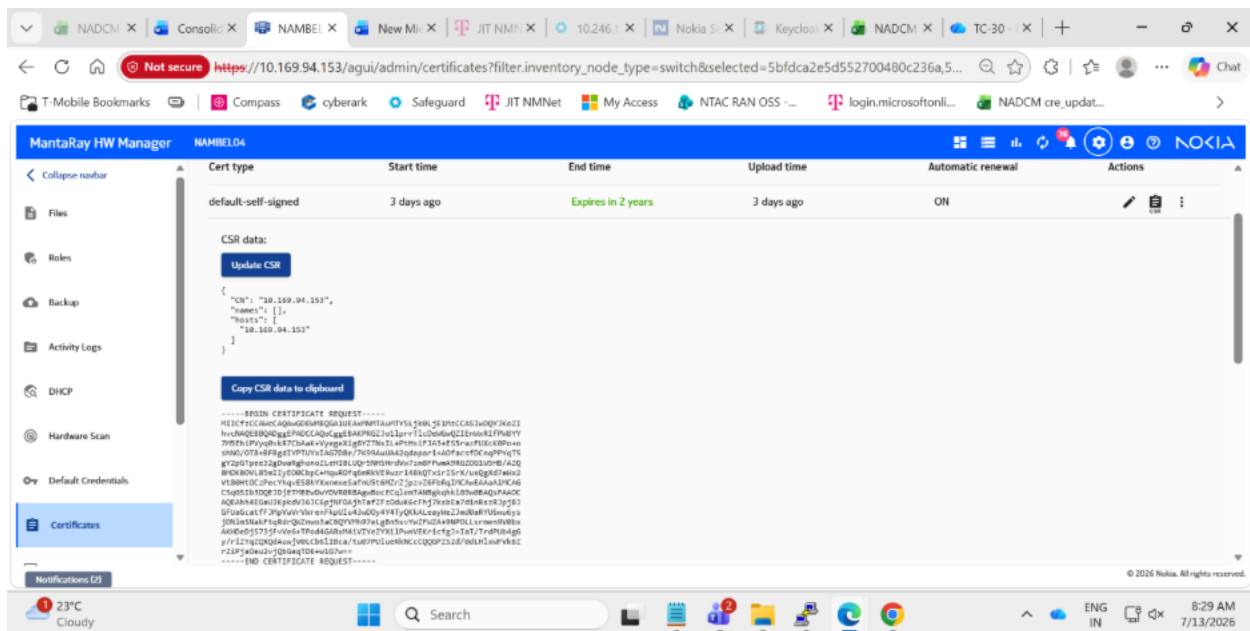
61. Possibility to update/modify HW Manager interface certificate CSR data

Steps: -

1. In NADCM GUI go to **Administration**.
2. Select **Certificates** and go to MantaRay HW Manager interface certificates tab.
3. Click on CSR icon and update CSR is visible.

Result: -

Update CSR option is available.



62. Support for cancelling upload/download of a file

Steps: -

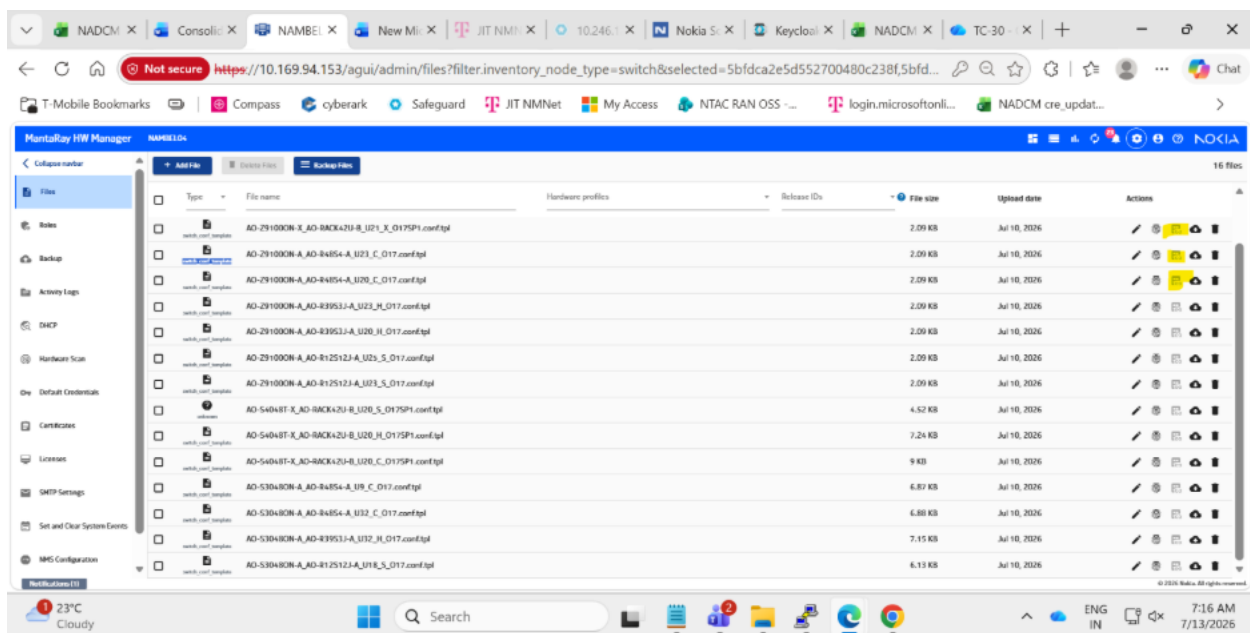
1. Login NADCM and select “Administration → Files”.
2. Click on add files and select the upload file.

Result: - Verified the support of cancelling upload/ download.

The screenshot shows the NADCM web interface with the 'Files' section selected. A notification at the top states: "Transfer completed NAMBLE dev conf_files_upload_conf_access_nbscert_ExportData_2026-07-13T07:11:43.394Z.png 379.2 KB". Below this, a table lists the uploaded files:

Type	File name	Hardware profiles	Release IDs	File size	Upload date	Actions
Files	NAMBLE dev conf_files_upload_conf_access_nbscert_ExportData_2026-0...			379.17 KB	Jul 13, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-X_AO-8ACX420-8_U23_X_017SP1.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-X_AO-8ACX420-8_U21_X_017SP1.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-84IS4-A_U23_C_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-84IS4-A_U20_C_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-83953-A_U23_H_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-83953-A_U20_H_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-812512-A_U29_S_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-291000N-A_AO-812512-A_U23_S_017.conf.tpl			2.09 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-54048T-X_AO-8ACX420-8_U20_S_017SP1.conf.tpl			4.52 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-54048T-X_AO-8ACX420-8_U20_H_017SP1.conf.tpl			7.24 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-54048T-X_AO-8ACX420-8_U20_C_017SP1.conf.tpl			9 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-530480N-A_AO-84IS4-A_U9_C_017.conf.tpl			6.87 KB	Jul 10, 2026	[Edit] [Delete] [Download]
switch_conf_templates	AO-530480N-A_AO-84IS4-A_U32_C_017.conf.tpl			6.88 KB	Jul 10, 2026	[Edit] [Delete] [Download]

The bottom status bar shows the temperature as 23°C, the time as 7:12 AM, and the date as 7/13/2026.

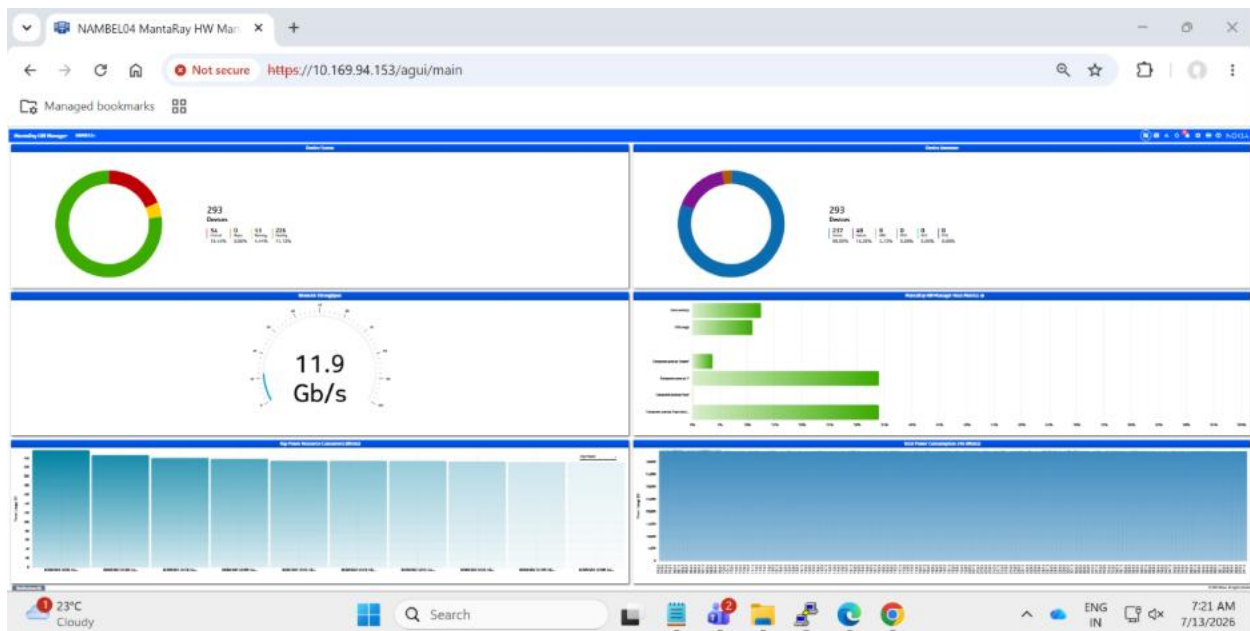


64. User session cookies in browser are not persistent anymore

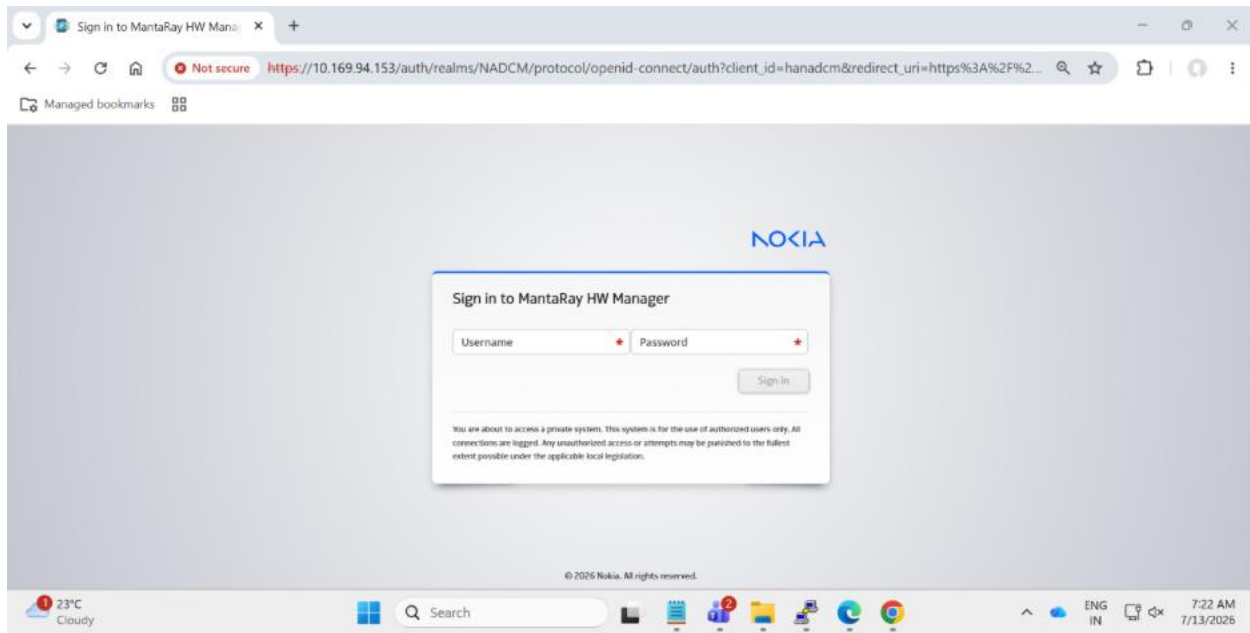
Steps: -

1. Login the NADCM GUI.
2. Close the browser.
3. Previous session will be expired and need to login again the NADCM GUI on same browser.

Result: -



Close and reopen the browser



65. Download WBX Switch Configuration using NADCM

Steps: -

1. In NADCM, go to Devices.
2. Select the HW Switch → Click on Menu
3. Download switch configuration as shown below.

Result: - Successfully downloaded the switch configuration from NADCM.

The screenshot displays the NADCM web interface. On the left, a table lists various network devices. The right pane shows the details for a selected switch, including its status, configuration, and a rack topology diagram.

Type	Name	Status	Hardware
Switch	SRF001-1224-MSMT-001	OK	SRF001-1224
Switch	SRF001-1224-MSMT-002	OK	SRF001-1224
Switch	SRF001-1224-MSMT-003	OK	SRF001-1224
Switch	SRF001-1224-MSMT-004	OK	SRF001-1224
Switch	SRF001-1224-MSMT-005	OK	SRF001-1224
Switch	SRF001-1224-MSMT-006	OK	SRF001-1224
Switch	SRF001-1224-MSMT-007	OK	SRF001-1224
Switch	SRF001-1224-MSMT-008	OK	SRF001-1224
Switch	SRF001-1224-MSMT-009	OK	SRF001-1224
Switch	SRF001-1224-MSMT-010	OK	SRF001-1224
Switch	SRF001-1224-MSMT-011	OK	SRF001-1224
Switch	SRF001-1224-MSMT-012	OK	SRF001-1224
Switch	SRF001-1224-MSMT-013	OK	SRF001-1224
Switch	SRF001-1224-MSMT-014	OK	SRF001-1224
Switch	SRF001-1224-MSMT-015	OK	SRF001-1224
Switch	SRF001-1224-MSMT-016	OK	SRF001-1224
Switch	SRF001-1224-MSMT-017	OK	SRF001-1224
Switch	SRF001-1224-MSMT-018	OK	SRF001-1224
Switch	SRF001-1224-MSMT-019	OK	SRF001-1224
Switch	SRF001-1224-MSMT-020	OK	SRF001-1224
Switch	SRF001-1224-MSMT-021	OK	SRF001-1224
Switch	SRF001-1224-MSMT-022	OK	SRF001-1224
Switch	SRF001-1224-MSMT-023	OK	SRF001-1224
Switch	SRF001-1224-MSMT-024	OK	SRF001-1224
Switch	SRF001-1224-MSMT-025	OK	SRF001-1224
Switch	SRF001-1224-MSMT-026	OK	SRF001-1224
Switch	SRF001-1224-MSMT-027	OK	SRF001-1224
Switch	SRF001-1224-MSMT-028	OK	SRF001-1224
Switch	SRF001-1224-MSMT-029	OK	SRF001-1224
Switch	SRF001-1224-MSMT-030	OK	SRF001-1224
Switch	SRF001-1224-MSMT-031	OK	SRF001-1224
Switch	SRF001-1224-MSMT-032	OK	SRF001-1224
Switch	SRF001-1224-MSMT-033	OK	SRF001-1224
Switch	SRF001-1224-MSMT-034	OK	SRF001-1224
Switch	SRF001-1224-MSMT-035	OK	SRF001-1224
Switch	SRF001-1224-MSMT-036	OK	SRF001-1224
Switch	SRF001-1224-MSMT-037	OK	SRF001-1224
Switch	SRF001-1224-MSMT-038	OK	SRF001-1224
Switch	SRF001-1224-MSMT-039	OK	SRF001-1224
Switch	SRF001-1224-MSMT-040	OK	SRF001-1224
Switch	SRF001-1224-MSMT-041	OK	SRF001-1224
Switch	SRF001-1224-MSMT-042	OK	SRF001-1224
Switch	SRF001-1224-MSMT-043	OK	SRF001-1224
Switch	SRF001-1224-MSMT-044	OK	SRF001-1224
Switch	SRF001-1224-MSMT-045	OK	SRF001-1224
Switch	SRF001-1224-MSMT-046	OK	SRF001-1224
Switch	SRF001-1224-MSMT-047	OK	SRF001-1224
Switch	SRF001-1224-MSMT-048	OK	SRF001-1224
Switch	SRF001-1224-MSMT-049	OK	SRF001-1224
Switch	SRF001-1224-MSMT-050	OK	SRF001-1224

Device Details:

- Name: SRF001-1224-MSMT-001
- Management IP: 10.10.10.1
- Management MAC: 34-34-34-34-34-34
- Hardware profile: SRF001-1224-MSMT-001
- Product name: SRF001-1224-MSMT-001
- Product serial: 10100002
- Device ID: 545251967a7a30049682c99

Device Status:

- Power: ON
- Configuration: OK
- Configuration file: SRF001-1224-MSMT-001.conf

Rack Topology Check:

The rack topology diagram shows a 48-port switch with ports 1-48. Ports 1-24 are labeled 'P1' and ports 25-48 are labeled 'P2'. The diagram indicates that the switch is connected to a power source and has a valid configuration.

NADCM x Consolix x NAMBEI x New Mi x JTT NM x 10.246. x Nokia S x Keycloak x NADCM x TC-30 x + -

Not secure https://10.169.94.153/agui/devices?filter.inventory_node_type=switch&selected=5d52579e7a2ca30049682c99

T-Mobile Bookmarks Compass cyberark Safeguard JTT NIMNet My Access NTAC RAN OSS login.microsoftl... NADCM cre, updat...

MantaRay I/W Manager NAMBLE104

TypoProfile x Name x Status x Location x Tags x HW Labels x FW versions x Other x 48 Items

Type	Name	Status	Hardware p...
☐	SONAT001-L23-MGMT-001	✓	S4040T-001
☐	SONAT001-L23-001	✓	Nuage Net
☐	SONAT001-L23-002	✓	Nuage Net
☐	SONAT001-L23-MGMT-002	✓	S4040T-001
☐	SONAT001-L23-MGMT-001	✓	S4040T-001
☒	SONAT001-001	✓	29-100-001
☐	SONAT001-002	✓	29-100-001
☐	SONAT001-L23-001	✓	S4040T-001
☐	SONAT001-L23-002	✓	S4040T-001
☐	SONAT001-L23-003	✓	S4040T-001
☐	SONAT001-L23-004	✓	S4040T-001
☐	SONAT001-L23-005	✓	S4040T-001
☐	SONAT001-L23-006	✓	S4040T-001
☐	SONAT001-L23-007	✓	S4040T-001
☐	SONAT001-L23-008	✓	S4040T-001
☐	SONAT001-L23-009	✓	S4040T-001
☐	SONAT001-L23-010	✓	S4040T-001
☐	SONAT001-L23-011	✓	S4040T-001
☐	SONAT001-L23-012	✓	S4040T-001
☐	SONAT001-L23-013	✓	S4040T-001
☐	SONAT001-L23-014	✓	S4040T-001
☐	SONAT001-L23-015	✓	S4040T-001
☐	SONAT001-L23-016	✓	S4040T-001
☐	SONAT001-L23-017	✓	S4040T-001
☐	SONAT001-L23-018	✓	S4040T-001
☐	SONAT001-L23-019	✓	S4040T-001
☐	SONAT001-L23-020	✓	S4040T-001
☐	SONAT001-L23-021	✓	S4040T-001
☐	SONAT001-L23-022	✓	S4040T-001
☐	SONAT001-L23-023	✓	S4040T-001
☐	SONAT001-L23-024	✓	S4040T-001
☐	SONAT001-L23-025	✓	S4040T-001
☐	SONAT001-L23-026	✓	S4040T-001
☐	SONAT001-L23-027	✓	S4040T-001
☐	SONAT001-L23-028	✓	S4040T-001
☐	SONAT001-L23-029	✓	S4040T-001
☐	SONAT001-L23-030	✓	S4040T-001
☐	SONAT001-L23-031	✓	S4040T-001
☐	SONAT001-L23-032	✓	S4040T-001
☐	SONAT001-L23-033	✓	S4040T-001
☐	SONAT001-L23-034	✓	S4040T-001
☐	SONAT001-L23-035	✓	S4040T-001
☐	SONAT001-L23-036	✓	S4040T-001
☐	SONAT001-L23-037	✓	S4040T-001
☐	SONAT001-L23-038	✓	S4040T-001
☐	SONAT001-L23-039	✓	S4040T-001
☐	SONAT001-L23-040	✓	S4040T-001
☐	SONAT001-L23-041	✓	S4040T-001
☐	SONAT001-L23-042	✓	S4040T-001
☐	SONAT001-L23-043	✓	S4040T-001
☐	SONAT001-L23-044	✓	S4040T-001
☐	SONAT001-L23-045	✓	S4040T-001
☐	SONAT001-L23-046	✓	S4040T-001
☐	SONAT001-L23-047	✓	S4040T-001
☐	SONAT001-L23-048	✓	S4040T-001
☐	SONAT001-L23-049	✓	S4040T-001
☐	SONAT001-L23-050	✓	S4040T-001

Device status

Select a device

SONAT001-001 IP: 10.143.8.253

PSUs: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Other interfaces

Back Topology Check

Download Device Data

Set Initial Setup Configuration

Upload Configuration

Delete Device

Download Configuration

Refresh Device Data from HW

Device Details

Name: SONAT001-001

Management IP: 10.143.8.253

Management MAC: 8a:0d:8f:41

Hardware profile: LON-01-05-24

Product name: SONAT001-001

Product serial: 001

Device ID: 001

Identification and operations: SONAT001-001, 10.143.8.253

Power management: ON

Firmware management: Selected devices: 1

23°C Cloudy

Search

7:27 AM 7/13/2026

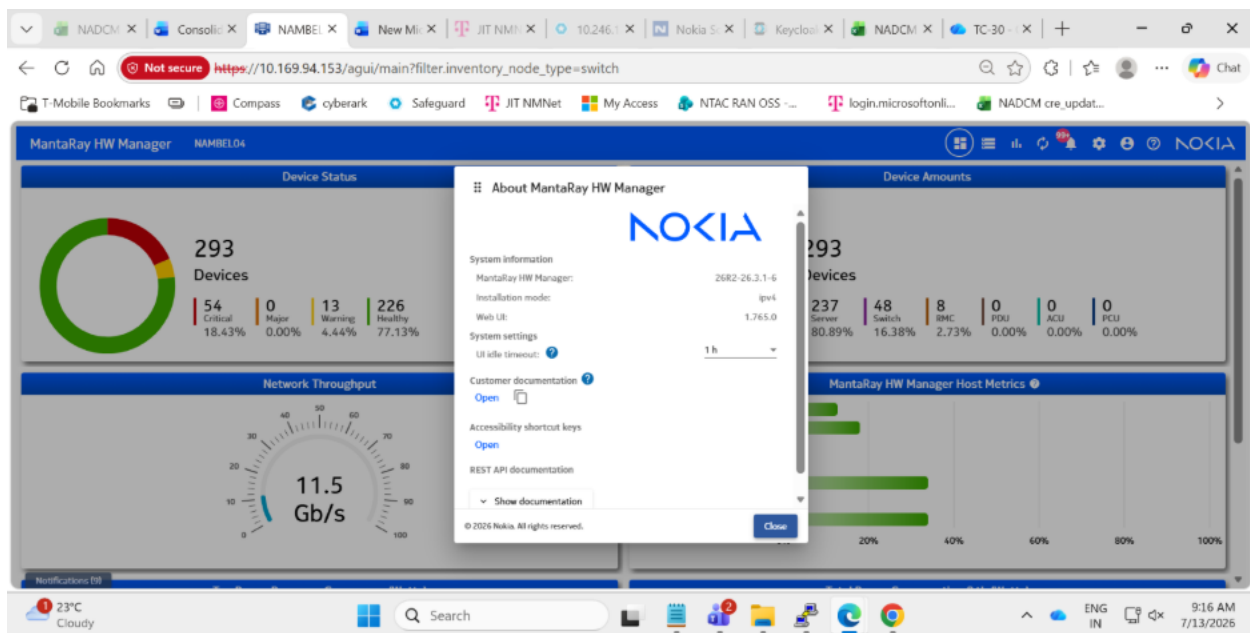
66. HWM activity and audit logging improvements

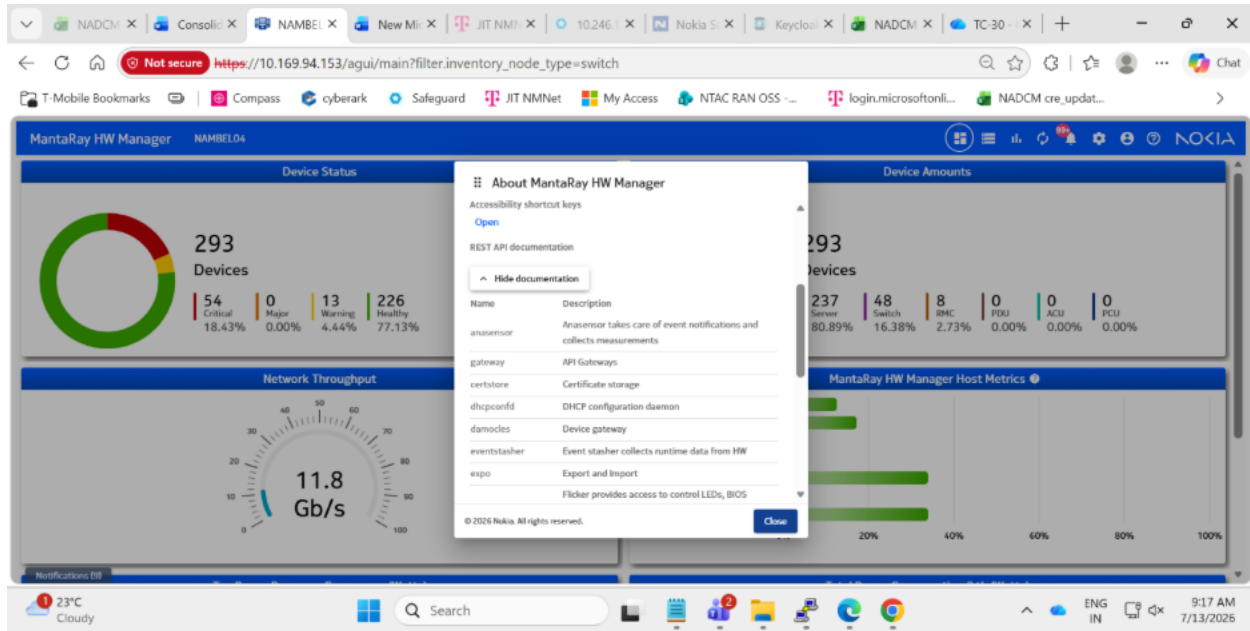
Steps: -

1. Login to NADCM GUI
2. Go to About MantaRay HW Manager tab
3. Click on **REST API documentation** and scroll down
4. Click on **Export and Import** and generate new tab Swagger UI

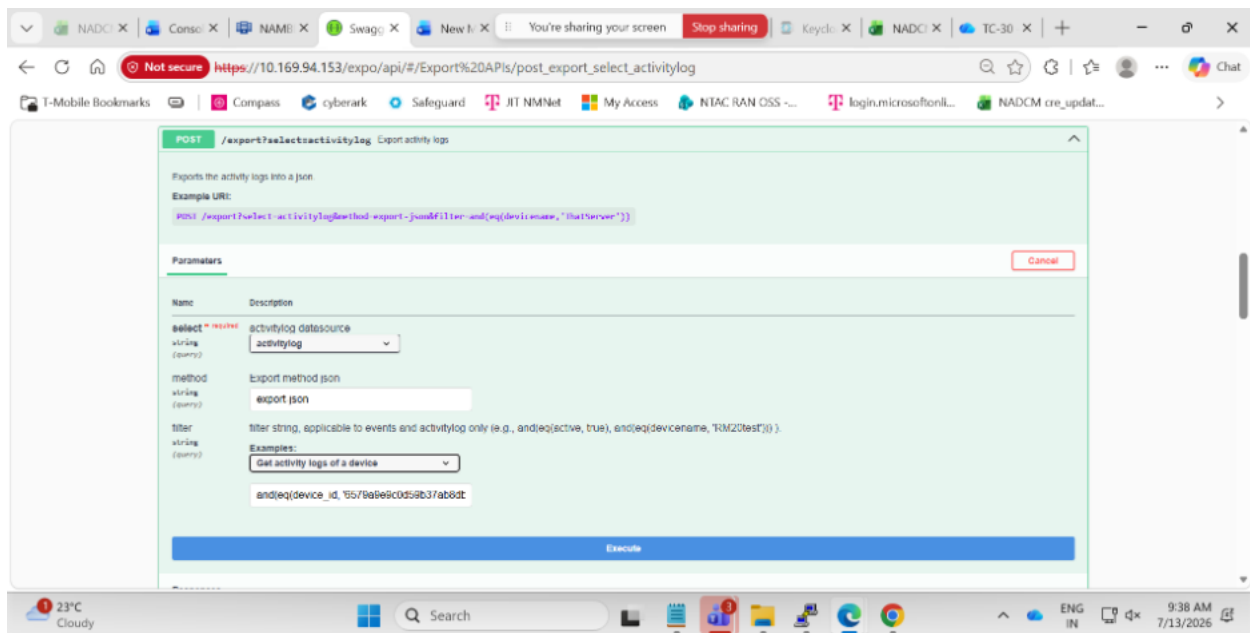
Result: -

1. Login the NADCM→ About

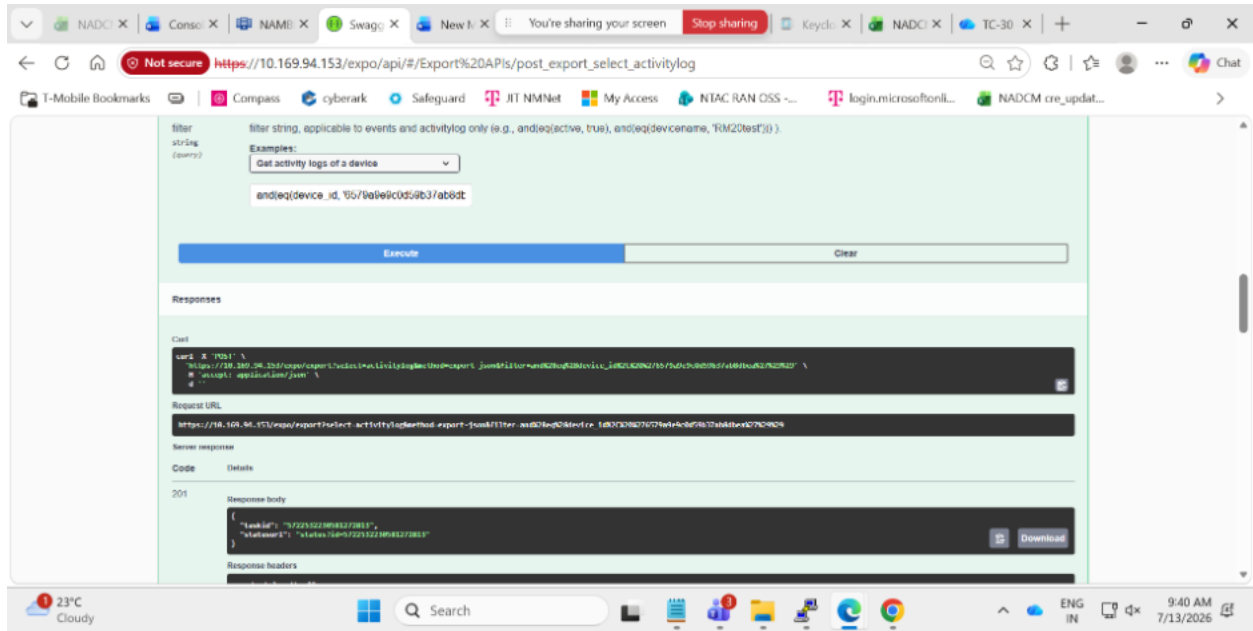




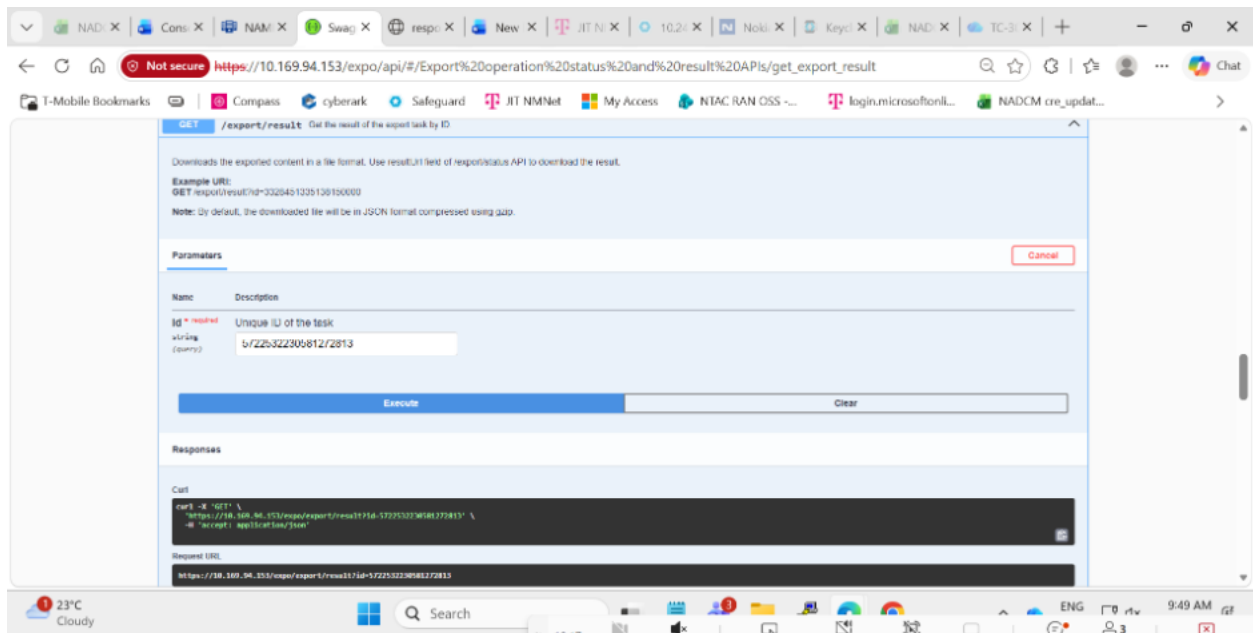
2. Search **Post export activity log** option and select **Modified value** and put **device id** and then **Execute**



3. After Execution got response output and copy device id .



4. Go to **GET** export device ID and download the file

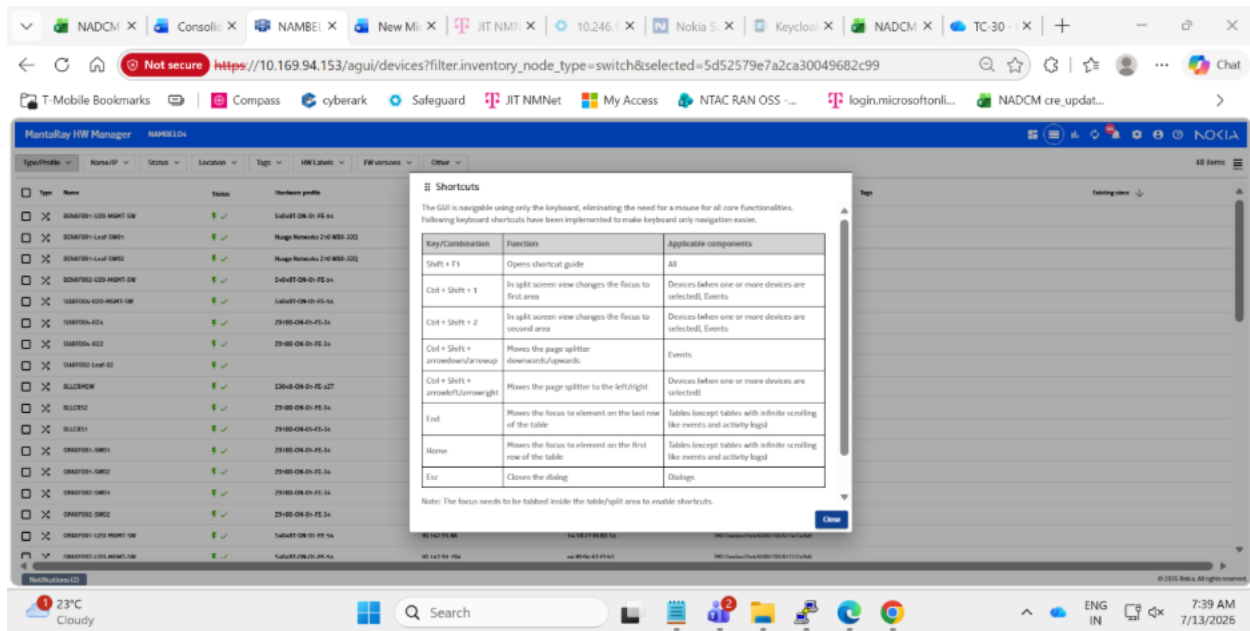


67. HWM GUI accessibility enhancement phase 5

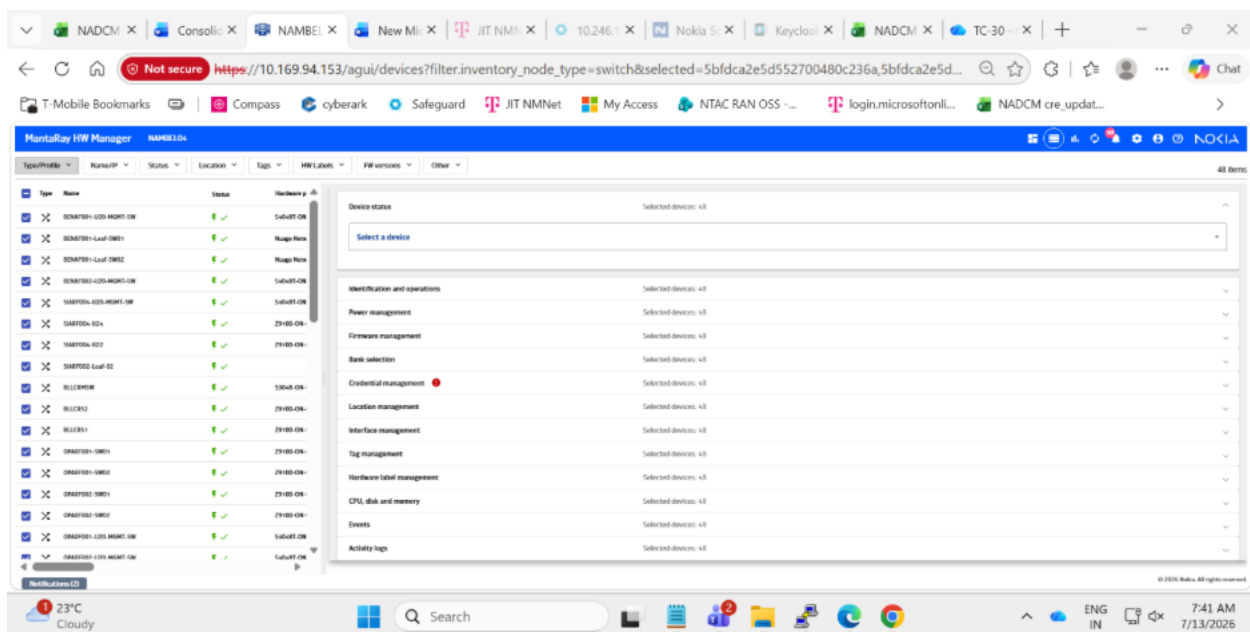
Steps: -

Accessibility shortcut keys are added. Menu can be accessed using - Shift + F1 or Select About view and select Accessibility shortcut keys.

Result: - 1. Shift+f1:



2. Ctrl+shift+2 :



3.Ctrl+shift+right arrow and left arrow:

Right arrow:

The screenshot shows the Meritway HW Manager web application. The main area displays a table of devices with columns for Type, Name, Status, Hardware profile, and IP. The table lists various network devices like switches and routers. On the right, there is a sidebar with a 'Device status' section and a list of management categories such as Identification and operations, Power management, Firmware management, Basic selection, Credential management, Location management, Interface management, Tag management, Hardware label management, CPU, disk and memory, Events, and Activity logs. The interface is running in a browser window with multiple tabs open, including NADCM, Consolix, NAMBEI, and others. The Windows taskbar at the bottom shows the date and time as 7:45 AM on 7/13/2026.

Left arrow:

This screenshot is identical to the one above, showing the Meritway HW Manager interface. It displays the same table of devices and the same sidebar with navigation options. The browser window and Windows taskbar are also visible, showing the same date and time (7:46 AM on 7/13/2026).

3.Ctrl+shift+right Up and Down arrow:

Up arrow:

Browser tabs: NADCM, Consolix, NAMBEI, New Mi, JIT NMNet, 10.246, Nokia S, Keycloak, NADCM, TC-30.

Address bar: Not secure https://10.169.94.153/agu/events/main?filter.inventory_node_type=switch&selected=5bfdca2e5d52700480c236a,5bfdca...

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoft, NADCM cre, updat...

MantaRay I/W Manager NAMBEI104

Events | Event & Notification Rules

Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Read)	Actions
Critical	BELFD031	Device can not be managed. Reason gres: - other error-unexpected amount of notifications: 0			2026/07/13 07:51:12	
Critical	BELFD031	Device can not be managed. Reason gres: - other error-unexpected amount of notifications: 0			2026/07/13 07:50:34	
Warning	BENAF002-4339M-Comp	Voltage Abnormal	VIAT_JHM	Lower Critical	2026/07/13 06:08:05	
Info	NADCMhost	Login failed from IP:10.147.179.96	Login Info		2026/07/13 05:03:37	
Critical	OPNAF004-Compute-U16R	Device can not be managed. Reason ahref: - cred error-Unauthorised ipri: - cred error-Unauthori...			2026/07/12 09:53:45	
Warning	BENAF001-4339M-Compute	Voltage Abnormal	VIAT_JHM	Lower Critical	2026/07/12 07:51:11	
Critical	SWAF004-U14	Device can not be managed. Reason ipri: - cred error-Unauthorised			2026/07/16 14:23:59	
Critical	OPNAF001_Master-1-025	Device can not be managed. Reason ipri: - cred error-Unauthorised			2026/07/16 14:23:49	
Critical	BENEF001-U141-Comp	Device can not be managed. Reason redflsh: - other error-GetBoot error: invalid character "0" look...			2026/07/16 14:23:44	
Critical	BENAF001-U16-Compute	Device can not be managed. Reason redflsh: - other error-GetBoot error: invalid character "0" look...			2026/07/16 14:23:41	
Critical	NADCMhost	Device can not be managed. Reason redflsh: - cred error-Unauthorised			2026/07/16 14:23:24	
Critical	BENEF001-U14R-Usr_cloud	Device can not be managed. Reason redflsh: - cred error-Unauthorised			2026/07/16 14:23:22	
Critical	OPNAF006-LEAF01	/system/tdp/interface(name=ethernet-1/239), admin enabled (all down)	/system/tdp/interface(name=ethernet-1/239)	admin enabled (all down)	2026/07/16 13:18:09	
Critical	OPNAF006-LEAF01	/system/tdp/interface(name=ethernet-1/239/41), admin enabled (all down)	/system/tdp/interface(name=ethernet-1/239/41)	admin enabled (all down)	2026/07/16 13:18:09	
Warning	Server36	Device is not reachable			2026/07/16 12:16:25	
Critical	BENAF002-4332B-RMC02	Device can not be managed, smp/sd used			2026/07/16 10:47:48	
Critical	BENAF001-U141-Compute	Device can not be managed. Reason redflsh: - other error-GetBoot error: invalid character "0" look...			2026/07/16 10:43:47	

23°C Cloudy | Search | 7:52 AM 7/13/2026

Down arrow:

Browser tabs: NADCM, Consolix, NAMBEI, New Mi, JIT NMNet, 10.246, Nokia S, Keycloak, NADCM, TC-30.

Address bar: Not secure https://10.169.94.153/agu/events/main?filter.inventory_node_type=switch&selected=5bfdca2e5d52700480c236a,5bfdca...

Navigation: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoft, NADCM cre, updat...

MantaRay I/W Manager NAMBEI104

Events | Event & Notification Rules

779 Events

Critical 755

Major 0

Warning 24

Info 0

Cleared 0

Active Events

All Events

Clear selected events (108)

Select devices

No data available

Severity	Device identification	Description	Sensor name/type	Status/Value	Date (Read)	Actions
Critical	BELFD031	Device can not be managed. Reason gres: - other error-unexpected amount of notifications: 0			2026/07/13 07:51:12	
Critical	BELFD031	Device can not be managed. Reason gres: - other error-unexpected amount of notifications: 0			2026/07/13 07:50:34	
Warning	BENAF002-4339M-Comp	Voltage Abnormal	VIAT_JHM	Lower Critical	2026/07/13 06:08:05	
Info	NADCMhost	Login failed from IP:10.147.179.96	Login Info		2026/07/13 05:03:37	
Critical	OPNAF004-Compute-U16R	Device can not be managed. Reason ahref: - cred error-Unauthorised ipri: - cred error-Unauthori...			2026/07/12 09:53:45	
Warning	BENAF001-4339M-Compute	Voltage Abnormal	VIAT_JHM	Lower Critical	2026/07/12 07:51:11	
Critical	SWAF004-U14	Device can not be managed. Reason ipri: - cred error-Unauthorised			2026/07/16 14:23:59	
Critical	OPNAF001_Master-1-025	Device can not be managed. Reason ipri: - cred error-Unauthorised			2026/07/16 14:23:49	
Critical	BENEF001-U141-Comp	Device can not be managed. Reason redflsh: - other error-GetBoot error: invalid character "0" look...			2026/07/16 14:23:44	
Critical	BENAF001-U16-Compute	Device can not be managed. Reason redflsh: - other error-GetBoot error: invalid character "0" look...			2026/07/16 14:23:41	

23°C Cloudy | Search | 7:52 AM 7/13/2026

End:

Browser tabs: NADCM, Consolix, NAMBEI, New Mi, JIT NM, 10.246, Nokia S, Keycloak, NADCM, TC-30, +

Address bar: Not secure https://10.169.94.153/agu/admin/activity?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a5bfdc...

Browser bookmarks: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoftonli..., NADCM cre, updat...

MantaRay HW Manager NAMBEI04

Activity Logs

Result	Action	Details	Device	User	Time
❌	GET CONFIGURATION	Task failed (Step: 1/2) show details	SIARF004-U24	automatic@nokia.com	28 minutes ago
✅	EXPORT	Targets: inventory, eventrules, systemconfig, notificationrules, certificates, hwallocations, gwcerts. Format: export-json	NADCMsystem	automatic@nokia.com	42 minutes ago
✅	CHECK INTERFACE INFO	Task completed show details	OPARF002-SW01	automatic@nokia.com	an hour ago
❌	CHECK INTERFACE INFO	Task failed (Step: 1/2) show details	SIARF004-U24	automatic@nokia.com	an hour ago
✅	CHECK INTERFACE INFO	Task completed show details	OPARF001-U20-HIGHT-SW	automatic@nokia.com	3 hours ago
❌	GET CONFIGURATION	Task failed (Step: 1/2) show details	BENAF001-4x4-SW01	automatic@nokia.com	3 hours ago
❌	GET CONFIGURATION	Task failed (Step: 1/2) show details	BENAF001-4x4-SW01	automatic@nokia.com	3 hours ago
✅	CHANGE DEVICE ACCESS CREDENTIALS	MANAGEMENT credentials were changed.	BENAF001-U20L-Compule	automatic@nokia.com	3 hours ago
✅	CHANGE DEVICE ACCESS CREDENTIALS	MANAGEMENT, SHARPSAP credentials were changed.	BENAF001-U20L-Compule	automatic@nokia.com	3 hours ago

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Windows taskbar: 23°C Cloudy, Search, ENG IN, 7:54 AM 7/13/2026

Home:

Browser tabs: NADCM, Consolix, NAMBEI, New Mi, JIT NM, 10.246, Nokia S, Keycloak, NADCM, TC-30, +

Address bar: Not secure https://10.169.94.153/agu/admin/activity?filter.inventory_node_type=switch&selected=5bfdca2e5d552700480c236a5bfdc...

Browser bookmarks: T-Mobile Bookmarks, Compass, cyberark, Safeguard, JIT NMNet, My Access, NTAC RAN OSS, login.microsoftonli..., NADCM cre, updat...

MantaRay HW Manager NAMBEI04

Activity Logs

Result	Action	Details	Device	User	Time
❌	GET CONFIGURATION	Task failed (Step: 1/2) hide details <pre>{ "HW access user is not privileged, stopping." }</pre>	SIARF004-U24	automatic@nokia.com	31 minutes ago
✅	EXPORT	Targets: inventory, eventrules, systemconfig, notificationrules, certificates, hwallocations, gwcerts. Format: export-json	NADCMsystem	automatic@nokia.com	an hour ago
✅	CHECK INTERFACE INFO	Task completed show details	OPARF002-SW01	automatic@nokia.com	an hour ago
❌	CHECK INTERFACE INFO	Task failed (Step: 1/2) show details	SIARF004-U24	automatic@nokia.com	an hour ago
-	-	Task completed	-	-	-

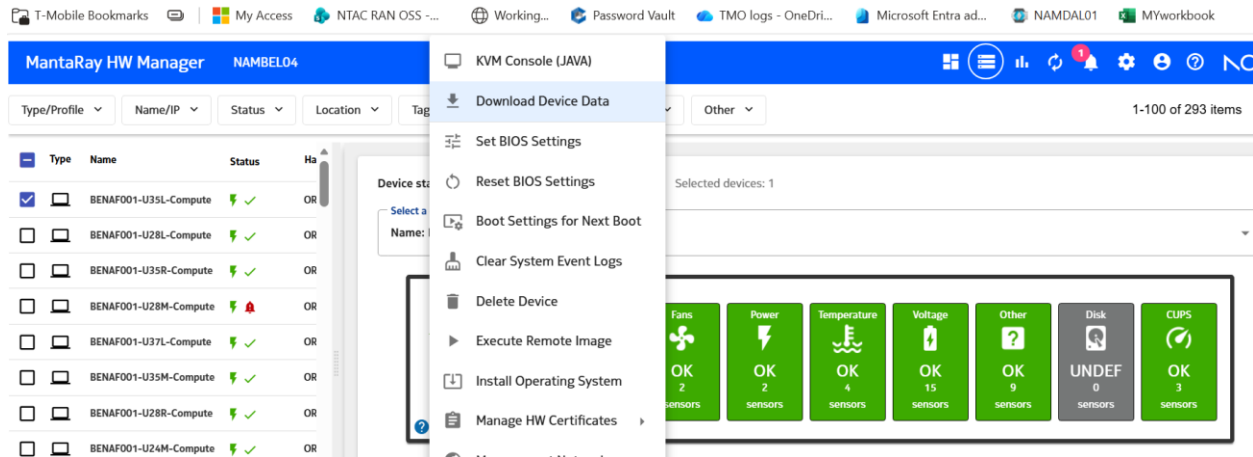
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Windows taskbar: 23°C Cloudy, Search, ENG IN, 7:57 AM 7/13/2026

69.HWM support for Redfish events

Steps: -

- 1.Choose any device
2. Download Device data
3. events.json contains Redfish events at device dump



active_events.json	7/17/2026 3:45 PM	JSON File	1 KB
activity_logs.json	7/17/2026 3:45 PM	JSON File	5 KB
bios_current_settings.json	7/17/2026 3:45 PM	JSON File	4 KB
bios_supported_settings.json	7/17/2026 3:45 PM	JSON File	48 KB
bmc_current_settings.json	7/17/2026 3:45 PM	JSON File	1 KB
bmc_supported_settings.json	7/17/2026 3:45 PM	JSON File	1 KB
events.json	7/17/2026 3:45 PM	JSON File	2 KB